



## DESIGN AND BEHAVIOUR GUIDE

# Airwave Style Guide

Airwave, Airwave Hub and Airwave Companion, with the Android app and the Discord bot: one Aqua system, its rules, every screen, and how to keep it true.

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### PROFILE

snow-leopard-itunes-9

### COVERAGE

73 surfaces · 13 journeys · 85 rules

### SOURCE FINGERPRINT

14aac31baf1723ec

### INTERACTIVE VERSION

<docs/design/styleguide.html>

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# Airwave Style Guide

snow-leopard-itunes-9 · three apps · 237 tokens checked · 73 surfaces · source 14aac31baf1723ec

One guide for the three apps of Airwave — Airwave, the player (a page, also inside the Android app); Airwave Hub, the admin window the container serves; and Airwave Companion, the Windows app — and for the words the Discord bot uses. All three are a reconstruction of Apple's 2009–2010 interface, and each is built from a design file in `design/frontends/`. Every value here is read from `tokens.json` as this page is built, the windows are drawn with the stylesheets and kits the products ship, and every rule, screen and journey is read from `design/` — so the page cannot describe one thing and show another.

## The rules that outrank the visuals

A styleguide that opens with colour teaches the wrong lesson. These four decide whether a control is drawn at all, and each has overruled the aesthetics here at least once.

UX-PRIN-001

### A control that cannot act is not drawn

No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state. The equalizer window ships without traffic lights: three circles that close nothing would be decoration.

UX-PRIN-002

### An unavailable capability says so where the choice is made

Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting. The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.

UX-PRIN-003

### Say what the code does, not what the feature is called

Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes. Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.

UX-PRIN-004

### A disabled control is still legible about why

Greyed, not gone. The reason is part of the accessible name, not only a tooltip. “500 Hz band, not used by 528 Hz (MI)”.

## Brand and voice

Airwave is a self-hosted music suite: a player that works on the files you already have, a hub you run yourself, and a companion that brings a Windows library to it. Its look is a careful reconstruction of Apple's 2009–2010 interface — the Snow Leopard window, the iTunes 9 list, the iPod's Now Playing screen — chosen for its density and its honesty about state. The reference is `docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md`.



#### Airwave

music-player/public/icon.svg

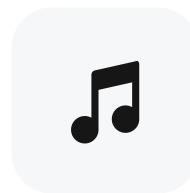
Also the Android launcher (brand colour #3b74c8) and the maskable PWA icon.



#### Airwave Companion

windows-companion/resources/icon.svg

The same tile with a folder behind the note: the one app that reaches the filesystem.



#### The note

packages/aqua-ui/src/icons/glyphs.tsx

Both icons draw this exact path; a unit test fails if they drift apart again.

## USING THE MARK

- The tile is a rounded square (radius 112 on a 512 grid), a three-stop blue gradient and a soft gloss over the top half. Keep all three; the gloss is the period detail.
- Never recolour the note or redraw it by hand. Use the SVG or the `note` glyph.
- The blue tile is the product's mark. Inside the interface blue is never decoration — it means selected, active or default (see Colour).
- ● **PROPOSED** Clear space and minimum size have not been specified. Until they are, keep a quarter of the tile's width clear around it and do not use it below 16px; the tray icon is its own simplified drawing.

## VOICE

Plain, specific and a little dry. The product explains what it is doing and what it cannot do, in the place you are looking. It does not apologise, cheer or use exclamation marks, and it never claims more than the code does (UX-PRIN-003).

### VOICE-001

#### Empty states say what to do next ● OBSERVED

An empty list says what is missing and what to do next, in a few plain words: one sentence inside the list in the Airwave windows, a title and a line in the component library ("Nothing queued"). No apologies, no exclamation marks. "No folders yet — add the one this PC keeps these in." · "No guides yet — the guide will list channels without programmes."

### VOICE-002

#### Errors explain the problem and the way out ● ADOPTED

Name what failed in the person's terms and what fixes it ("Add a bot token first (Admin → Discord → Token)"). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

### VOICE-003

#### One product name, set in one place ● ADOPTED

The suite is Airwave and its apps are Airwave, Airwave Hub and Airwave Companion; the names come from BRANDING in TypeScript, and "Now Playing" is the name of the player's home page only (DEC-025). Native resources that cannot import BRANDING (Android strings.xml, the SVG icons' aria-labels, the Electron builder config) repeat it and must be changed together. Machine identifiers keep their spelling.

## WORDS WE USE

| SAY                | MEANING                                                                                                                                                       | NOT                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Airwave</b>     | The suite, and the player's own name. The other two apps are Airwave Hub and Airwave Companion; in running text, "the player", "the hub" and "the companion". | Now Playing as a product name; names invented per screen |
| <b>Now Playing</b> | The player's home page: what is playing now. Nothing else.                                                                                                    | The suite, the hub or the companion                      |

| SAY                            | MEANING                                                   | NOT                       |
|--------------------------------|-----------------------------------------------------------|---------------------------|
| <b>Solo / Shared listening</b> | The two modes on the status bar.                          | Party mode, sync mode     |
| <b>Group</b>                   | People listening to one shared queue on a hub.            | Room, session             |
| <b>Pair</b>                    | Connecting a device to a hub, confirmed by a fingerprint. | Log in, register a device |
| <b>Shared link</b>             | A public link to a track or playlist the hub serves.      | Share URL, public page    |
| <b>Fetch</b>                   | Running yt-dlp or spotDL on this machine.                 | Rip, grab                 |
| <b>Hub library</b>             | Files that live on the hub's data volume.                 | Cloud, server music       |

● OBSERVED The vocabulary above is how the code and the docs already talk; it has not been reviewed as a glossary.

REAL COPY

|                                                                                                                                   |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>No folders yet</b></p> <p>— add the one this PC keeps these in.</p> <p>Companion · Library ▶ Saved Music</p>                | <p><b>Nothing needs you.</b></p> <p>The whole of an empty “Needs attention” list.</p> <p>Hub · Overview</p>                                                           |
| <p><b>A link that doesn't answer</b></p> <p>keeps what it last held, and is tried again by itself.</p> <p>Companion · Live TV</p> | <p><b>A stream never implies a download.</b></p> <p>Where a provider doesn't allow saving, there's no button — not one that fails.</p> <p>Hub · Music ▶ Downloads</p> |

Three apps, one system

One period, one set of rules, three apps — and each app's interface comes from its own design file in `design/frontends/`. The design file is the authority for how its app looks; this guide is the authority for what the three share: the rules, the words, the ledger of screens, and the checks that keep them in step.

| APP                                                        | DESIGN                               | WHAT IT WEARS                                                                                                                                                                                                                                                                                                                 | WHERE TO LOOK HERE                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Airwave</b><br>music-player · android · local-helper    | airwave-<br>now-<br>playing.h<br>tml | The design file itself, served as the shell: <code>music-player/index.html</code> is generated from it by <code>make-shell.py</code> , with its own custom properties, its own script for the drawn controls, and <code>src/shell/bridge.ts</code> behind its seams (DEC-019). It imports nothing from the component library. | The served shell in Elements (screenshots of the production build), the <code>player-shell-*</code> rows in Every screen and the <code>NP-*</code> rules. The page skin is the component library's version of the same look, kept in <code>music-player/src</code> but not served. |
| <b>Airwave Hub</b><br>docker-container/src/web             | airwave-<br>hub.html                 | <code>airwave-window.css</code> and <code>airwave-hub.css</code> (generated from the designs by <code>pnpm build:window-css</code> ), AquaArt's drawings, its own kit <code>ui.tsx</code> , and <code>styles.css</code> for what the design left to its script.                                                               | The Airwave window in Elements, and its screens in the mockups.                                                                                                                                                                                                                    |
| <b>Airwave Companion</b><br>windows-companion/src/renderer | airwave-<br>companion<br>.html       | <code>airwave-window.css</code> , the same AquaArt, its own kit <code>ui.tsx</code> , and <code>styles.css</code> for what a real window needs (it fills its frame; its chrome is the title bar).                                                                                                                             | The Airwave window in Elements, and its screens in the mockups.                                                                                                                                                                                                                    |

|                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>The Airwave window</b></p> <p><code>--win-*</code>, <code>--chrome-*</code>, <code>--ink*</code>, <code>--aq-*</code> · <code>airwave-window.css</code>, <code>airwave-hub.css</code> · Lucida Grande · light only (DEC-003) · Airwave Hub, Airwave Companion</p> | <p><b>The component library</b></p> <p><code>--aqua-*</code>, <code>--np-*</code>, <code>--lib-*</code> · <code>aqua.css</code>, <code>now-playing.css</code> · <code>packages/aqua-ui</code> React components · imported by the player's React source, the gallery and this guide; no served product (DEC-026)</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

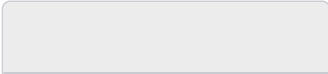
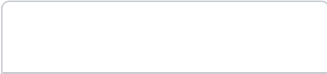
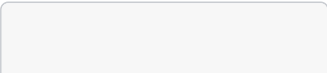
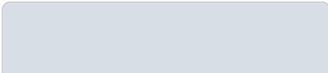
## Colour

Chosen by role, never by picking a hex. Blue means *selected*, *active*, or *the default action* — it is not a brand colour and it is never decoration. Three palettes follow, straight from the token file: the Airwave window's (what the hub and the companion wear), then the component library's two.

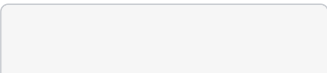
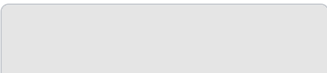
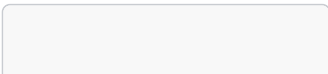
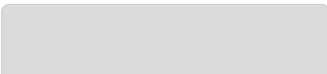
### THE AIRWAVE WINDOW — 56 COLOURS

Owned by the design files: `airwave-window.css` and `airwave-hub.css` are generated from them, and `tokens.json` records the values so `pnpm styleguide:check` notices when a design changes one. The last group is the hub's own: the design's `--ink-3`, `--success` and `--danger` fail 4.5:1 as text on white, so the hub reads sentences in darker inks of the same hues (DEC-022).

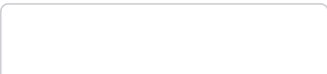
#### WINDOW AND CONTENT

|                                                                                                              |                                                                                                            |                                                                                                             |                                                                                                            |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <br>desktop<br>#6D737A      | <br>winOutline<br>#747474 | <br>winBody<br>#ECECEC    | <br>content<br>#FFFFFF  |
| <br>contentMuted<br>#F7F7F7 | <br>rowStripe<br>#F1F6FB  | <br>rowDivider<br>#D8DEE5 | <br>wellEdge<br>#9A9A9A |

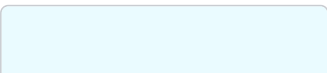
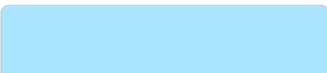
#### CHROME AND HEADERS

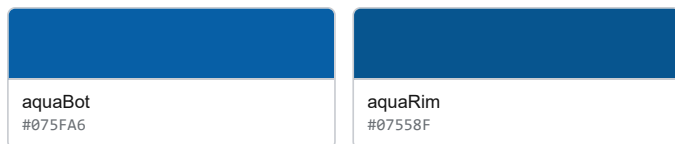
|                                                                                                             |                                                                                                              |                                                                                                               |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <br>chromeTop<br>#F6F6F6  | <br>chromeUpper<br>#E5E5E5 | <br>chromeLower<br>#C4C4C4 | <br>chromeBottom<br>#AFAFAF |
| <br>chromeSep<br>#858585 | <br>chromeHi<br>#FFFFFF   | <br>headTop<br>#F8F8F8    | <br>headBot<br>#DADADA     |
| <br>headRule<br>#ADADAD  | <br>headInk<br>#3A3D41    |                                                                                                               |                                                                                                                 |

#### SELECTION AND FOCUS

|                                                                                                          |                                                                                                          |                                                                                                           |                                                                                                               |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <br>selTop<br>#69B5F3 | <br>selMid<br>#378DDA | <br>selBot<br>#1764B2 | <br>selBorder<br>#0F4E8B |
| <br>selInk<br>#FFFFFF | <br>focus<br>#3F9FE8  | <br>aqRing<br>#7EA8E0 |                                                                                                               |

#### AQUA GEL

|                                                                                                            |                                                                                                           |                                                                                                            |                                                                                                               |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <br>aquaSpec<br>#EAFBFF | <br>aquaTop<br>#A9E5FF | <br>aquaMid<br>#38B6F7 | <br>aquaLower<br>#0D83D6 |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|



## INK



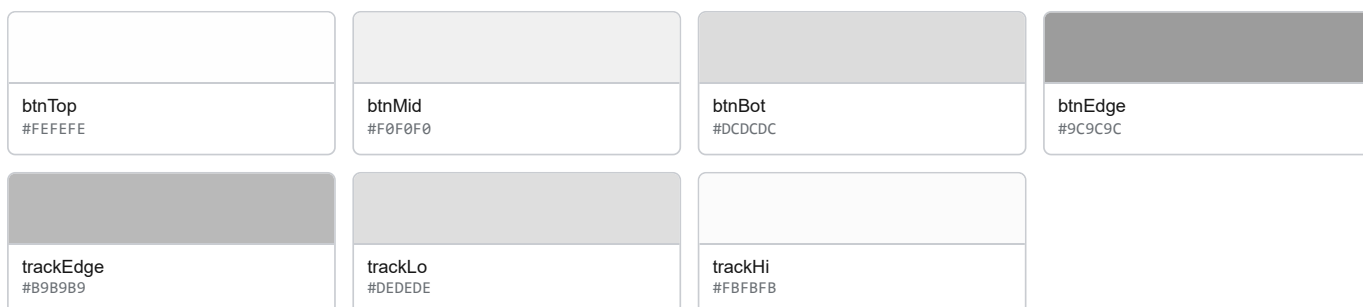
## STATUS AND THE FIRST-RUN GATE



## LCD PLATE



## FACES AND TRACKS



## CONTRAST INKS (HUB WEB, DEC-022)



WINDOW AND CONTENT

|                                                 |                                                  |                                               |                                            |
|-------------------------------------------------|--------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| <div></div> <div>desktop<br/>#6D737A</div>      | <div></div> <div>windowOutline<br/>#747474</div> | <div></div> <div>windowBody<br/>#ECECEC</div> | <div></div> <div>content<br/>#FFFFFF</div> |
| <div></div> <div>contentMuted<br/>#F7F7F7</div> | <div></div> <div>rowStripe<br/>#F1F6FB</div>     | <div></div> <div>rowDivider<br/>#D8DEE5</div> |                                            |

CHROME

|                                                    |                                                    |                                                |                                                 |
|----------------------------------------------------|----------------------------------------------------|------------------------------------------------|-------------------------------------------------|
| <div></div> <div>chromeTop<br/>#F6F6F6</div>       | <div></div> <div>chromeUpper<br/>#E5E5E5</div>     | <div></div> <div>chromeLower<br/>#C4C4C4</div> | <div></div> <div>chromeBottom<br/>#AFAFAF</div> |
| <div></div> <div>chromeSeparator<br/>#858585</div> | <div></div> <div>chromeHighlight<br/>#FFFFFF</div> |                                                |                                                 |

SOURCE LIST

|                                               |                                                  |                                                   |                                                     |
|-----------------------------------------------|--------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|
| <div></div> <div>sidebarTop<br/>#DCE8F4</div> | <div></div> <div>sidebarBottom<br/>#BCEEE0</div> | <div></div> <div>sidebarDivider<br/>#93A3B2</div> | <div></div> <div>sidebarGroupText<br/>#536473</div> |
|-----------------------------------------------|--------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|

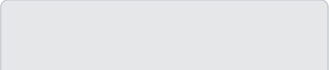
SELECTION AND FOCUS

|                                                  |                                                 |                                                    |                                                    |
|--------------------------------------------------|-------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| <div></div> <div>selectionTop<br/>#69B5F3</div>  | <div></div> <div>selectionMid<br/>#378DDA</div> | <div></div> <div>selectionBottom<br/>#1764B2</div> | <div></div> <div>selectionBorder<br/>#0F4E8B</div> |
| <div></div> <div>selectionText<br/>#FFFFFF</div> | <div></div> <div>focus<br/>#3F9FE8</div>        |                                                    |                                                    |

AQUA GEL

|                                                 |                                            |                                            |                                              |
|-------------------------------------------------|--------------------------------------------|--------------------------------------------|----------------------------------------------|
| <div></div> <div>aquaSpecular<br/>#EAFBFF</div> | <div></div> <div>aquaTop<br/>#A9E5FF</div> | <div></div> <div>aquaMid<br/>#38B6F7</div> | <div></div> <div>aquaLower<br/>#0D83D6</div> |
| <div></div> <div>aquaBottom<br/>#075FA6</div>   | <div></div> <div>aquaRim<br/>#07558F</div> |                                            |                                              |

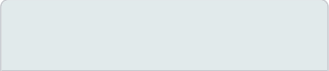
## GRAPHITE

|                                                                                   |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| graphiteTop<br>#E5E7E9                                                            | graphiteMid<br>#AEB4B9                                                            | graphiteBottom<br>#737A80                                                          |

## TEXT

|                                                                                   |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| text<br>#161616                                                                   | textSecondary<br>#565B60                                                          | textDisabled<br>#96999C                                                            |

## LCD

|                                                                                   |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| lcdTop<br>#E1EAE8                                                                 | lcdBottom<br>#AEBCC0                                                              | lcdText<br>#172326                                                                 |

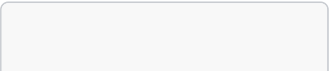
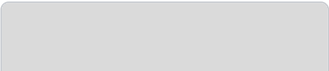
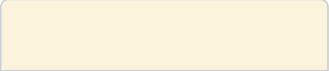
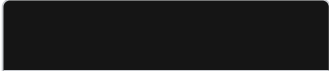
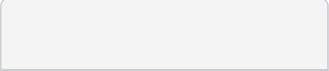
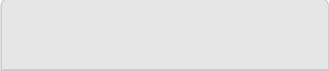
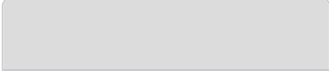
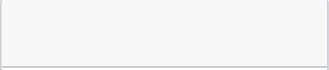
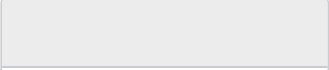
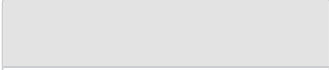
## STATUS

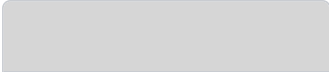
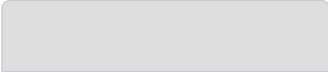
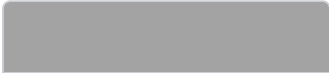
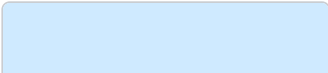
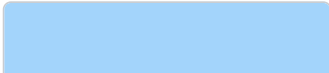
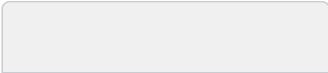
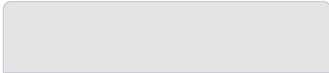
|                                                                                   |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| danger<br>#D64A44                                                                 | warning<br>#D9A431                                                                | success<br>#4E9D47                                                                 |

## TRAFFIC LIGHTS

|                                                                                     |                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |  |
| trafficClose<br>#E9635B                                                             | trafficMinimize<br>#E3B64B                                                          | trafficZoom<br>#58A953                                                               |

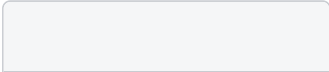
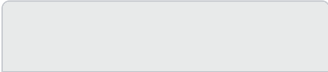
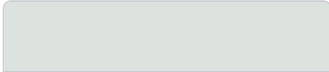
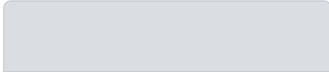
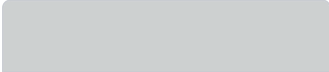
## OTHER

|                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |
| headerTop<br>#F8F8F8                                                                | headerBottom<br>#DADADA                                                             | headerRule<br>#ADADAD                                                                | headerText<br>#3A3D41                                                                 |
|  |  |  |  |
| warningBg<br>#F8F3DC                                                                | warningEdge<br>#D8B75E                                                              | ctlInk<br>#151515                                                                    | ctlFace1<br>#FDFDFD                                                                   |
|  |  |  |  |
| ctlFace2<br>#F4F4F4                                                                 | ctlFace3<br>#E6E6E6                                                                 | ctlFace4<br>#DCDCDC                                                                  | ctlFaceHover1<br>#FFF                                                                 |
|  |  |  |  |
| ctlFaceHover2<br>#F8F8F8                                                            | ctlFaceHover3<br>#ECECEC                                                            | ctlFaceHover4<br>#E3E3E3                                                             | ctlFaceActive1<br>#C9C9C9                                                             |

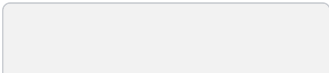
|                                                                                                                                |                                                                                                                         |                                                                                                                                |                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <br><b>ctIFaceActive2</b><br>#D6D6D6          | <br><b>ctIFaceActive3</b><br>#DEDEDE   | <br><b>ctIFaceDisabled1</b><br>#FBFBFB       | <br><b>ctIFaceDisabled2</b><br>#F1F1F1      |
| <br><b>ctIBorder</b><br>#A3A3A3               | <br><b>ctIBorderBottom</b><br>#8D8D8D  | <br><b>ctIBorderDisabled</b><br>#C9C9C9      | <br><b>ctIFocusRing</b><br>#0D61A7          |
| <br><b>ctIDefaultInk</b><br>#06203A           | <br><b>ctIDefault1</b><br>#CFEAFB      | <br><b>ctIDefault2</b><br>#A3D4FB            | <br><b>ctIDefault3</b><br>#74B9F5           |
| <br><b>ctIDefault4</b><br>#5AA8EF             | <br><b>ctIDefaultBorder</b><br>#4684C8 | <br><b>ctIDefaultBorderBottom</b><br>#356DB0 | <br><b>ctIBoxInk</b><br>#1A2733             |
| <br><b>ctIBox1</b><br>#FDFDFD                 | <br><b>ctIBox2</b><br>#F0F0F0          | <br><b>ctIBox3</b><br>#E4E4E4                | <br><b>ctIBoxBorder</b><br>#A0A0A0          |
| <br><b>ctIBoxOn1</b><br>#7DB4E6               | <br><b>ctIBoxOn2</b><br>#4F92D6        | <br><b>ctIBoxOn3</b><br>#3F81C6              | <br><b>ctIBoxOnBorder</b><br>#3D76AD        |
| <br><b>ctIFieldBg</b><br>#FFF               | <br><b>ctIFieldBorder</b><br>#9A9A9A | <br><b>ctIFieldBorderTop</b><br>#7D7D7D    | <br><b>ctIFieldFocusBorder</b><br>#1768A7 |
| <br><b>ctIFieldInvalidBorder</b><br>#B8433D | <br><b>ctIPlaceholder</b><br>#8B9095 | <br><b>dangerInk</b><br>#7A1712            |                                                                                                                                |

## COMPONENT LIBRARY, PAGE SKIN — 25 TOKENS

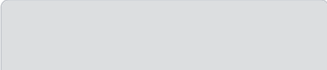
### STATUS BAR

|                                                                                                                              |                                                                                                                   |                                                                                                                    |                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <br><b>barTop</b><br>#F5F6F7              | <br><b>barUpper</b><br>#E8EAEA | <br><b>barLower</b><br>#DCE2DE | <br><b>barBottom</b><br>#D9DFE1 |
| <br><b>barEdge</b><br>RGB(31 41 45 / 22%) |                                                                                                                   |                                                                                                                    |                                                                                                                      |

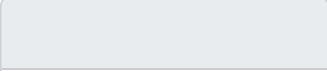
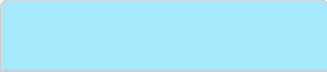
### FIELD

|                                                                                                                   |                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <br><b>fieldTop</b><br>#F2F2F2 | <br><b>fieldBottom</b><br>#FFFFFF |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|

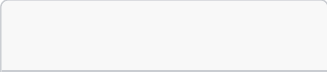
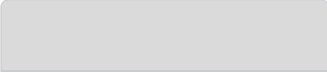
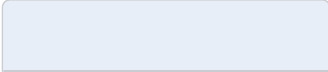
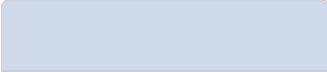
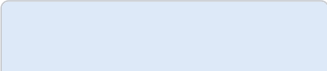
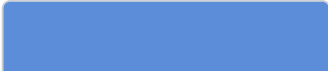
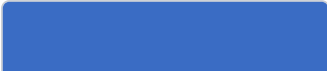
## MODE SWITCH

|                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| modeFaceTop<br>#FDFDFD                                                            | modeFaceBottom<br>#DCDEE0                                                         | modeOnTop<br>#767C85                                                               | modeOnBottom<br>#A9B0B8                                                             |

## SCRUBBER

|                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| railTop<br>#E9ECEE                                                                | railBottom<br>#FFFFFF                                                             | railFillTop<br>#8FB9E4                                                             | railFillMid<br>#2B9BEC                                                              |
|  |                                                                                   |                                                                                    |                                                                                     |
| railFillBottom<br>#A5E9FD                                                         |                                                                                   |                                                                                    |                                                                                     |

## LIST

|                                                                                    |                                                                                    |                                                                                     |                                                                                      |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|   |   |   |   |
| listHeaderTop<br>#F8F8F8                                                           | listHeaderBottom<br>#DADADA                                                        | listSortTop<br>#E8EEF7                                                              | listSortBottom<br>#CFDBEB                                                            |
|  |  |  |  |
| listStripe<br>#F2F5F9                                                              | listPlaying<br>#DDE9F8                                                             | listSelectTop<br>#5B8DD9                                                            | listSelectBottom<br>#3A6CC4                                                          |

## MARKS

|                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |
| live<br>#CC1027                                                                     | star<br>#F0A422                                                                     |

## DARK

The page skin has a full dark palette and it is opt-out: every dark rule is guarded `@media (prefers-color-scheme: dark) { :root:not([data-np-theme='light']) { ... } }`, so the system preference wins unless a page pins itself light. Redefine *only* tokens inside that block. Neither the Airwave window nor the library's window skin has a dark palette, because Snow Leopard had none (DEC-003); the player's shell carries its own dark values (NPD-002).

The page skin's colours that change in the dark scheme, read from the shipped stylesheet when this page was built (88 properties). Light on the left, dark on the right.

|                                                          |                                                                                                                                                                         |                                                                |                                                                                                                                                                             |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --ctx-bg<br>rgb(247 247 250 / 88%) → rgb(40 42 48 / 90%) |   | --lib-bg<br>#fff → #16181c                                     |   |
| --ctx-ink<br>#1b1c1f → #eef0f3                           |   | --lib-border<br>#ababab → #08090b                              |   |
| --ctx-sep<br>rgb(0 0 0 / 12%) → rgb(255 255 255 / 12%)   |   | --lib-head-bot<br>#dadada → #26292e                            |   |
| --lib-accent<br>#2b6fd6 → #6aa3f2                        |   | --lib-head-emboss<br>rgb(255 255 255 / 80%) → rgb(0 0 0 / 55%) |   |
| --lib-alt<br>#f2f5f9 → #1b1f26                           |   | --lib-head-hot-bot<br>#cfcfcf → #303338                        |   |

|                                                 |                                                                                     |                                                |                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| --lib-head-hot-top                              |    | --np-bar-4                                     |    |
| #efefef → #45484f                               |                                                                                     | #dfe4e1 → #212429                              |                                                                                       |
| --lib-head-ink                                  |    | --np-bar-5                                     |    |
| #3a3d41 → #c6cbd3                               |                                                                                     | #dce2de → #1e2126                              |                                                                                       |
| --lib-head-rule                                 |    | --np-bar-bottom                                |    |
| #adadad → #0f1114                               |                                                                                     | #d9dfe1 → #1a1d22                              |                                                                                       |
| --lib-head-sep                                  |    | --np-bar-edge                                  |    |
| #ccc → #17191d                                  |                                                                                     | rgb(31 41 45 / 22%) → rgb(0 0 0 / 60%)         |                                                                                       |
| --lib-head-top                                  |    | --np-bar-emboss                                |    |
| #f8f8f8 → #3a3d44                               |                                                                                     | rgb(255 255 255 / 55%) → transparent           |                                                                                       |
| --lib-icon                                      |    | --np-bar-top                                   |    |
| #8a8f98 → #8d939c                               |                                                                                     | #f5f6f7 → #34373d                              |                                                                                       |
| --lib-ink                                       |    | --np-field-bottom                              |    |
| #000 → #f2f3f5                                  |                                                                                     | #fff → #24272d                                 |                                                                                       |
| --lib-np                                        |    | --np-field-hint                                |    |
| #dde9f8 → #22354f                               |                                                                                     | #ababab → #7c8087                              |                                                                                       |
| --lib-soft                                      |    | --np-field-icon                                |    |
| #222326 → #d4d8df                               |                                                                                     | #8e8e91 → #85898f                              |                                                                                       |
| --lib-sort-bot                                  |    | --np-field-lip                                 |    |
| #cfdbeb → #233043                               |                                                                                     | rgb(255 255 255 / 70%) → rgb(255 255 255 / 8%) |                                                                                       |
| --lib-sort-hot-bot                              |  | --np-field-mid                                 |  |
| #c2d2e6 → #2b3b52                               |                                                                                     | #fbfbfb → #1b1e23                              |                                                                                       |
| --lib-sort-hot-top                              |  | --np-field-ring                                |  |
| #dde7f4 → #36485f                               |                                                                                     | rgb(0 0 0 / 20%) → rgb(0 0 0 / 70%)            |                                                                                       |
| --lib-sort-top                                  |  | --np-field-shadow                              |  |
| #e8eef7 → #2c3a4e                               |                                                                                     | rgb(0 0 0 / 32%) → rgb(0 0 0 / 75%)            |                                                                                       |
| --lib-star                                      |  | --np-field-top                                 |  |
| #f0a422 → #f4b241                               |                                                                                     | #f2f2f2 → #141619                              |                                                                                       |
| --np-accent                                     |  | --np-ink                                       |  |
| #2b6fd6 → #6aa3f2                               |                                                                                     | #1b1c1f → #f1f2f4                              |                                                                                       |
| --np-avatar-face                                |  | --np-key-ink                                   |  |
| #f3f3f3 → #2b2e34                               |                                                                                     | #1b1c1f → #ccd1d8                              |                                                                                       |
| --np-avatar-lip                                 |  | --np-key-muted                                 |  |
| rgb(255 255 255 / 70%) → rgb(255 255 255 / 10%) |                                                                                     | #8b9099 → #7f858e                              |                                                                                       |
| --np-avatar-ring                                |  | --np-key-on                                    |  |
| rgb(0 0 0 / 26%) → rgb(0 0 0 / 55%)             |                                                                                     | #2f6ec0 → #5fa0ea                              |                                                                                       |
| --np-bar-1                                      |  | --np-list-alt                                  |  |
| #eef0f1 → #2f3238                               |                                                                                     | #f2f5f9 → #1b1f26                              |                                                                                       |
| --np-bar-2                                      |  | --np-list-badge                                |  |
| #e8eaea → #2a2d33                               |                                                                                     | #dfe3e9 → #2f333b                              |                                                                                       |
| --np-bar-3                                      |  | --np-list-badge-ink                            |  |
| #e2e6e5 → #25282e                               |                                                                                     | #3b3f47 → #d6dae1                              |                                                                                       |

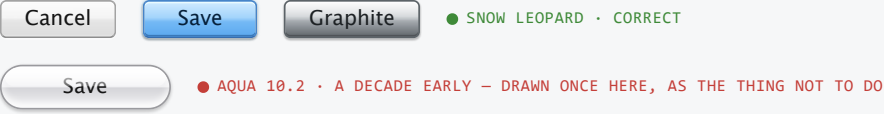
|                                                |  |                                                |  |
|------------------------------------------------|--|------------------------------------------------|--|
| --np-list-border                               |  | --np-vol-b                                     |  |
| #ababab → #08090b                              |  | #b7bdc4 → #878e96                              |  |
| --np-list-icon                                 |  | --np-vol-face                                  |  |
| #8a8f98 → #8d939c                              |  | #fff → #1b1e23                                 |  |
| --np-live                                      |  | --srch-art-bot                                 |  |
| #cc1027 → #ff4d62                              |  | #e3e3e3 → #24262b                              |  |
| --np-live-halo                                 |  | --srch-art-ink                                 |  |
| rgb(204 16 39 / 32%) → rgb(255 77 98 / 36%)    |  | #a9a9a9 → #8d939c                              |  |
| --np-meta                                      |  | --srch-art-ring                                |  |
| #575757 → #9ba0a8                              |  | rgb(0 0 0 / 14%) → rgb(255 255 255 / 10%)      |  |
| --np-page                                      |  | --srch-art-top                                 |  |
| #fff → #0e0f12                                 |  | #fbfbfb → #33363d                              |  |
| --np-rail-line                                 |  | --srch-bg                                      |  |
| rgb(0 0 0 / 16%) → rgb(0 0 0 / 72%)            |  | #f9f9f9 → #26282e                              |  |
| --np-seg-border                                |  | --srch-border                                  |  |
| rgb(0 0 0 / 36%) → rgb(0 0 0 / 70%)            |  | rgb(0 0 0 / 25%) → rgb(0 0 0 / 65%)            |  |
| --np-seg-divider                               |  | --srch-chev                                    |  |
| rgb(0 0 0 / 30%) → rgb(0 0 0 / 55%)            |  | #8f8f8f → #9aa0a9                              |  |
| --np-seg-face-bottom                           |  | --srch-dot                                     |  |
| #dcdee0 → #292c31                              |  | #c9c9c9 → #4a4e56                              |  |
| --np-seg-face-mid                              |  | --srch-dot-on                                  |  |
| #f0f1f2 → #33363c                              |  | #7c7c7c → #b8bdc5                              |  |
| --np-seg-face-top                              |  | --srch-head-bot                                |  |
| #fdfdfd → #3a3d44                              |  | #eee → #25272d                                 |  |
| --np-seg-ink                                   |  | --srch-head-top                                |  |
| #43464b → #c3c7cf                              |  | #f6f6f6 → #2e3138                              |  |
| --np-seg-lip                                   |  | --srch-hot                                     |  |
| rgb(255 255 255 / 65%) → rgb(255 255 255 / 8%) |  | #ececec → #32353c                              |  |
| --np-seg-on-bottom                             |  | --srch-ink                                     |  |
| #a9b0b8 → #23262c                              |  | #2c2c2c → #eef0f3                              |  |
| --np-seg-on-mid                                |  | --srch-lip                                     |  |
| #939aa3 → #1c1f24                              |  | rgb(255 255 255 / 60%) → rgb(255 255 255 / 7%) |  |
| --np-seg-on-top                                |  | --srch-rule                                    |  |
| #767c85 → #14161a                              |  | #dedede → rgb(0 0 0 / 50%)                     |  |
| --np-vol-a                                     |  | --srch-soft                                    |  |
| #9aa1a9 → #6e747c                              |  | #9b9b9b → #8d939c                              |  |

## Aqua highlights — where the blue is allowed to shine

The specimens from here to Motion are the component library's controls. The principles are the Airwave window's too — the same light source, the same restraint about blue — but its controls are AquaArt's drawings, shown under The Airwave window.

Snow Leopard had flattened most of Aqua's gel by 2009, but not all of it: selection, the default action, the scroller and a few controls kept the glass. Those are exactly the places this system keeps it, and this is each one, live.

## THE DEFAULT ACTION



The one button on a form that carries the Aqua blue: a pale cyan cap over a saturated body, a specular hairline at the top, a dark rim, and a slow pulse. The shape stays the Snow Leopard rounded rect — no pill, no glass lozenge, no gloss step across the middle.

## CONTROLS THAT KEEP THE GEL



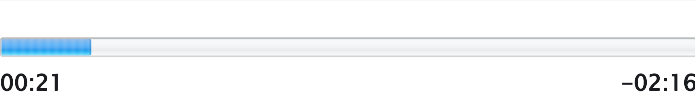
The checked box, the pop-up's end cap, the aqua-tinted segmented control and the indeterminate progress bar's barber pole are the last four places a form is blue.

## THE OVERLAY SCROLLER



The full five-stop Aqua gel survives at strength in exactly one place: the list's overlay scroller, which fades in while you scroll and out 900 ms after you stop.

## THE IPOD RAIL



Not gel, but the other blue on the page: the iPod Classic's progress bar, 12 px, near-square ends, and a fill that brightens downward — sky at the top through a saturated dip at 62% to a cyan glow along the lower lip. No knob: the device had none.

And the focus ring — -aqua-focus #3F9FE8 with a soft outer glow — is blue everywhere, because the thing that has the keyboard is the one thing on screen that must never be ambiguous.

## Type

Two families, one scale. The window skin — the Airwave window and the library's alike — sets Lucida Grande, as OS X did; the Airwave window runs 12px body text with 11px secondary lines and 10px chips. The page skin sets **Helvetica** — the face the iPod classic drew its Now Playing

screen in, which is what the hero is modelled on — with Arial as its metric-compatible stand-in where Helvetica is absent. Each specimen is at its true size.

PAGE SKIN · HELVETICA

|      |                                                 |                    |
|------|-------------------------------------------------|--------------------|
| 13px | Sundress — A\$AP Rocky                          | --aqua-font-system |
| 12px | Marlow & the Tidewater — Quiet Arithmetic       | --aqua-font-view   |
| 11px | Searches this device only until a hub is paired | --aqua-font-small  |
| 10px | PREAMP · 32 · 64 · 125 · 250 · 500 · 1K · 2K    | --aqua-font-label  |
| 9px  | Mini controls only                              | --aqua-font-mini   |

WINDOW SKIN · LUCIDA GRANDE

|      |                                         |                    |
|------|-----------------------------------------|--------------------|
| 13px | Airwave Hub                             | --aqua-font-system |
| 12px | Copper Meridian — Orbital Cartographers | --aqua-font-view   |
| 11px | 1,240 songs, 2.6 hours, 245 MB          | --aqua-font-small  |
| 10px | LIBRARY · PLAYLISTS · CONNECTED         | --aqua-font-label  |
| 9px  | Mini controls only                      | --aqua-font-mini   |

Weights are **400** and **700**. There is no 500 and no 600: the profile predates variable UI fonts, and a half-weight reads as a rendering fault beside a true bold. Numbers that change in place — timers, dB values, tempo, track times — take font-variant-numeric: tabular-nums, so a changing digit does not shift the ones beside it.

Space

One scale, no arbitrary values. Each bar is drawn at its true width.

|                                 |      |   |
|---------------------------------|------|---|
| --aqua-space-micro              | 2px  |   |
| --aqua-space-tight              | 4px  | ■ |
| --aqua-space-inline             | 6px  | ■ |
| --aqua-space-control            | 8px  | ■ |
| --aqua-space-small-group        | 10px | ■ |
| --aqua-space-group              | 12px | ■ |
| --aqua-space-frame-top          | 14px | ■ |
| --aqua-space-strong-group       | 16px | ■ |
| --aqua-space-window-edge        | 20px | ■ |
| --aqua-space-major              | 24px | ■ |
| --aqua-space-bottom-bar-regular | 32px | ■ |

SIZES

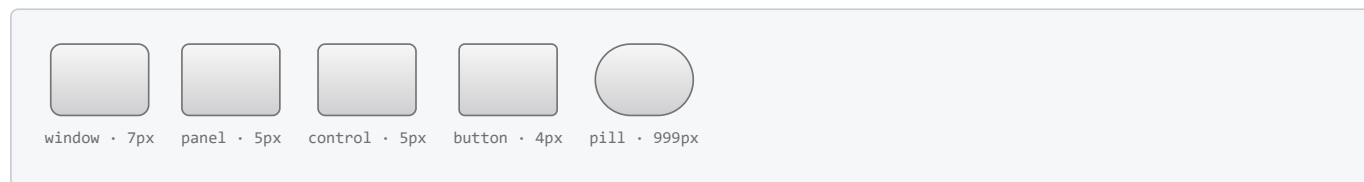
|                              |                            |                       |                  |
|------------------------------|----------------------------|-----------------------|------------------|
| visualControlRegular<br>22PX | visualControlSmall<br>19PX | searchRegular<br>22PX | tableRow<br>20PX |
| listRow<br>18PX              | railHeight<br>12PX         | transportKey<br>48PX  | playKey<br>64PX  |
| barHeight<br>40PX            | sidebarDefault<br>196PX    |                       |                  |

**32 px** (--aqua-hit) is the floor for a pointer target, and **44 px** on a coarse pointer, which is what both platform guidelines ask for. A 12 px rail and an 18 px row are period-correct and stay that size — they get their hit area from an invisible ::after that expands past the visual. Never shrink a hit target to match a visual.

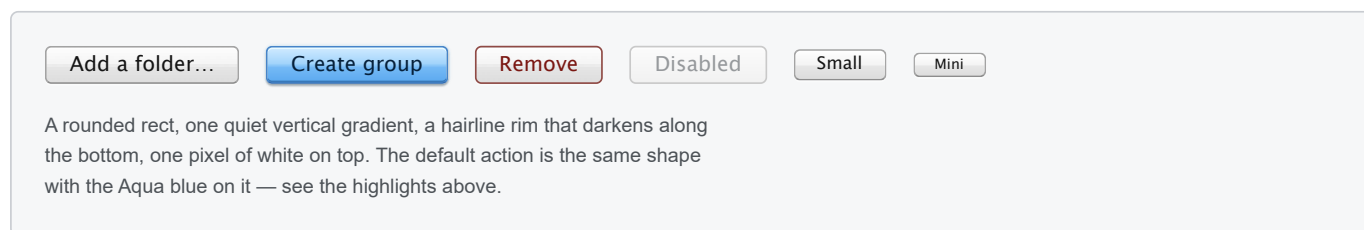
## Shape and material

One light source, above and slightly forward. Raised things are light at the top and dark at the bottom; recessed things are the reverse. Every shadow follows from that, and a rule that contradicts it looks broken even when the colours are right.

### RADII

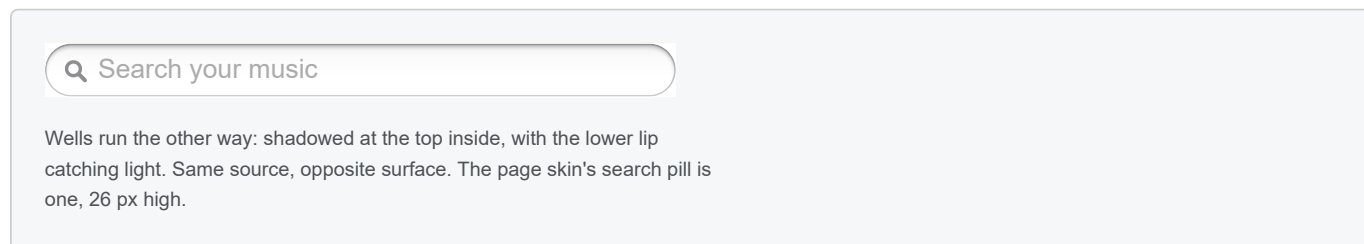


### THE PUSH BUTTON

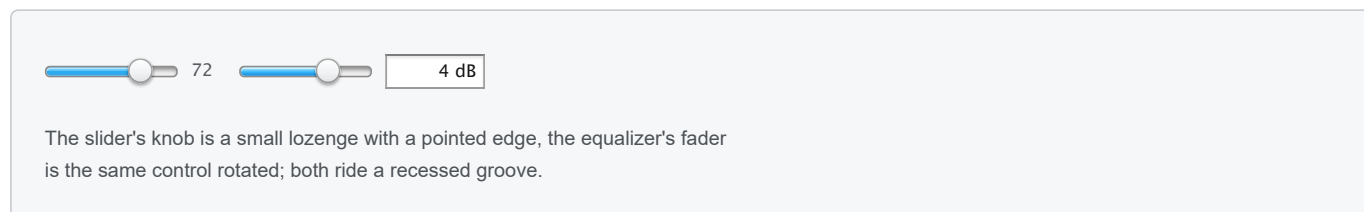


```
background: linear-gradient(to bottom, #fdfdfd 0%, #f4f4f4 47%, #e6e6e6 53%, #dcdcdc 100%);
border: 1px solid #a3a3a3;
border-bottom-color: #8d8d8d;
border-radius: 4px;
box-shadow:
  inset 0 1px 0 rgb(255 255 255 / 90%), /* the pixel of light on the top edge */
  0 1px 0 rgb(255 255 255 / 55%);      /* the surface beneath catching it */
```

### RECESSED FIELDS

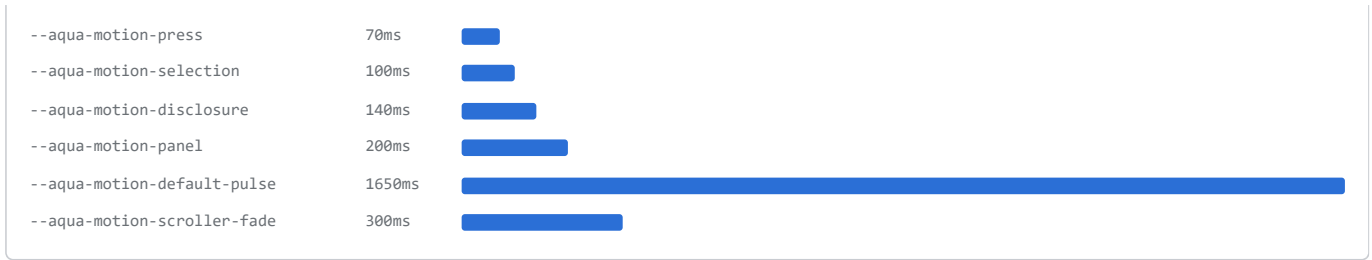


### FADERS



## Motion

Confirmation, never decoration: nothing animates that a person did not just cause. Everything eases on `cubic-bezier(0.2, 0.7, 0.2, 1)`. Bars are drawn in proportion.



**Reduced motion is a contract, not a courtesy.** Under `prefers-reduced-motion`: reduce the live dot stops pulsing, the marquee does not scroll (the cell keeps its ellipsis), the sheet's travel drops to zero and it fades only, the constellation opens as a table rather than an animated field, and the 3D stage stops its idle drift. Two variables carry most of it — `--aqua-anim-state: paused` and `--aqua-sheet-travel: 0` — so a new animation should read one of them rather than inventing a third.

What a listener opens is `music-player/index.html`, the Airwave design served as the shell (DEC-019), and what Android and the local helper ship too. It is one document with its own stylesheet and scripts, so it cannot be put inside one of this guide's frames; it is shown here as it is, photographed from the production build. The page skin in the next chapter is the component library's version of the same look — the React interface kept in `music-player/src` — and is not what is served.

**Music, playing** Eight songs added through the shell's own import path, Gantry playing: the jewel case on the stage, the iPod rail, the transport, and the iTunes 10 list with the playing row selected.

The image displays two versions of the iTunes application interface side-by-side, illustrating a redesign. Both versions are titled "Now Playing" and show the "Gantry" album by "Alder Quartet" with "First Light" as the track. The left version, representing the original iTunes, features a large circular radar chart, a progress bar, and a list of radio stations for Chicago. The right version, representing a modernized version, features a smaller radar chart, a more compact progress bar, and a similar station list. The modernized version includes additional UI elements like a search bar and a "Now Playing" header.

**Original iTunes (Left):**

- Header: Now Playing
- Album: Gantry, Alder Quartet, First Light
- Progress bar: 00:08 to -01:52
- Controls: Play/Pause, Previous, Next, Repeat, Volume
- Radio Stations List:

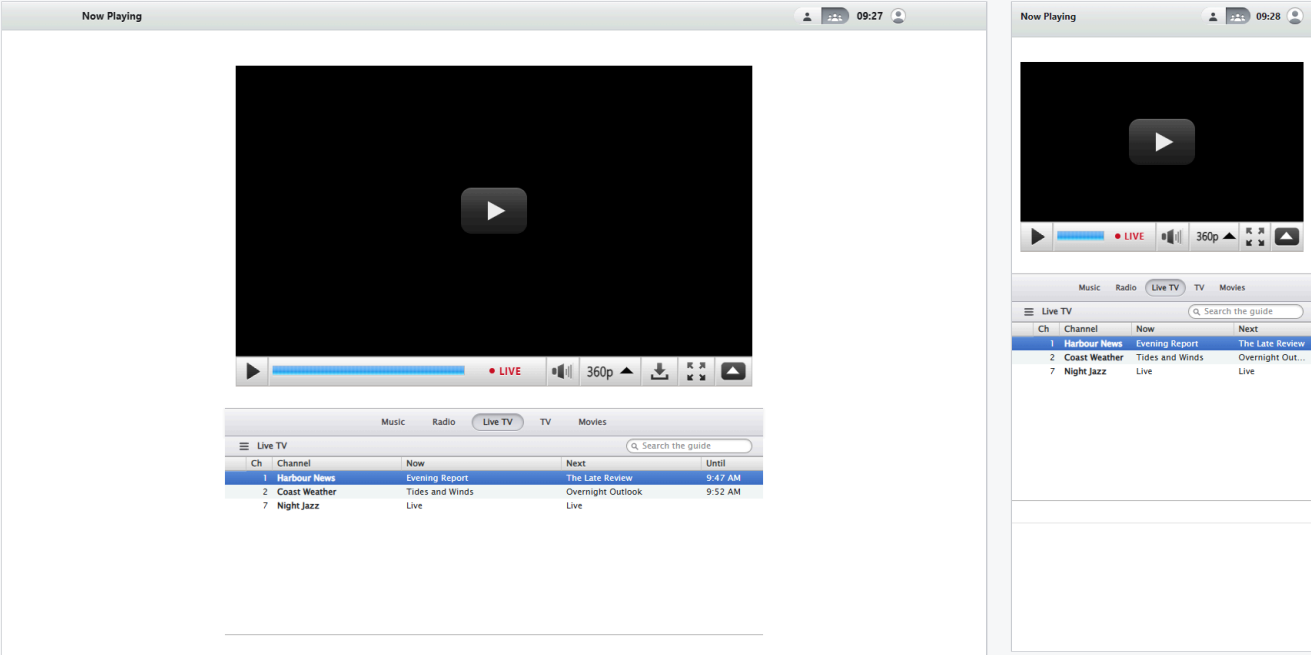
| Station               | Song               | Artist | Kbps | Where       |
|-----------------------|--------------------|--------|------|-------------|
| 107.5 WGGI            | Urban contemporary | —      | —    | Chicago, IL |
| 103.5 KISS FM         | Pop                | —      | —    | Chicago, IL |
| 93.9 Lite FM          | Soft rock          | —      | —    | Chicago, IL |
| Rock 95.5 WCHH        | Rock               | —      | —    | Chicago, IL |
| Big 95.5 Country      | Country            | —      | —    | Chicago, IL |
| Inspiration 1390      | Gospel             | —      | —    | Chicago, IL |
| WLS 94.7 FM           | Classic hits       | 80     | —    | Chicago, IL |
| WFMT 98.7 Classical   | Classical          | 130    | —    | Chicago, IL |
| WDCB 90.9 Jazz        | Jazz               | 128    | —    | Chicago, IL |
| Smooth Jazz WNJA      | Jazz               | 128    | —    | Chicago, IL |
| WBZZ 91.5             | News               | 64     | —    | Chicago, IL |
| Moody Radio WMBI 90.1 | Talk               | 96     | —    | Chicago, IL |
| WCPT 820              | Talk               | 64     | —    | Chicago, IL |
| FUBU Radio            | Hip hop            | 192    | —    | Chicago, IL |
| 011.fm House of Hair  | Rock               | 192    | —    | Chicago, IL |
| Radio DePaul          | Pop                | 56     | —    | Chicago, IL |

**Modernized iTunes (Right):**

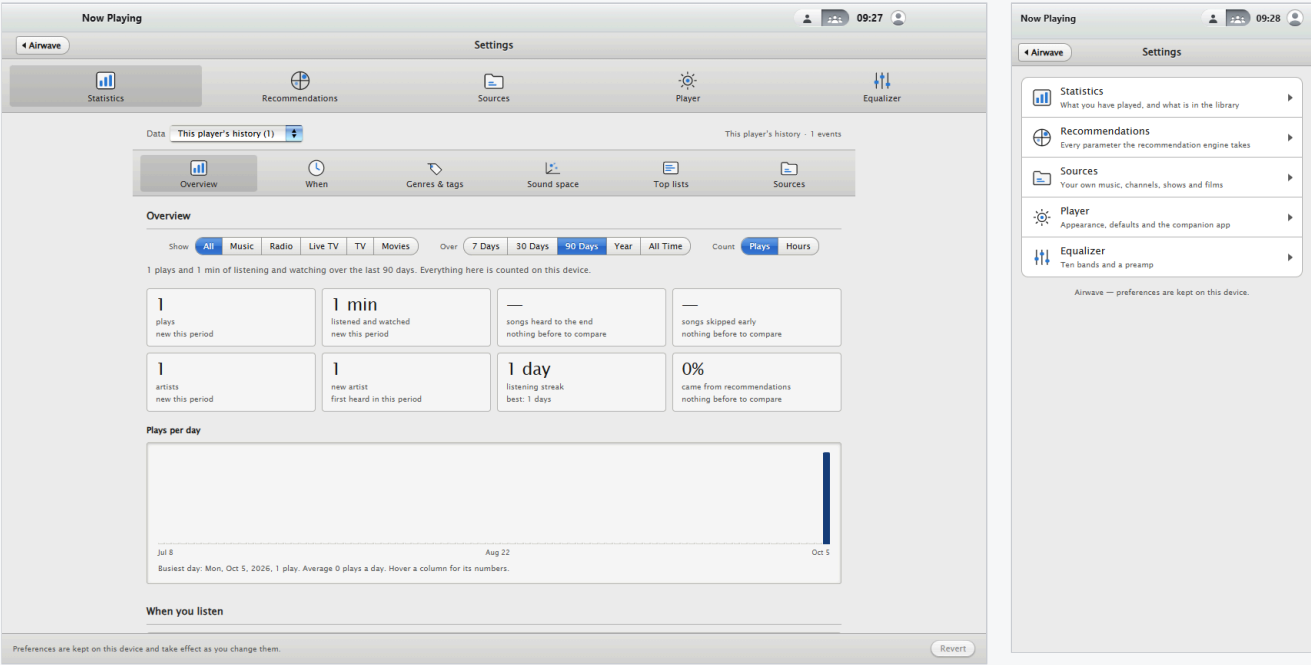
- Header: Now Playing
- Album: Gantry, Alder Quartet, First Light
- Progress bar: 00:06 to -01:54
- Controls: Play/Pause, Previous, Next, Repeat, Volume
- Radio Stations List:

| Station          | Song               | Artist |
|------------------|--------------------|--------|
| 107.5 WGGI       | Urban contemporary | —      |
| 103.5 KISS FM    | Pop                | —      |
| 93.9 Lite FM     | Soft rock          | —      |
| Rock 95.5 WCHH   | Rock               | —      |
| Big 95.5 Coun... | Country            | —      |
| Inspiration 1390 | Gospel             | —      |
| WLS 94.7 FM      | Classic hits       | —      |
| WFMT 98.7 Cl...  | Classical          | —      |
| WDCB 90.9 Jazz   | Jazz               | —      |
| Smooth Jazz ...  | Jazz               | —      |
| WBZZ 91.5        | News               | —      |
| Moody Radio ...  | Talk               | —      |
| WCPT 820         | Talk               | —      |
| FUBU Radio       | Hip hop            | —      |
| 011.fm House...  | Rock               | —      |

**Live TV** Channels and guide from the companion's helper on this machine (with no companion, a paired hub's copy fills the same guide): the QuickTime bar over the picture, Now and Next from the guide, and "Live" where a channel has none.



**Settings** Settings is a page with its own address. On a desktop it opens on Statistics with the section tabs across the top; on a phone it opens on the list of sections.



The page — what the player is made of

The music player's skin: a sticky status bar, a section strip, the hero with the jewel case on its stage, the iPod rail and a transport of eleven keys, and the iTunes 10 list under it. Everything below is live — press play and the case opens; pick a row and the stage follows it.

Now Playing

09:28

Q

Search your music

Music Library8

Queue6

Playlists

Listening history

Midnight Set, Side B

Fennel Grove

Long Wave Sessions, Vol. 2

00:23

-02:35

| # |  | Song                           | Artist                 | Time | BPM |  | ★ | Album                      |
|---|--|--------------------------------|------------------------|------|-----|--|---|----------------------------|
| 1 |  | Copper Meridian                | Orbital Cartographers  | 5:05 | —   |  | ★ | Copper Meridian            |
| 2 |  | Ein außerordentlich langer ... | Cassette Bloom         | 6:21 | 104 |  | ★ | Live from Pier 9           |
| 3 |  | Harbour Lights                 | Orbital Cartographers  | 4:24 | —   |  | ★ | Copper Meridian            |
| 4 |  | Lantern Road                   | Marlow & the Tidewater | 3:24 | 118 |  | ★ | Quiet Arithmetic           |
| 5 |  | Long Wave                      | Fennel Grove           | 5:16 | 122 |  | ★ | Long Wave Sessions, Vol. 2 |
| 6 |  | Midnight Set, Side B           | Fennel Grove           | 2:58 | 110 |  | ★ | Long Wave Sessions, Vol. 2 |
| 7 |  | Pier 9 (Live)                  | Cassette Bloom         | 4:11 | —   |  | ★ | Live from Pier 9           |
| 8 |  | Quiet Arithmetic               | Marlow & the Tidewater | 3:51 | 96  |  | ★ | Quiet Arithmetic           |

The transport is centred on the rail. Play sits on the progress rail's centre line at every width, with previous and next either side of it on the same line, and the keys never run past the rail's ends. The volume line sits beside the keys only when it fits on both sides; otherwise it goes beneath, centred, as it does here. When the keys do not fit on one line, previous, play and next take the first line on their own and the other eight keys share the line below. The row measures itself to decide this, because the room it needs depends on how many keys a product carries (DEC-016).

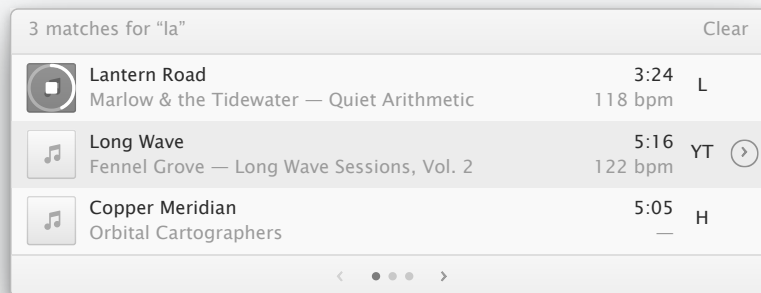
THE MUSIC LIST

| # |  | Song                          | Artist                | Time | BPM |  | ★ | Album            |
|---|--|-------------------------------|-----------------------|------|-----|--|---|------------------|
| 1 |  | Copper Meridian               | Orbital Cartographers | 5:05 | —   |  | ★ | Copper Meridian  |
| 2 |  | Ein außerordentlich langer... | Cassette Bloom        | 6:21 | 104 |  | ★ | Live from Pier 9 |
| 3 |  | Harbour Lights                | Orbital Cartographers | 4:24 | —   |  | ★ | Copper Meridian  |

| # |  | Song                 | Artist                 | Time | BPM |  | ★ | Album                      |
|---|--|----------------------|------------------------|------|-----|--|---|----------------------------|
| 4 |  | Lantern Road         | Marlow & the Tidewater | 3:24 | 118 |  | ★ | Quiet Arithmetic           |
| 5 |  | Long Wave            | Fennel Grove           | 5:16 | 122 |  | ★ | Long Wave Sessions, Vol. 2 |
| 6 |  | Midnight Set, Side B | Fennel Grove           | 2:58 | 110 |  | ★ | Long Wave Sessions, Vol. 2 |
| 7 |  | Pier 9 (Live)        | Cassette Bloom         | 4:11 | —   |  | ★ | Live from Pier 9           |
|   |  | Quiet Arithmetic     | Marlow & the Tidewater | 3:51 | 96  |  | ★ | Quiet Arithmetic           |

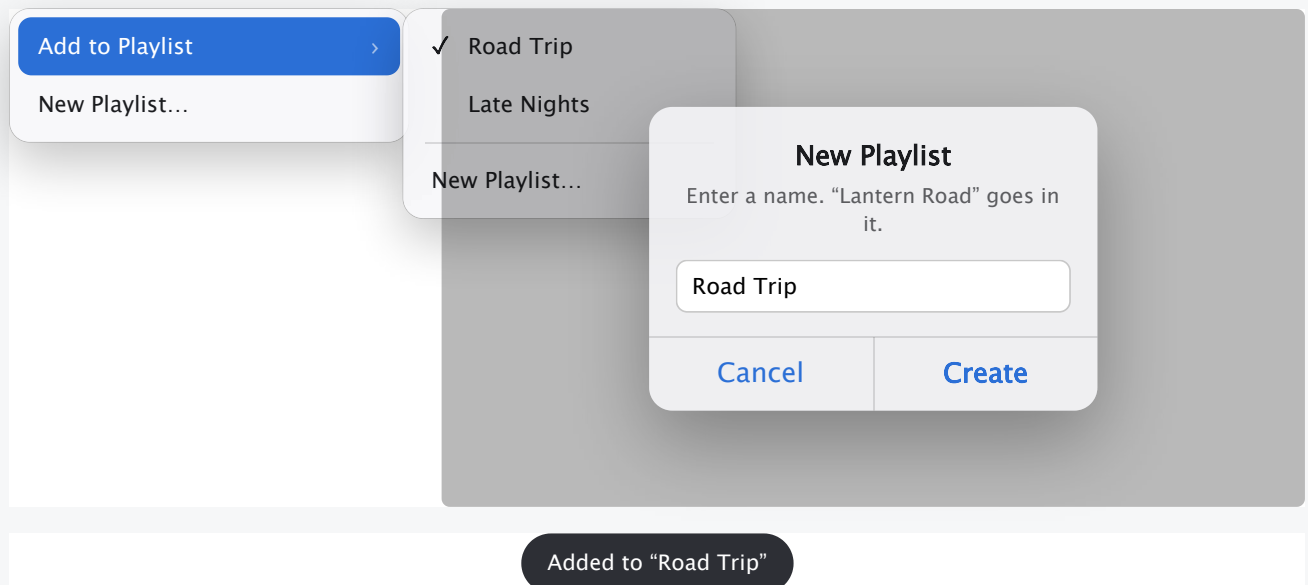
The same component the player renders, with the reference's stylesheet block: nine columns, 18 px rows, the stripe, no rules, an embossed header with the sorted column tinted, monochrome source badges read from each track's locator, icon columns for offline and star, the playing row in bold with a speaker glyph and a marquee, a right-click menu, and the gel scroller that fades in while you scroll.

## THE SEARCH POPOVER



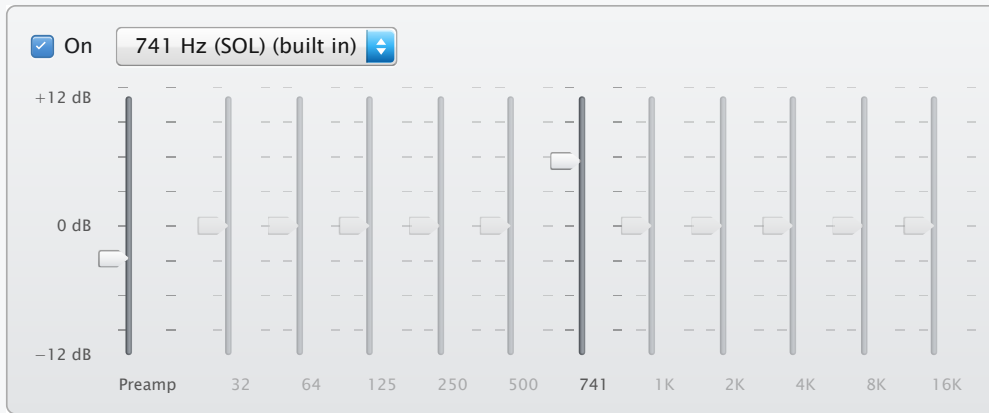
The iTunes 11 results popover: a gradient header with the count and Clear, 40 px rows whose artwork tile is a fifteen-second audition button with a countdown ring, a circled chevron on the hot row, and a pager. Arrow keys walk the rows and page off the end.

## CONTEXT MENU, SHEET AND TOAST



The desktop context menu with its playlist submenu; the New Playlist sheet; and the toast that answers a change. All three are the reference's, classes and all.

## THE EQUALIZER WINDOW



The iTunes equalizer: On beside the preset pop-up, then a preamp and the ten graphic centres on a  $\pm 12$  dB scale with tick dashes flanking each rail. The window hugs its rails at 600 px (DEC-014) rather than stretching to the page. A one-band preset is chosen here so the greyed rail is visible — those faders name the preset that is not using them.

## SHARED MODE, AND THE PAGE'S FURNITURE

**Friday Listening Club** ● Jalon ● Ash ● Rowan

2 listening

The hub keeps the queue; your player follows it. Proposals go to the hub and come back accepted or refused.

**Music** 412 tracks indexed from folders on this device

▶ Play all 🔀 Shuffle all + Add a folder... Choose files...

412 tracks · 6 queued · playing from this device Flat 0:23 / 3:24

The strip under the hero when you are in a group; a section's head and toolbar row; and the status line the bottom bar used to carry.

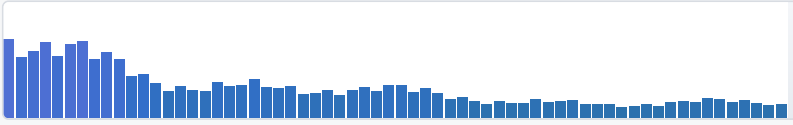
## A LIVE BROADCAST

12:34 ● LIVE

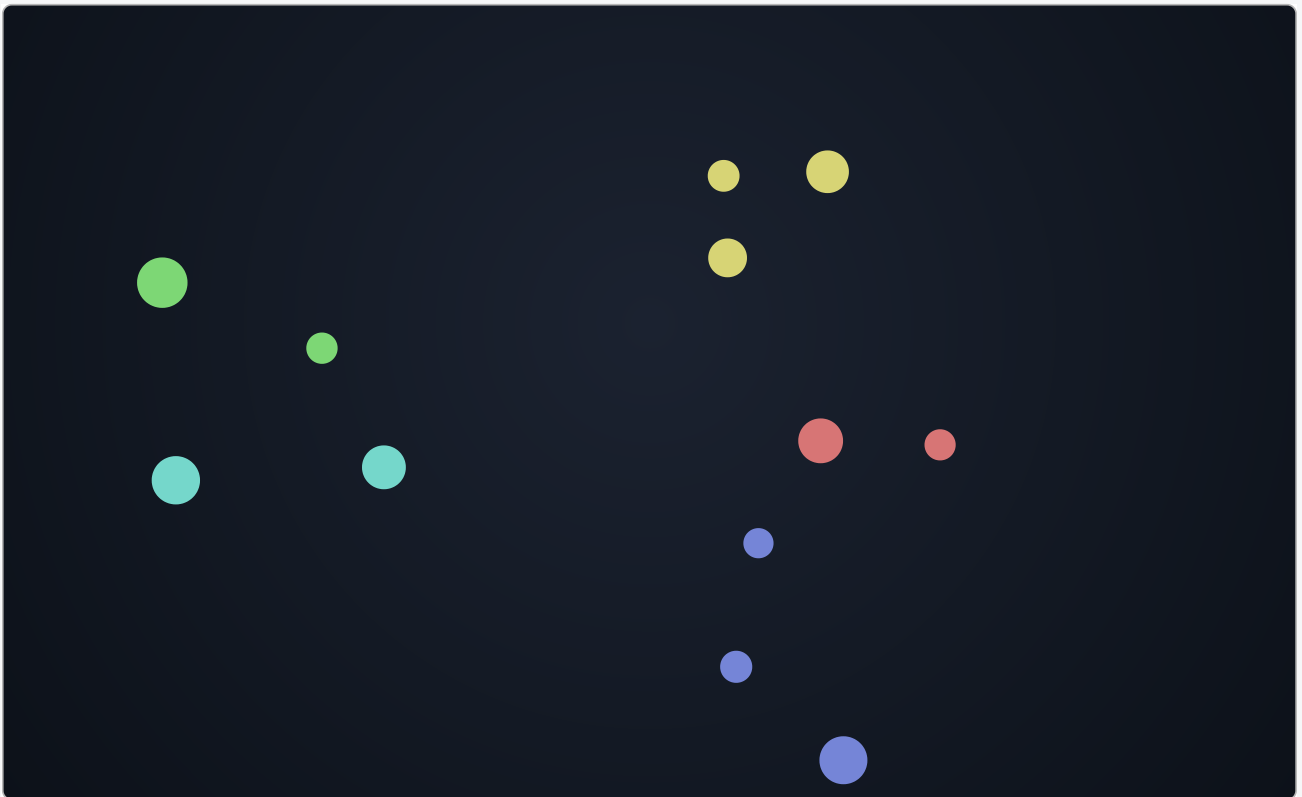
In shared mode the rail is not yours to drag: the duration stamp becomes the LIVE marker, the fill sits at the live edge, and the rail drops its slider role rather than announcing a control that refuses every change.

## THE VISUALISERS — SPECTRUM, STAGE AND CONSTELLATION

The player draws three visualisers, and all three follow one rule: a visualiser must carry the data, not hide it. The two below are drawn here by the player's own code — `drawSpectrum` and `layoutStars` in `music-player/src/lib/`, the functions the views call — in the player's own `.player-spectrum` and `.player-constellation` styles. The third, the **jewel case**, is live on the hero at the top of this chapter.



The spectrum, under “How this is being played” in the Now playing section. Each analyser frame is averaged into 64 bars; a bar’s hue and lightness rise with its level, so a loud band reads brighter as well as taller. It draws only while a song plays — paused, the loop stops rather than repaint a still picture — and under reduced motion it redraws four times a second instead of every frame. There is no audio here, so the specimen is fed a generated frame; print shows one fixed frame.



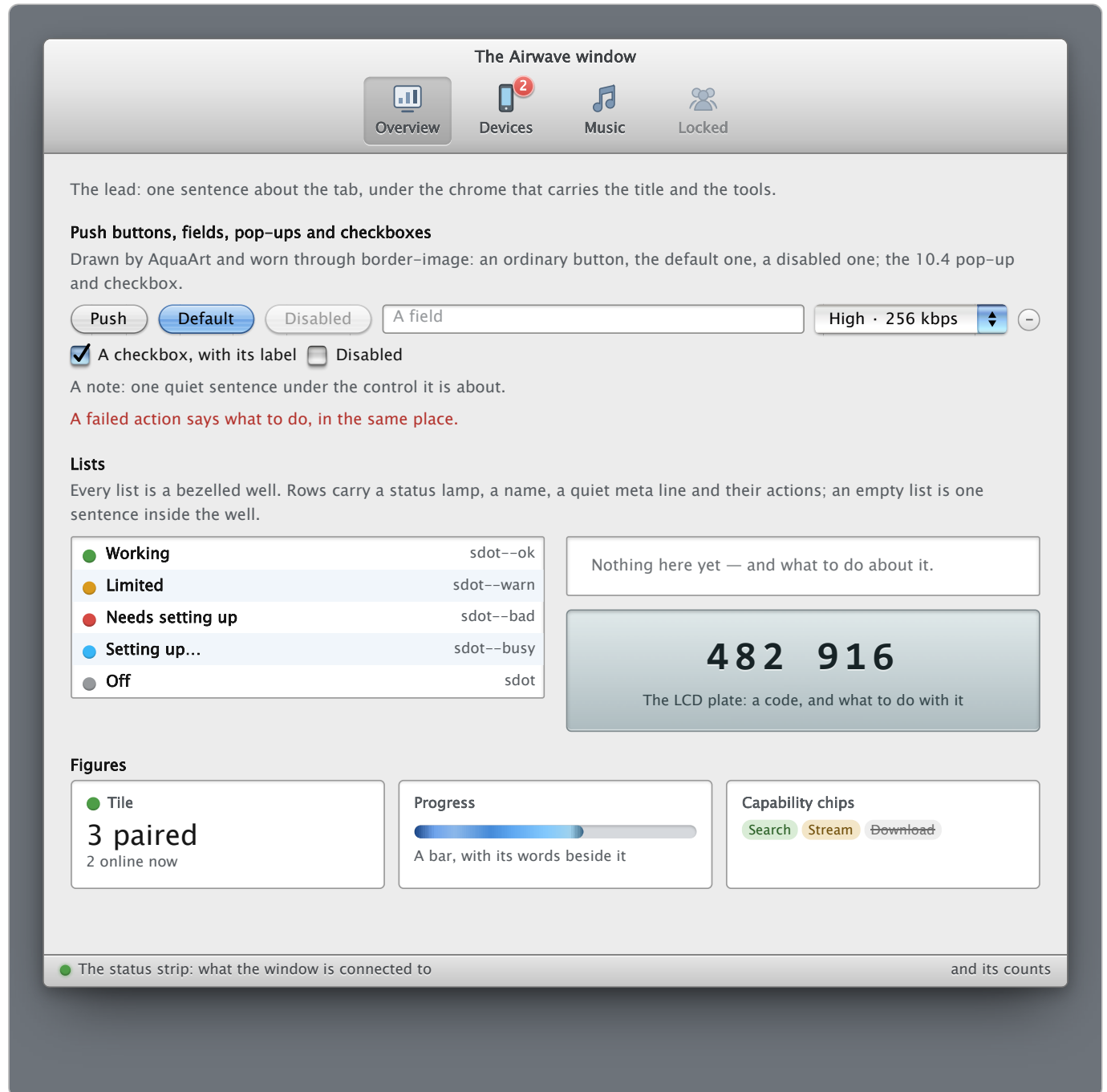
The constellation: the library as a star field, one star per album. Albums by one artist share an angular sector, a star’s size is the album’s length, and its colour is the artist’s hue, so the layout is information, never scatter. The app renders it with WebGL and a slow rotation; this is the same placement projected through the same camera ( $55^\circ$ ,  $z = 42$ ), as SVG so it also prints. Hover a star for its album.

The **jewel case** is `packages/aqua-ui/src/stage/jewel-case.ts`: the case and disc built from primitives, sleeve and tray-card textures generated from the track’s real cover and running order, a drag with momentum, and a pose remembered per device. It stands in for Cover Flow (DEC-006). The stage and the constellation load Three.js lazily and share the chunk, so a listener who opens neither never downloads it. All three keep their meaning without motion: the spectrum slows to a level, the stage stops its idle drift and keeps the CSS cover, and the constellation opens as the same albums in a table, with identical selection and keyboard behaviour (UX-MOTION-001, UX-STATE-001). Both sections also appear whole in the mockups: *Now playing (audio chain)* and *Constellation*.

## The Airwave window — what the hub and the companion are made of

A centred Snow Leopard window: one sheet of chrome carrying the title and a row of icon tools, a pane that scrolls beneath it, and a status strip at the foot. Airwave Hub has six tools (Overview, Devices, Music, Groups, Sharing, System) and Airwave Companion four (Library, Live TV, Remote, Settings). There is no source list, no traffic lights and no media toolbar: the tools are the navigation, a roving-tabindex tab strip (UX-KEY-001).

The frame below is not a drawing of it. It holds exactly what the hub's page loads — `airwave-window.css`, `airwave-hub.css` and the hub's `styles.css`, with none of the component library's stylesheets — AquaArt is installed in it as the products' `main.tsx` installs it, and the controls are the hub's own kit.



### WHERE EACH PART COMES FROM

- **The stylesheets are generated.** `packages/aqua-ui/scripts/make-window-css.py` copies the `<style>` of `airwave-companion.html` into `airwave-window.css` and every rule the hub design adds into `airwave-hub.css`. Change the design file and run `pnpm build:window-css`; never edit the two by hand.
- **The controls are drawings.** AquaArt (`packages/aqua-ui/src/airwave/aqua-art.ts`, imported as `@now-playing/aqua-ui/airwave-art`) draws the Snow Leopard push button and the 10.4 pop-up and checkbox as SVG and publishes them as custom properties (`--aq-btn-22`, `--aq-def-22`, `--aq-pop-22`, `--aq-cb-14` and their pressed, disabled and checked forms). One implementation, shared by both products (NPD-021, DEC-026).

- **The kits are per product.** `docker-container/src/web/ui.tsx` and `windows-companion/src/renderer/ui.tsx` write the same classes — `.push`, `.field`, `.pop`, `.chk`, `.well`, `.rows`, `.sdot` — with different props and, for the sheet, different markup, so they are not one component (DEC-026). By product, below, lists what each exports.
- **Each product's `styles.css` is the remainder.** The hub's holds the layout the design wrote inline, the states it never drew (sign-in, the sheet, loading and failed lists) and the contrast inks. The companion's makes the window fill its frame, makes the chrome the title bar and drag region (DEC-020), and adds the song list and the sheet.

#### STATES EVERY LIST AND ACTION OWES

- **Loading, empty and failed are one quiet line inside the well** — never an illustration, so nothing jumps when rows arrive (UX-STATE-001). A failed list offers Try Again in the same row.
- **A failed action says what to do** in a sentence under the control that failed (`.note--bad`), not in a toast.
- **A disabled button says why** on hover and to assistive technology; a busy one keeps its width.
- **Anything that cannot be undone asks first, in a sheet** that drops over the pane, with Cancel as the safe default (UX-SAFE-001, DEC-021). Both sheets are in the mockups.
- **Colour never carries the meaning alone:** every status lamp has its word beside it or in its accessible name.

### Library controls

The component library's controls, in both of its skins. The player's React source imports them; the hub and the companion do not — their controls are the Airwave window's, above.

Buttons: neutral / default / graphite / destructive / busy / disabled / small / mini / ellipsis / link

Cancel

Save

Graphite

⚠ Delete Playlist

⚙ Importing

Disabled

Small

Mini

Export...

Pressed

⬇ Download · 84 MB

Icon buttons: framed / plain / capsule / pressed / menu / disabled / large

⚙

☆

📁

🔗

⋮

⬇

🚗

Checkbox / text field / validation / pop-up

☒ Show explicit content
 ☐ Some folders
 ☐ Disabled

Display name

Jalon

Shown to group members

Reference tuning

528

● Reference must be between 400 and 480 Hz

Disabled

Default format

Original

Sliders / progress / segmented

4 dB

0

42%

Importing 18 of 94 songs...

60%

Paused

Syncing artwork...

List

Grid

Artist

👤

👥

### Overlays

Sheets attach to their window; alerts stand alone; toasts confirm what just happened and get out of the way. Press the buttons. These are the component library's overlays; the Airwave windows' confirmation sheets are drawn in the mockups.

Sheet / alert / toast

New Playlist...

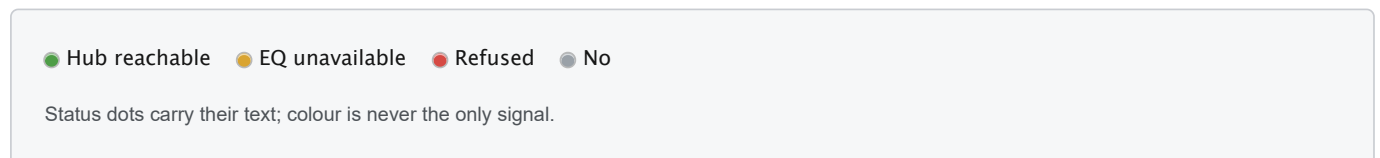
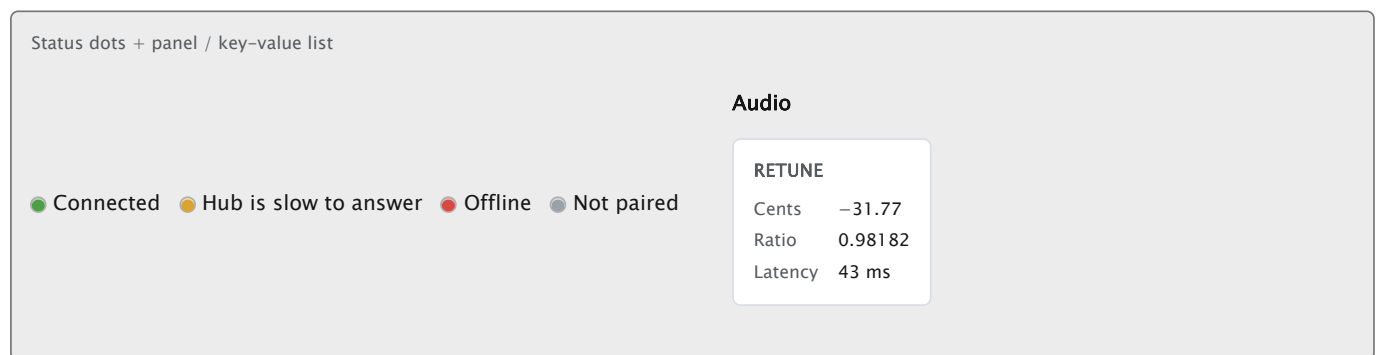
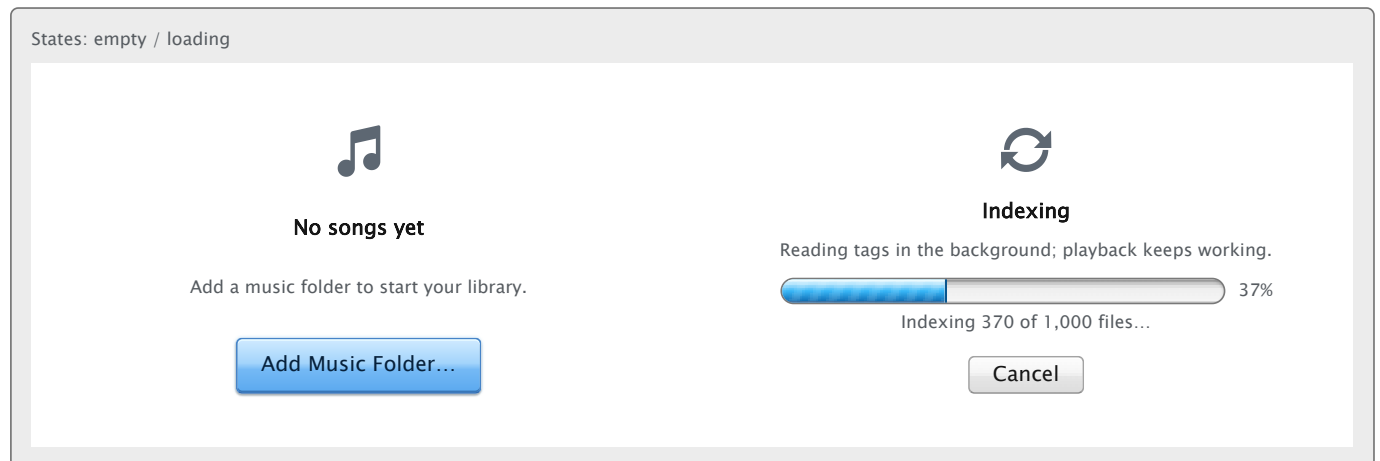
Show Alert...

Toast

Error toast

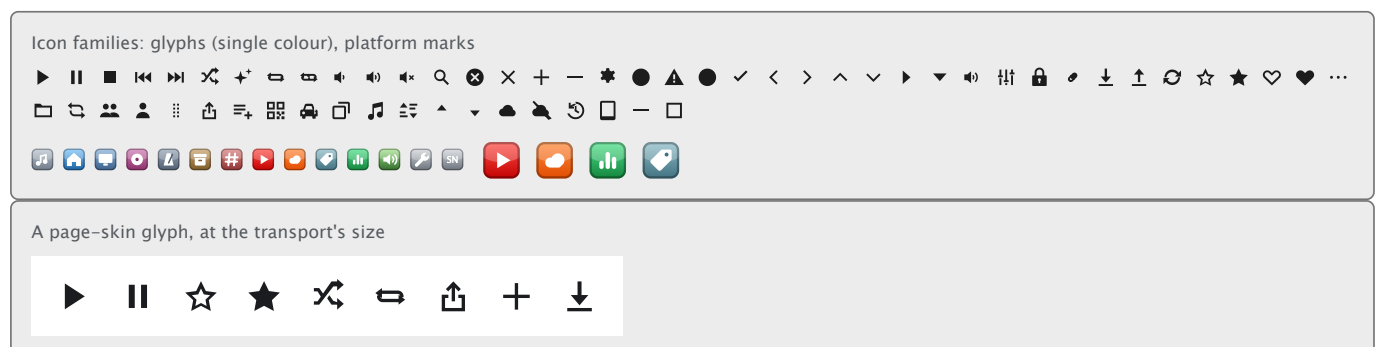
## States — the group people skip

Every screen owes an answer for empty, loading, offline, partial, refused and out-of-date. The component library draws two of them as panels, `EmptyState` and `LoadingState`; the rest are said in the view's own words where the choice is made. In the Airwave windows the same answers are one quiet line inside the list's well (see The Airwave window).



## Icons

Two families in the component library: single-colour glyphs for every control, and the platform marks that say where a track comes from. The Airwave windows draw their own: six 26-pixel tools for the hub (`docker-container/src/web/icons.tsx`) and four tools and three row glyphs for the companion (`windows-companion/src/renderer/icons.tsx`), as their designs drew them; both sets are in the mockups.



## By product

What each product is built from today, read from its source when this page was built — so a product that moves on cannot leave a tick behind.

## THE AIRWAVE WINDOW KITS

| PIECE                                      | AIRWAVE HUB | AIRWAVE COMPANION |
|--------------------------------------------|-------------|-------------------|
| airwave-window.css (generated)             | •           | •                 |
| airwave-hub.css (generated)                | •           | —                 |
| AquaArt — @now-playing/aqua-ui/airwave-art | •           | •                 |
| ActionError                                | •           | —                 |
| Ago                                        | •           | —                 |
| Check                                      | •           | •                 |
| ConfirmProvider                            | —           | •                 |
| EmptyCells                                 | •           | —                 |
| EmptyRow                                   | •           | •                 |
| Field                                      | •           | —                 |
| Group                                      | •           | —                 |
| HubUiProvider                              | •           | —                 |
| LoadingRow                                 | —           | •                 |
| Note                                       | •           | —                 |
| Option                                     | —           | •                 |
| Pop                                        | •           | •                 |
| Progress                                   | —           | •                 |
| Push                                       | •           | •                 |
| Remove                                     | —           | •                 |
| Rows                                       | —           | •                 |
| Sdot                                       | •           | —                 |
| Status                                     | —           | •                 |
| SubHead                                    | •           | —                 |

Rows after the first three are what each product's `ui.tsx` exports. A name in both columns is two components that write the same classes, not one shared component: `Push` takes `primary` in the hub and `isDefault` in the companion, `Check` reports a boolean in one and passes input props through in the other, and the confirmation sheet is a `role="alertdialog"` layer in the hub and a modal `<dialog>` in the companion (DEC-026).

## THE COMPONENT LIBRARY

`packages/aqua-ui` exports 38 components. The hub and the companion import none of them. All 38 are imported by the player's React source in `music-player/src`, which is kept and tested but is not what the player serves (DEC-019). A component no product imports is removed from the library rather than kept as legacy.

### Imported by the player's React source (38)

AquaTable BarClock BarSearch Button ButtonLink Checkbox EmptyState Glyph Hero HeroArt IconButton InlineValidation JewelStage KeyButton KeyTransport KeyValueType LevelSlider LoadingState ModeSwitch MusicList PageBar Panel PanelSection PopUpMenu ProfileButton ProgressBar ProviderArtworkProvider ProviderMark SectionStrip SegmentedControl Sheet Slider SourceBadge Spinner StatusDot TextField ToastProvider TrackScrubber

The player's shell also draws its search popover, sheets, toast and equalizer window from its own markup rather than through a component.

# Interaction and content rules

Every rule has a stable ID. Cite it in a change that touches the behaviour, and change the rule, the owning code and its test together. **Adopted** rules are the standard; **observed** ones describe what the code does today and have not been ratified. The source is `design/ux-rules.json`.

## PRINCIPLES THAT OUTRANK THE VISUALS

|                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>UX-PRIN-001</div> <div><b>A control that cannot act is not drawn</b></div> <div>● ADOPTED</div>                     | <p>No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state.</p> <p>The equalizer window ships without traffic lights: three circles that close nothing would be decoration.</p> <p><b>OWNED BY</b> <code>music-player/src/views/Equalizer.tsx</code></p> <p><b>CHECKED BY</b> <span>reviewer</span> Judgement call, checked in review (<code>docs/AQUA_CONFORMANCE.md</code>).<br/><code>docs/DEVIATIONS.md</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <div>UX-PRIN-002</div> <div><b>An unavailable capability says so where the choice is made</b></div> <div>● ADOPTED</div> | <p>Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting.</p> <p>The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.</p> <p><b>OWNED BY</b> <code>packages/aqua-ui/src/components/PageBar.tsx</code> <code>packages/aqua-ui/src/components/States.tsx</code><br/><code>windows-companion/src/renderer/views/Library.tsx</code> <code>windows-companion/src/renderer/views/Hub.tsx</code></p> <p><b>CHECKED BY</b> <code>packages/aqua-ui/tests/dom/page-chrome.test.tsx</code> — “reports the reason instead of switching when a mode is unavailable”<br/><code>music-player/tests/e2e/np/livevol.spec.ts</code> — “the radio transport: Previous and Next read as unavailable, Play does not, and they say why”<br/><code>windows-companion/tests/dom/companion-shell.dom.test.tsx</code> — “with no hub paired, Send to Hub says where to pair one”</p> |
| <div>UX-PRIN-003</div> <div><b>Say what the code does, not what the feature is called</b></div> <div>● ADOPTED</div>     | <p>Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes.</p> <p>Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.</p> <p><b>OWNED BY</b> <code>music-player/src/views/Equalizer.tsx</code> <code>music-player/src/views/Settings.tsx</code></p> <p><b>CHECKED BY</b> <code>music-player/tests/e2e/np/connect.spec.ts</code> — “the companion, the container and backup”<br/><code>music-player/tests/e2e/helper.spec.ts</code> — “a link the helper will not fetch from is refused, with the reason”</p>                                                                                                                                                                                                                                                                                                                           |
| <div>UX-PRIN-004</div> <div><b>A disabled control is still legible about why</b></div> <div>● ADOPTED</div>              | <p>Greyed, not gone. The reason is part of the accessible name, not only a tooltip.</p> <p>“500 Hz band, not used by 528 Hz (MI)”.</p> <p><b>OWNED BY</b> <code>music-player/src/views/Equalizer.tsx</code></p> <p><b>CHECKED BY</b> <code>music-player/tests/e2e/np/connect.spec.ts</code> — “the companion, the container and backup”</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## KEYBOARD, FOCUS AND TOUCH

UX-KEY-001 **One tab stop per group; arrows move inside it**

● ADOPTED

The section strip, the mode switch, tab strips, tables and menus use a roving tabIndex. Home/End jump, Enter activates, type-ahead selects. Focus travels with the selection. The tools of an Airwave window are such a strip: arrows, Home and End move the selection, and a locked tool is skipped.

**OWNED BY** packages/aqua-ui/src/components/SectionStrip.tsx packages/aqua-ui/src/components/AquaTable.tsx  
packages/aqua-ui/src/components/MusicList.tsx docker-container/src/web/App.tsx  
windows-companion/src/renderer/App.tsx windows-companion/src/renderer/views/Library.tsx

**CHECKED BY** docker-container/tests/e2e/a11y.spec.ts — “@a11y the tab strip is navigable with the keyboard alone”  
packages/aqua-ui/tests/dom/table.test.tsx — “uses a roving tabIndex, arrow navigation, Space selection and Enter activation”  
packages/aqua-ui/tests/dom/page-chrome.test.tsx — “has one tab stop, moves with arrows and Home/End, chooses with Enter, and answers type-ahead”  
docker-container/tests/dom/admin-shell.dom.test.tsx — “moves between tabs with the arrow keys, Home and End, keeping one tab stop”  
windows-companion/tests/dom/companion-shell.dom.test.tsx — “moves along the tools with the arrow keys, choosing as it goes”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “End reaches the last song, fetching only its stretch; Shift+Home chooses everything between by asking for the ids”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “Ctrl+A chooses every song, and Send to Hub sends them in batches”

UX-FOCUS-001 **Focus is always visible**

● ADOPTED

Every focusable element shows --aqua-focus-shadow or a 2px --np-accent outline at 2px offset, including when the ring is drawn on an ancestor via :focus-within.

**OWNED BY** packages/aqua-ui/src/styles/aqua.css packages/aqua-ui/src/styles/now-playing.css  
windows-companion/src/renderer/styles.css

**CHECKED BY** music-player/tests/e2e/a11y.spec.ts — “@a11y focus is visible wherever it lands”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “dims only the pane under the toolbar while a sheet is open, and keeps it modal”

UX-KEY-002 **Every action is reachable with the keyboard alone**

● ADOPTED

Every section and every transport control can be reached and operated without a pointer. Space toggles playback without scrolling the page.

**OWNED BY** music-player/src/App.tsx docker-container/src/web/App.tsx windows-companion/src/renderer/App.tsx

**CHECKED BY** music-player/tests/e2e/a11y.spec.ts — “@a11y the transport is reachable and named from the keyboard alone”  
music-player/tests/e2e/player.spec.ts — “space toggles playback and does not scroll the page”  
docker-container/tests/e2e/a11y.spec.ts — “@a11y the tab strip is navigable with the keyboard alone”

UX-TOUCH-001 **Nothing interactive is under 44px on a coarse pointer**

● ADOPTED

Under (pointer: coarse) controls grow; where the visual cannot grow, a transparent ::after takes the taps. Desktop keeps its 18px rows. Hit targets are never shrunk to match a visual; --aqua-hit is 32px on desktop, 44px on touch.

**OWNED BY** packages/aqua-ui/src/styles/aqua.css packages/aqua-ui/src/styles/now-playing.css

**CHECKED BY** music-player/tests/e2e/responsive.spec.ts — “nothing overflows, no text under 12px, no target under 44px”  
music-player/tests/e2e/responsive.spec.ts — “the desktop keeps the design it was drawn with”  
packages/aqua-ui/tests/e2e/styleguide.spec.ts — “every screen of every product fits at every size”

UX-TOUCH-002 **No text under 12px on touch; field text is 16px**

● ADOPTED

The touch layer steps the scale up (15/14/13/12/11px). A field's text is 16px because iOS zooms the page when a smaller field takes focus.

**OWNED BY** packages/aqua-ui/src/styles/aqua.css packages/aqua-ui/src/styles/now-playing.css

**CHECKED BY** music-player/tests/e2e/responsive.spec.ts — “settings, where every panel of explanatory text lives”

UX-FORM-001 **A hub settings form sends nothing until Save, and Revert puts back what the hub has**

● ADOPTED

In the hub's Network, Backup, Recommendations and Discord forms nothing is sent while typing or choosing. Values the hub would refuse are checked first and said under the form; Save sends what changed together and says so; Revert puts back what the hub holds. The companion is different on purpose: its preferences are written as they change and the foot of the pane says so (UX-FEED-001).

**OWNED BY** docker-container/src/web/views/Network.tsx docker-container/src/web/views/Backup.tsx  
docker-container/src/web/views/Recommendations.tsx docker-container/src/web/views/Discord.tsx

**CHECKED BY** docker-container/tests/dom/admin-forms.dom.test.tsx — “sends nothing until Save, checks the address first, and Revert puts it back”  
docker-container/tests/dom/admin-forms.dom.test.tsx — “saves the schedule, the parts and Keep together, and Revert puts them back”  
docker-container/tests/dom/admin-forms.dom.test.tsx — “saves the edit, and Revert puts back what the hub has”  
docker-container/tests/dom/admin-forms.dom.test.tsx — “saves the edited settings, then starts the bot”  
docker-container/tests/dom/admin-forms.dom.test.tsx — “refuses a number outside its range before sending it”  
docker-container/tests/e2e/hub.spec.ts — “network settings wait for Save, check the address first, and say Saved.”  
docker-container/tests/e2e/hub.spec.ts — “backup settings are saved with Save, the schedule says when the next one runs, and an archive downloads”

## STATES, FEEDBACK AND MOTION

UX-STATE-001 **Every data view answers empty, loading and error**

● ADOPTED

A view that fetches says it is loading, says what failed and offers a retry, and when it has nothing says what to do next. In the Airwave windows each of those is one quiet line inside the list's well, so nothing jumps when rows arrive (listState, EmptyRow and EmptyCells in the hub's kit; LoadingRow and EmptyRow in the companion's); a section that throws is caught by the hub's ViewBoundary and the window stays up. In the component library they are LoadingState and EmptyState; a failure is said in the view's own words with its retry. Blank panes are defects.

**OWNED BY** docker-container/src/web/views/common.tsx docker-container/src/web/ViewBoundary.tsx  
packages/aqua-ui/src/components/States.tsx docker-container/src/web/ui.tsx windows-companion/src/renderer/ui.tsx  
docker-container/src/web/lib/overview.tsx

**CHECKED BY** music-player/tests/e2e/np/stats.spec.ts — “before any play: nothing recorded, and no generated source on offer”  
music-player/tests/e2e/player.spec.ts — “opens with no music, says so, and raises no errors”  
docker-container/tests/dom/admin-shell.dom.test.tsx — “keeps the window alive when a section fails to render”  
windows-companion/tests/dom/companion-shell.dom.test.tsx — “shows no sample channels: two empty lists in the design's words”  
docker-container/tests/dom/admin-shell.dom.test.tsx — “draws the badge and the Needs attention list from one reading, so they always agree”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “draws a window of rows, not 50,000, and counts them all”

Nothing animates that a person did not just cause. Under prefers-reduced-motion the live dot stops, marquees hold their ellipsis, sheets fade without travel, and the 3D stage stops drifting. New animations read --aqua-anim-state or --aqua-sheet-travel.

**OWNED BY** packages/aqua-ui/src/styles/aqua.css packages/aqua-ui/src/styles/now-playing.css packages/aqua-ui/src/components/Marquee.tsx

**CHECKED BY** music-player/tests/e2e/a11y.spec.ts — “@a11y respects reduced motion: nothing loops that a person did not start” packages/aqua-ui/tests/unit/aqua-conformance.test.ts — “§17.5”

Progress bars carry a name and a value; an indeterminate bar is aria-busy and has text. Live regions are polite and carry one sentence. The Airwave windows have no toast: what the window just did is said in the status strip (hub) or at the foot of the pane and beside the control (companion).

**OWNED BY** packages/aqua-ui/src/components/ProgressBar.tsx packages/aqua-ui/src/components/Toast.tsx docker-container/src/web/ui.tsx windows-companion/src/renderer/ui.tsx docker-container/src/web/views/Downloads.tsx windows-companion/src/renderer/views/Transfers.tsx

**CHECKED BY** packages/aqua-ui/tests/dom/controls.test.tsx — “progress has a name and value; indeterminate is busy with text” windows-companion/tests/dom/companion-shell.dom.test.tsx — “writes a preference back as it changes, and says so at the foot of the pane”

yt-dlp, spotDL and (on Windows) FFmpeg are set up without a prompt, in the background, into the app's own tools folder, and only after the file matches the SHA-256 its project published for that release. Each downloader row reads Setting up... with a named progress bar, Ready (adding "Set up automatically" when the app installed it), or Couldn't set up: <reason> beside a Try Again button; the manual install advice appears only when the tool cannot be set up on this machine. Tools are set up one at a time, and one waiting its turn behind another also reads Setting up... — it is queued, not missing, so it shows no install advice and no badge. Only failed or unsupported tools count as missing — in the companion's Settings badge and on the player's Connections card, where a tool on its way reads "setting up". Where FFmpeg's absence explains an empty Tempo column, the Library says whether it is on its way or could not be set up.

Settings ▶ Downloaders: "FFmpeg — Setting up... 42%" with a progress bar; "spotDL — Couldn't set up: GitHub did not answer (503)." and Try Again. Music: "Setting up FFmpeg — tempos appear once it finishes."

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | local-helper/src/provision.ts   local-helper/src/install.ts   windows-companion/src/main/helper.ts<br>windows-companion/src/renderer/views/Settings.tsx   windows-companion/src/renderer/views/Library.tsx<br>windows-companion/src/renderer/App.tsx   music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CHECKED BY | windows-companion/tests/dom/companion-shell.dom.test.tsx — "Settings shows setting up with progress, ready and set up automatically, and a failure with Try Again"<br><br>windows-companion/tests/dom/companion-shell.dom.test.tsx — "Settings keeps the install advice for a tool this PC cannot set up"<br><br>windows-companion/tests/dom/companion-shell.dom.test.tsx — "the Library says tempos are on their way while ffmpeg is being set up"<br><br>windows-companion/tests/dom/companion-shell.dom.test.tsx — "the Library points to Settings when ffmpeg could not be set up, and says nothing once it is ready"<br><br>windows-companion/tests/integration/main-process.test.ts — "sets missing downloaders up by itself, reports how that went, and tries again on request"<br><br>local-helper/tests/unit/install.test.ts — "refuses bytes that do not match the published digest, and leaves nothing behind"<br><br>local-helper/tests/unit/provision.test.ts — "records a failure and waits six hours before trying again, unless told to try now"<br><br>windows-companion/tests/dom/companion-shell.dom.test.tsx — "a tool setup has not reached yet is on its way, not missing: no badge, no install advice"<br><br>windows-companion/tests/dom/companion-controls.dom.test.tsx — "Check asks about that tool; one found behind says which version is out, wears the badge, and Update installs it"<br><br>windows-companion/tests/dom/companion-controls.dom.test.tsx — "a missing tool offers Install, and Check waits for it; a busy helper says why Update must wait"<br><br>windows-companion/tests/dom/companion-controls.dom.test.tsx — ""Update downloaders automatically" is on by default and writes its preference" |

UX-DL-001 **A download carries clean tags and its cover, or says what it lacks**

ADOPTED

A file the helper or the companion downloads with yt-dlp has the thumbnail embedded as cover art (JPEG) and clean tags: a YouTube upload's title is cleaned ("(Official Video)", "(4K Remaster)", frame rates and a trailing "- Official ..." go; "Artist - Song" splits; the channel is the artist when the title names none), featured artists are written into the artist tag as "Artist feat. Guest" (more: "Artist feat. Guest & Other") with the main artist alone as album artist, a source's own album and genre are kept (a site category is not a genre), and the date is the whole release or upload date, YYYY-MM-DD. Each song of a playlist gets its own. The cleaning happens on the link's description in the helper, never on a command line. Without FFmpeg nothing can be embedded: the file is saved as the site sent it and the finished job says so in words. spotDL embeds its own tags and art. The library keeps the whole date beside the year.

ffprobe of a downloaded file: title=Big Buck Bunny, artist=Blender, album\_artist=Blender, date=2014-11-10, and an attached picture. Without FFmpeg: "Saved as the site sent it: FFmpeg is not on this PC, so there is no cover art and the title, artist and date were not written into the file."

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | local-helper/src/jobs.ts packages/domain/src/media-tags.ts packages/domain/src/titles.ts packages/contracts/src/entities/library.ts music-player/src/lib/library.ts windows-companion/src/main/library.ts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CHECKED BY | local-helper/tests/unit/download-tags.test.ts — "embeds the thumbnail as JPEG cover art and writes the tags when FFmpeg is here"<br>local-helper/tests/unit/download-tags.test.ts — "describes, cleans, then downloads from the cleaned description"<br>local-helper/tests/unit/download-tags.test.ts — "writes features into the artist tag as "A feat. B", and the album artist alone"<br>local-helper/tests/unit/download-tags.test.ts — "without FFmpeg, downloads in one go and says what the file lacks"<br>packages/domain/tests/unit/media-tags.test.ts — "takes the date tag first, then the release date, then the original release date"<br>packages/domain/tests/unit/titles.test.ts — "takes "4K Remaster", frame rates and a trailing " - Official ..." off as noise" |

UX-A11Y-001 **Roles are true and colour is never the only signal**

ADOPTED

A rail that cannot be dragged is role=img, not a slider that refuses changes. Icon buttons have names. The playing row has a glyph as well as a tint; status dots carry text.

|            |                                                                                                                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | packages/aqua-ui/src/components/Hero.tsx packages/aqua-ui/src/components/IconButton.tsx packages/aqua-ui/src/components/Badge.tsx                                                                                       |
| CHECKED BY | packages/aqua-ui/tests/dom/page-chrome.test.tsx — "goes read-only for a live broadcast and says why"<br>packages/aqua-ui/tests/dom/controls.test.tsx — "icon buttons require names; menu affordance sets aria-haspopup" |

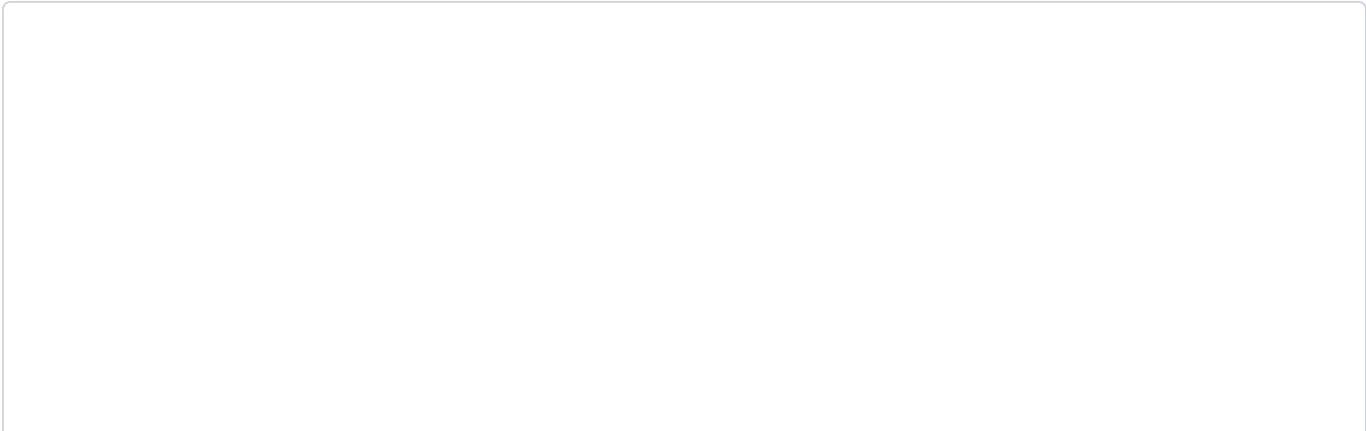
UX-LINK-001 **A download is a link**

ADOPTED

Downloads and destinations in another tab use ButtonLink (an <a> wearing the button face), never a <button> that assigns location.

|            |                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | packages/aqua-ui/src/components/Button.tsx                                                                                |
| CHECKED BY | music-player/tests/e2e/np/stats.spec.ts — "after real plays: the page counts them, and export and import round-trip them" |

THE AIRWAVE WINDOW



An Airwave window has one sheet of chrome carrying the product's name, from **BRANDING**, and a row of tools — nothing else. The window is the same in every state: signed out and on first run the hub draws its form or its gate inside it, with the tools it cannot open locked and the reason said. In the companion the chrome is the title bar: the operating system's own is hidden but the frame is kept, Windows draws minimise, maximise and close over the top-right corner, the chrome is the drag region and the tools are not, and the application menu is removed (DEC-020).

"Airwave Companion" over Library · Live TV · Remote · Settings; "Airwave Hub" over six tools, five of them locked until a real password is set.

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OWNED BY</b>   | <code>docker-container/src/web/App.tsx</code> <code>windows-companion/src/renderer/App.tsx</code> <code>windows-companion/src/renderer/styles.css</code><br><code>windows-companion/src/main/index.ts</code> <code>windows-companion/src/shared/identity.ts</code><br><code>packages/aqua-ui/src/styles/airwave-window.css</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>CHECKED BY</b> | <code>windows-companion/tests/contract/app-identity.test.ts</code> — "hides the OS title bar but never the frame"<br><code>windows-companion/tests/contract/app-identity.test.ts</code> — "asks Windows to draw its buttons over the chrome, at the height of the design's title strip"<br><code>windows-companion/tests/contract/app-identity.test.ts</code> — "names the window once, from the shared constant"<br><code>windows-companion/tests/contract/app-identity.test.ts</code> — "makes the chrome the drag region and leaves the tools clickable"<br><code>windows-companion/tests/contract/app-identity.test.ts</code> — "wears the design's stylesheet, not a copy of it"<br><code>windows-companion/tests/dom/companion-shell.dom.test.tsx</code> — "has one title, four tools and nothing else in its chrome"<br><code>docker-container/tests/dom/admin-shell.dom.test.tsx</code> — "renders the window with the six tabs, the status strip and Sign Out"<br><code>docker-container/tests/dom/admin-shell.dom.test.tsx</code> — "is drawn inside the same window, with every tab locked and the reason said" |

## TRUST, SAFETY AND DESTRUCTIVE ACTIONS

The button names the consequence ("Forget This Hub...", "Remove Folder", "Revoke"), sits apart from the default action, and the hub and the companion ask before doing it — in a sheet over the window it belongs to, never a second window. The safe answer has the keyboard: Cancel takes the focus in the hub, and is the default button for a destructive action in the companion; Escape cancels and focus returns to what asked (DEC-021).

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|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OWNED BY</b>   | <code>packages/aqua-ui/src/components/Sheet.tsx</code> <code>docker-container/src/web/views/Devices.tsx</code><br><code>windows-companion/src/renderer/views/Folders.tsx</code> <code>docker-container/src/web/ui.tsx</code><br><code>windows-companion/src/renderer/ui.tsx</code> <code>docker-container/src/web/views/Backup.tsx</code><br><code>docker-container/src/web/views/Discord.tsx</code> <code>docker-container/src/web/views/Downloads.tsx</code><br><code>docker-container/src/web/views/Groups.tsx</code> <code>docker-container/src/web/views/Library.tsx</code><br><code>docker-container/src/web/views/Profiles.tsx</code> <code>docker-container/src/web/views/Shares.tsx</code><br><code>windows-companion/src/renderer/App.tsx</code> <code>windows-companion/src/renderer/views/Backup.tsx</code><br><code>windows-companion/src/renderer/views/Hub.tsx</code> <code>windows-companion/src/renderer/views/Streaming.tsx</code><br><code>windows-companion/src/renderer/views/About.tsx</code>                                                                                                                                                                    |
| <b>CHECKED BY</b> | <code>packages/aqua-ui/tests/dom/controls.test.tsx</code> — "marks the destructive action, separates it from the default, and words it as what it does"<br><code>docs/AQUA_CONFORMANCE.md</code><br><div>reviewer</div> Each new destructive action is reviewed for wording and confirmation.<br><code>docker-container/tests/dom/admin-shell.dom.test.tsx</code> — "asks before revoking a device, and Cancel leaves it alone"<br><code>docker-container/tests/e2e/a11y.spec.ts</code> — "@a11y the confirmation sheet has no detectable violations, holds the keyboard, and Escape cancels"<br><code>windows-companion/tests/dom/companion-shell.dom.test.tsx</code> — "asks before removing one, says the files are not touched, and does nothing on Cancel"<br><code>windows-companion/tests/dom/companion-shell.dom.test.tsx</code> — "asks before forgetting the hub, and says the music stays"<br><code>windows-companion/tests/dom/companion-shell.dom.test.tsx</code> — "asks before restoring the defaults"<br><code>windows-companion/tests/dom/companion-controls.dom.test.tsx</code> — "shows the cache, asks before clearing it, and clears it through the main process" |

UX-SAFE-002    **The bootstrap password must be replaced before anything else**

ADOPTED

admin/admin signs in once, into the same window: Overview, with the amber gate asking for a real password and the other five tabs locked with the reason. Every setup-gated route refuses until then, and the hub stays on loopback outside a container.

- OWNED BY

docker-container/src/web/App.tsx    docker-container/src/auth/service.ts
- CHECKED BY

docker-container/tests/integration/setup-and-pairing.test.ts — “refuses every setup-gated route until the password is changed”

docker-container/tests/e2e/a11y.spec.ts — “@a11y the sign-in screen has no detectable violations”

docker-container/tests/dom/admin-shell.dom.test.tsx — “shows the gate in the window, with only Overview open and the other tabs locked with the reason”

docker-container/tests/dom/admin-shell.dom.test.tsx — “explains what stays off until the password is changed”

UX-SAFE-003    **Pairing is confirmed by comparing a fingerprint**

ADOPTED

A device receives a scoped credential only after the administrator confirms the fingerprint shown on both screens. Codes are never listed again after they are issued.

- OWNED BY

docker-container/src/web/views/Devices.tsx    docker-container/src/pairing/service.ts
- CHECKED BY

docker-container/tests/integration/setup-and-pairing.test.ts — “issues a scoped credential only after the fingerprint is confirmed”

UX-SAFE-004    **Privacy claims are stated where they can be checked**

ADOPTED

Copy that says nothing is uploaded, copied or moved must be true of the code path beside it, and the player loads nothing from outside its own origin.

- OWNED BY

music-player/src/views/Settings.tsx    music-player/src/views/Metrics.tsx
- CHECKED BY

music-player/tests/e2e/helper.spec.ts — “fetching a link puts a real track in the library, and it plays from this device”

music-player/tests/e2e/player.spec.ts — “loads nothing from outside its own origin, including three.js”

A pasted M3U or XMLTV link is never kept on the strength of how it is spelled. The companion's main process reads it — http(s) only, public addresses only, redirects re-checked, a deadline, size caps on the wire and after unpacking — and keeps it only when a playlist holds channels or a guide holds programmes. Until that answer arrives the row reads “checking...”; a refusal is a sentence under the field, with what was typed left in it and nothing added. A kept link that stops answering keeps what it last held, says so, and offers Check Again. What is kept is served to the player on this PC through the helper's read-only routes /helper/v1/tv/channels and /helper/v1/tv/guide.

“✓ 112 channels” · “That address is on this PC or your own network. Live TV links need an address on the internet.”

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | windows-companion/src/main/live-tv/index.ts windows-companion/src/main/live-tv/fetch.ts windows-companion/src/main/live-tv/m3u.ts windows-companion/src/main/live-tv/xmltv.ts local-helper/src/server.ts windows-companion/src/renderer/views/LiveTv.tsx windows-companion/src/main/hub.ts docker-container/src/live-tv/service.ts docker-container/src/web/views/LiveTv.tsx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CHECKED BY | windows-companion/tests/integration/live-tv.test.ts — “reads the link, keeps it with how many channels it holds, and remembers it” windows-companion/tests/integration/live-tv.test.ts — “refuses what is not a playlist, in a sentence, and keeps nothing” windows-companion/tests/integration/live-tv.test.ts — “refuses this PC and the home network before anything is connected” windows-companion/tests/integration/live-tv.test.ts — “is refused by the reader itself, not only by the form in front of it” windows-companion/tests/integration/live-tv.test.ts — “stops at the size cap, on the wire and after unpacking” windows-companion/tests/integration/live-tv.test.ts — “keeps what a link last held when it stops answering, says so, and recovers” windows-companion/tests/integration/live-tv.test.ts — “serves both through the helper's routes, in the contract's shape” local-helper/tests/integration/helper-tv.test.ts — “gives a player page the channel list the companion keeps, in the contract's shape, with no token” local-helper/tests/integration/helper-tv.test.ts — “is read-only: nothing but GET is served on these routes” windows-companion/tests/dom/companion-shell.dom.test.tsx — “says why a link was not kept, keeps what was typed, and adds nothing” windows-companion/tests/integration/live-tv-to-hub.test.ts — “sends the channels and their guide while sharing is on, leaving out addresses with a sign-in” windows-companion/tests/integration/live-tv-to-hub.test.ts — “sends nothing while sharing is off, and removes what was sent once it is turned off” docker-container/tests/integration/hub-admin-controls.test.ts — “is accepted only from a companion holding library:share, and read only with library:read” docker-container/tests/e2e/hub.spec.ts — “Live TV from the companion is shown read-only in the Music tab” |

The companion's local helper listens on this PC alone by default. Switched on under Settings ▶ Network, other devices on the home network may use only its four token-free read routes; every other route still refuses them, even with the token, and a rebound name or a page from the internet is refused. What the switch exposes is said before it is turned on (DEC-032).

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | local-helper/src/security.ts local-helper/src/server.ts windows-companion/src/renderer/views/Settings.tsx                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| CHECKED BY | local-helper/tests/security/helper-lan.test.ts — “cannot be reached from the network at all” local-helper/tests/security/helper-lan.test.ts — “lists only the four token-free reads” local-helper/tests/security/helper-lan.test.ts — “refuses every other route from the network, even with the token” local-helper/tests/security/helper-lan.test.ts — “refuses a rebound name and a page from the internet” windows-companion/tests/dom/companion-controls.dom.test.tsx — “explains what the LAN switch exposes before it is turned on” |

UX-PRIV-001 **Asking for a new version needs no key and opens only the release page**

ADOPTED

The companion asks the project's public releases for the latest version, with no key, and only while the check is switched on (it is on by default and asks at once when switched on). It says in words when a newer version is out, when there is none and when GitHub does not answer. Download opens the fixed release page through the main process; no address from the answer is ever opened (DEC-033). Logs leave the PC only through Export, with secrets, tokens and folder paths taken out.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | windows-companion/src/main/updates.ts windows-companion/src/main/log.ts<br>windows-companion/src/renderer/views/Settings.tsx windows-companion/src/renderer/views/About.tsx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CHECKED BY | windows-companion/tests/unit/settings-plumbing.test.ts — “asks the project's releases without a key, says when a newer one is out, and opens only the release page”<br>windows-companion/tests/unit/settings-plumbing.test.ts — “says plainly when GitHub has nothing or does not answer, and shows nothing when switched off”<br>windows-companion/tests/unit/settings-plumbing.test.ts — “takes out secrets, tokens and folder paths before anything leaves this PC”<br>windows-companion/tests/dom/companion-controls.dom.test.tsx — “says a new version is out, and Download opens the release page through the main process”<br>windows-companion/tests/dom/companion-controls.dom.test.tsx — “asks GitHub at once when the new-version check is turned on, and says what it found”<br>windows-companion/tests/security/companion.test.ts — “opens the release page, the logs and the data folder without the page naming a place” |

SHARED LISTENING AND THE DISCORD BOT

UX-SHARED-001 **The shared queue rejects stale commands and says so**

ADOPTED

Every queue command carries the revision it was based on. A stale command is refused with the current revision, never applied over someone else's change.

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| OWNED BY   | docker-container/src/group/service.ts packages/domain/src/queue.ts |
| CHECKED BY | docker-container/tests/integration/groups-and-queue.test.ts        |

UX-DISCORD-001 **Slash and prefix commands behave identically**

ADOPTED

Both transports run through one CommandService and render the same templates; the only difference is how the reply is delivered.

|            |                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------|
| OWNED BY   | docker-container/src/group/command-service.ts docker-container/src/discord/gateway.ts                       |
| CHECKED BY | docker-container/tests/integration/discord-parity.test.ts — “produces identical outcomes for every command” |

UX-DISCORD-002 **Refusals are private and say what was needed**

ADOPTED

permissionDenied, noResults, unavailableSource, emptyQueue, wrongChannel and error replies are ephemeral and name the reason or the channel to use.

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | packages/domain/src/templates.ts docker-container/src/group/command-service.ts                                        |
| CHECKED BY | docker-container/tests/integration/discord-parity.test.ts — “applies the same permission decision on both transports” |

UX-DISCORD-003 **The bot never goes silent without a reason**

ADOPTED

A track the bot cannot play is skipped with the reason recorded; a failed decode advances as an error; the hub's Discord section (Sharing tab) shows the voice state and last error.

|            |                                                                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | docker-container/src/discord/voice.ts docker-container/src/discord/worker.ts<br>docker-container/src/web/views/Discord.tsx                                                                                            |
| CHECKED BY | docker-container/tests/unit/discord-voice.test.ts — “skips a track it cannot play instead of going silent”<br>docker-container/tests/integration/discord-worker.test.ts — “shows the worker's live status in the hub” |

UX-STREAM-001 **Streaming from the PC follows what the connection costs, and says why it paused**

ADOPTED

Two switches under Remote decide whether the companion streams on Wi-Fi and Ethernet and on metered connections (mobile data, a hotspot); both are on by default (DEC-031). The cost is read from what Windows says about the connection; on a connection that is switched off streaming pauses, the pane says why in a sentence, and it starts again when allowed. The switches are off while streaming itself is off.

**OWNED BY** windows-companion/src/main/network.ts windows-companion/src/main/awsp.ts  
windows-companion/src/renderer/views/Streaming.tsx

**CHECKED BY** windows-companion/tests/integration/awsp-supervisor.test.ts — “pauses on a metered connection when that is switched off, says why, and starts again when allowed”  
windows-companion/tests/unit/settings-plumbing.test.ts — “reads what Windows says a connection costs”  
windows-companion/tests/unit/settings-plumbing.test.ts — “decides whether streaming may serve, and says why not”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “has the two switches, writes each, and says why streaming is paused on this connection”  
windows-companion/tests/dom/companion-controls.dom.test.tsx — “keeps the switches off while streaming itself is off”

WORDS

VOICE-001 **Empty states say what to do next**

OBSERVED

An empty list says what is missing and what to do next, in a few plain words: one sentence inside the list in the Airwave windows, a title and a line in the component library (“Nothing queued”). No apologies, no exclamation marks.

“No folders yet — add the one this PC keeps these in.” · “No guides yet — the guide will list channels without programmes.”

**OWNED BY** music-player/src/views/Queue.tsx windows-companion/src/renderer/views/Transfers.tsx  
windows-companion/src/renderer/views/Folders.tsx windows-companion/src/renderer/views/LiveTv.tsx  
windows-companion/src/renderer/views/Library.tsx

**CHECKED BY** music-player/src/views/Queue.tsx  
windows-companion/tests/dom/companion-shell.dom.test.tsx — “shows no sample channels: two empty lists in the design’s words”

VOICE-002 **Errors explain the problem and the way out**

ADOPTED

Name what failed in the person’s terms and what fixes it (“Add a bot token first (Admin → Discord → Token)”). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

**OWNED BY** docker-container/src/app.ts docker-container/src/discord/service-interface.ts  
docker-container/src/web/views/Discord.tsx

**CHECKED BY** docker-container/src/discord/service-interface.ts  
docker-container/tests/dom/admin-forms.dom.test.tsx — “loads what the hub has and shows its warnings as plain sentences”  
docker-container/tests/dom/admin-forms.dom.test.tsx — “says what the hub said when it refuses”

VOICE-003 **One product name, set in one place**

ADOPTED

The suite is Airwave and its apps are Airwave, Airwave Hub and Airwave Companion; the names come from BRANDING in TypeScript, and “Now Playing” is the name of the player’s home page only (DEC-025). Native resources that cannot import BRANDING (Android strings.xml, the SVG icons’ aria-labels, the Electron builder config) repeat it and must be changed together. Machine identifiers keep their spelling.

**OWNED BY** packages/contracts/src/branding.ts android/app/src/main/res/values/strings.xml music-player/public/icon.svg  
windows-companion/resources/icon.svg windows-companion/src/shared/identity.ts

**CHECKED BY** packages/contracts/tests/branding.test.ts  
packages/contracts/src/branding.ts  
windows-companion/tests/contract/app-identity.test.ts — “names the window once, from the shared constant”

|             |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
|-------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| NP-PRIN-001 |  | <b>One component, one reference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ADOPTED |
|             |  | <p>Every component reconstructs one named Apple release (iPod classic, iTunes 10, OS X 10.6, iOS 5...) and is sampled from it, not averaged. Eras mix across components, never inside one. The reference is written beside the component's tokens.</p> <p>The contextual menu is OS X 10.6 (opaque, 5px radius, 20px rows, 13px text); the action sheet that replaces it on touch is iOS 5/6.</p>                                                                                                                                                                                                                                                                                     |         |
| OWNED BY    |  | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| CHECKED BY  |  | music-player/tests/e2e/np/menuverify.spec.ts — “era metrics, keyboard nav intact, escape closes, menu action still fires”<br>music-player/tests/e2e/np/menuverify.spec.ts — “dark menu opaque”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |
| NP-PRIN-002 |  | <b>Unavailable is shown and explained, never hidden</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ADOPTED |
|             |  | <p>A control that cannot act in this context stays visible and focusable, is marked aria-disabled, looks disabled, and says why when pressed or hovered. It is not removed and it does not silently do nothing.</p> <p>On a station, previous and next are greyed, titled 'Not available for radio', and pressing them says 'Pick another station from the list'.</p>                                                                                                                                                                                                                                                                                                                 |         |
| OWNED BY    |  | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| CHECKED BY  |  | music-player/tests/e2e/np/livevol.spec.ts — “the radio transport: Previous and Next read as unavailable, Play does not, and they say why”<br>music-player/tests/e2e/np/radio.spec.ts — “the tab opens on stations, not on a menu, and the panel is the song list’s own rules”<br>music-player/tests/e2e/np/func.spec.ts — “avatar gradient not black; the transport at rest is a slider with every key reachable”                                                                                                                                                                                                                                                                     |         |
| NP-PRIN-003 |  | <b>Measured, not assumed</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ADOPTED |
|             |  | <p>Numbers shown to the listener come from what actually happened. A track's seconds come from the transport's own clock (a seek is not a listen); stream time is measured from playing to pause/end; anything under five seconds is not a session; a film or song counts as finished only when it played to its end. No invented history, no sample figures — with one labelled exception: Statistics shows a generated demo year while this player has recorded nothing, and says so in the Data menu, the source line and Your data (NPD-023).</p> <p>Statistics opens on this player's own history once anything has played; before that, on a demo year that says it is one.</p> |         |
| OWNED BY    |  | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| CHECKED BY  |  | music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”<br>music-player/tests/e2e/np/prefs.spec.ts — “it is a page at both widths, and a section has its own address”<br>music-player/tests/e2e/np/discover.spec.ts — “a track’s seconds are measured, not assumed”                                                                                                                                                                                                                                                                                                                                           |         |
| NP-PRIN-004 |  | <b>Values live in tokens, once</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ADOPTED |
|             |  | <p>Colour, gradient and metric values are CSS custom properties on :root. Components read variables. The dark list is written twice (Auto and forced Dark) and the two copies must be identical.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| OWNED BY    |  | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| CHECKED BY  |  | <div>reviewer</div> Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |

NP-ERA-001 **Menus are opaque**

ADOPTED

Menus, sheets and popovers of this period had no vibrancy. Their backgrounds are opaque in both themes; only the iOS HUD is translucent.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/menuverify.spec.ts — “dark menu opaque”

NP-ERA-002 **Selection is a gradient, not a flat fill**

ADOPTED

Selected rows and highlighted menu items use the era's two-stop vertical gradient with white text: iTunes 10 blue in lists (--lib-sel-\*), Aqua blue in menus (--ctx-hi-\*), iPod blue on the iPod screen (--ipod-sel-\*).

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.

PLAYER SHELL — LIVE AND BROADCAST

NP-LIVE-001 **A live feed says LIVE where the time would be**

ADOPTED

On Live TV the video controller replaces the elapsed/duration readout with a red LIVE badge, from the moment the tab opens (not only after a channel plays). The badge is centred on the progress rail horizontally in its box and vertically on the rail's centre line at every width.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/livevol.spec.ts — “the Live TV tab is live before a channel is chosen, and a channel does not follow you on to TV”  
music-player/tests/e2e/np/devices.spec.ts — “: the LIVE readout sits right, idle and playing”  
music-player/tests/e2e/np/livevol.spec.ts — “LIVE on the video bar: the tab is live on arrival and a channel keeps it so”

NP-LIVE-002 **The progress of a broadcast does not move and cannot be seeked**

ADOPTED

During a live feed the scrubber is an image (role=img, no range ARIA), the channel loops so LIVE stays true, and pressing it explains instead of seeking.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/livevol.spec.ts — “LIVE on the video bar: the tab is live on arrival and a channel keeps it so”  
music-player/tests/e2e/np/func.spec.ts — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”  
music-player/tests/e2e/np/func.spec.ts — “avatar gradient not black; the transport at rest is a slider with every key reachable”

PLAYER SHELL — TRANSPORT AND SOUND

NP-TRANS-001 **One source sounds at a time**

ADOPTED

Starting a station stops the video; starting a channel or film stops the station. A film does not follow the listener onto the Live TV or TV tab.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/livevol.spec.ts — “one thing at a time: a channel stops the station, and tuning back stops the channel”  
music-player/tests/e2e/np/livevol.spec.ts — “the Live TV tab is live before a channel is chosen, and a channel does not follow you on to TV”

NP-TRANS-002 **The volume control owns every sound**

ADOPTED

The volume slider drives the station and the video as well as music; Home is silence and mutes, End is full, the level survives a change of station and a reload.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/livevol.spec.ts — “volume reaches the stream: the slider, Home and End, and a new station keeps the level”  
music-player/tests/e2e/np/livevol.spec.ts — “the level is remembered across a reload”

NP-TRANS-003 **Video starts at the highest quality**

ADOPTED

A channel, episode or film opens on its highest available rendition; the quality menu lets the viewer step down.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/media.spec.ts — “Live TV: the guide replaces the song list, the video player takes the top, and a channel plays”

NP-TRANS-004 **The equalizer never silences what it cannot process**

ADOPTED

Audio is routed through the equalizer only when the browser allows processing it (same-origin, blob:, data:, or the companion). Cross-origin streams play unequalized, and the pane says so.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”

PLAYER SHELL — MENUS AND COMMANDS

NP-MENU-001 **Right-click opens the item's commands; keys work inside**

ADOPTED

Right-clicking a song, station, channel, series or film opens an OS X contextual menu at the pointer with the commands for that kind of item. Arrow keys move, Return chooses, Escape closes and returns focus. Scrolling the page after opening does not close it instantly.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/menuverify.spec.ts — “era metrics, keyboard nav intact, escape closes, menu action still fires”  
music-player/tests/e2e/np/lists.spec.ts — “the song menu is untouched”  
music-player/tests/e2e/np/lists.spec.ts — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”

NP-MENU-002 **On touch, a long press raises an action sheet**

ADOPTED

Touch devices get the same commands as an iOS action sheet from a long press, with Cancel last.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/menuverify.spec.ts — “long press raises action sheet, listing all commands + playlists + cancel, and its action fires”

NP-MENU-003 **Confirmation is a HUD, not a dialog**

ADOPTED

A command that completes (added to a playlist, removed) confirms with a brief HUD naming what happened. Dialogs are only for commands that need more input.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/menuverify.spec.ts — “era metrics, keyboard nav intact, escape closes, menu action still fires”  
music-player/tests/e2e/np/menuverify.spec.ts — “long press raises action sheet, listing all commands + playlists + cancel, and its action fires”

NP-MENU-004 **One menu for several songs**

ADOPTED

Right-clicking inside a selection of several songs opens the contextual menu for all of them: its heading says how many, and each command names the count it will act on — Download N Songs (or Remove N Downloads when all are saved), Add N Songs to Group Queue, Add to Playlist ▶ (ticked only when the playlist holds every one), New Playlist from N Songs..., Deselect All. The selection survives the repaint a command causes.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/marquee.spec.ts — “dragging draws a selection box, and the menu for several acts on all of them”

PLAYER SHELL — LISTS AND PLAYLISTS

NP-LIST-001 **Every tab has playlists, behind the library button**

ADOPTED

Music, Radio, Live TV, TV and Movies each keep their own playlists. They are reached from the library (menu) button on the left of the scope bar, and filled from the item's contextual menu. They persist across reloads.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/lists.spec.ts — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”  
music-player/tests/e2e/np/lists.spec.ts — “the playlist is still there after a reload”

NP-LIST-002 **Sorting is a button**

ADOPTED

Sortable column headers contain buttons and expose aria-sort; the sorted column tints blue (iTunes 10). A set order (a playlist) shows no arrow until a column sort overrides it.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/func.spec.ts — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”  
music-player/tests/e2e/np/ipod.spec.ts — “no column may show a sort arrow while the rows are in set order”

NP-LIST-003 **Drag a box to pick several songs**

ADOPTED

On a pointer, pressing on a song row's empty space and dragging draws a rubber band; every row it touches is selected. Cmd/Ctrl-click adds or removes one song, Shift-click takes the range, Cmd/Ctrl-A takes the list, Escape clears. Picking is never playing: the click that ends a drag, and a modified click, start nothing. A plain click goes back to one song and plays it.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/marquee.spec.ts — “dragging draws a selection box, and the menu for several acts on all of them”

PLAYER SHELL — HEADER AND NAVIGATION

NP-CHROME-001 **The header search belongs to Music**

ADOPTED

The status bar's search pill searches songs and takes pasted song links, so it is shown only on Music. Other tabs search with the list's own field.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/hdr.spec.ts — “the header field is up on Music and gone on the other tabs, which keep their own field”

The avatar at the right of the status bar opens Settings as its own page (#settings), not a pop-up.

**OWNED BY**            music-player/index.html    music-player/scripts/make-shell.py

**CHECKED BY**        music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”

When Music or Radio is the tab and Settings covers the player, the header's search pill becomes a mini player: title and artist, time and duration, play/pause and skip, driving the real transport. Leaving a video tab — for Settings or another tab — pauses a film or episode, and disconnects Live TV (the stream is closed, not just muted).

**OWNED BY**            music-player/index.html    music-player/scripts/make-shell.py

**CHECKED BY**        music-player/tests/e2e/np/mini.spec.ts — “with Settings open, the header shows a mini player that drives the real transport”  
music-player/tests/e2e/np/mini.spec.ts — “video tabs: Settings from Live TV shows no mini player and disconnects; Music pauses a channel”

FINDING AND HEARING MUSIC BEFORE IT IS YOURS

In the header search popover, a result with a clip plays it on a click of its artwork tile — the whole clip, up to 30 seconds, with a countdown ring and a fade at the tail — and resting a fine pointer on the row for five seconds arms the same play with a visible fill in the selection blue. Leaving the row, a key, a scroll or another row cancels the hold; touch and reduced motion never arm; the click always works. The main track pauses for the audition and resumes after it. A row with no clip keeps its tile and says why on hover instead of dressing as a button. When the player is paired, rows lead with the hub's enriched answers (features, album, genre, tempo) and a match below confidence 0.5 shows the platform's own words; unpaired, the keyless chain is the whole story.

**OWNED BY**            music-player/index.html    music-player/scripts/make-shell.py

**CHECKED BY**        music-player/tests/e2e/np/preview.spec.ts — “paired, the hub answers first: album, features, genre chip and bpm on the row”  
music-player/tests/e2e/np/preview.spec.ts — “a click plays up to thirty seconds, and the main track waits its turn”  
music-player/tests/e2e/np/preview.spec.ts — “resting the pointer on a row arms it, fills the ring, and then plays”  
music-player/tests/e2e/np/preview.spec.ts — “reduced motion never arms; the click still works”  
music-player/tests/e2e/np/preview.spec.ts — “a row with no clip says so instead of pretending”

A YouTube, SoundCloud, Bandcamp or Spotify link pasted into the header search becomes a row at once and fills in from the best source that answers: the paired hub first, then the companion on this PC — its yt-dlp or spotDL reading the link with no account or key (GET /helper/v1/resolve) — then the site's oEmbed. A later, lesser answer only fills what a better one left empty. The details line under the title gains the release date (or, failing that, the upload date) as YYYY-MM-DD; the companion also supplies duration, album, features and art. A pasted playlist, set or album is replaced by its songs, each a row with its own address that can be added and fetched on its own, under a count that names the list — "Title: N songs" — and says "(the first 200 of M)" when the cap of 200 left some out. An entry the site lists by address alone shows its address as a name until its page of rows is on screen, when it is looked up. The companion answers a page it has vetted, or the token; it never answers another device on the network, because reading a link runs a tool.

Pasting <https://www.youtube.com/watch?v=dQw4w9WgXcQ>: "Never Gonna Give You Up" — "Rick Astley — YouTube · 2009-10-25", 3:33. Pasting a YouTube playlist: "Team Fortress 2 [2010 Version]: 29 songs".

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | music-player/scripts/make-shell.py music-player/index.html local-helper/src/resolve.ts local-helper/src/server.ts packages/contracts/src/api/local-helper.ts packages/domain/src/media-tags.ts packages/domain/src/titles.ts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CHECKED BY | music-player/tests/e2e/np/links.spec.ts — "a pasted YouTube link: the companion's answer lands over oEmbed's, with the date on the details line"<br>music-player/tests/e2e/np/links.spec.ts — "with no companion, a pasted link still resolves through oEmbed and nothing asks a local port"<br>music-player/tests/e2e/np/links.spec.ts — "a pasted playlist becomes its songs, each its own row, under a count that names the list and the cap"<br>music-player/tests/e2e/np/links.spec.ts — "a set listed by address alone shows each entry's own words once its page is on screen"<br>local-helper/tests/integration/helper-resolve.test.ts — "reads a YouTube video: clean title and artist, the whole date, the duration and the art"<br>local-helper/tests/integration/helper-resolve.test.ts — "lists a playlist's songs, each with its own address, and says the count and the cap"<br>local-helper/tests/integration/helper-resolve.test.ts — "reads a Spotify track through spotDL save, in a home of its own that is gone afterwards"<br>local-helper/tests/integration/helper-resolve.test.ts — "answers a vetted page without the token, and no page only with it"<br>local-helper/tests/integration/helper-resolve.test.ts — "refuses a host off the allowlist, a private address and a non-https link before starting anything"<br>local-helper/tests/security/helper-lan.test.ts — "refuses every other route from the network, even with the token" |

## RADIO: THE SONG ON THE AIR, AND KEEPING IT

The tuned station's song is read from what the station itself publishes: a feed it serves with CORS open, or its stream's ICY metadata read for the page by the paired container or the companion app (a browser never sees ICY). Only the tuned station reads its stream; the station list keeps to feeds, so a list never becomes a stream connection per row. With no feed and nothing to read the stream, the line shows the programme format the directory carries, never an invented title. Private and loopback addresses are never read on a page's behalf.

|            |                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | music-player/scripts/make-shell.py docker-container/src/api/routes/providers.ts local-helper/src/server.ts packages/domain/src/radio-node.ts                                                                                                                                                                                                                                                    |
| CHECKED BY | music-player/tests/e2e/np/radio-air.spec.ts — "a station with no feed shows the song its stream carries, read by the companion"<br>music-player/tests/e2e/np/radio-air.spec.ts — "with no companion and no container, a station without a feed invents nothing"<br>packages/domain/tests/integration/radio-node.test.ts — "refuses private and loopback addresses unless a test says otherwise" |

Right-clicking a station (a long press on touch) whose song is known adds an “On air” section to its menu: Add Song to Up Next, Add Song to Group Queue, and Add Song to Playlist with New Playlist.... The song becomes a library entry of this player’s own — filled in by the same search chain when it finds that song by that artist, labelled Radio when it does not — and it survives a reload. The group item names the group when there is one, lists them when there are several, and is disabled with the reason when the player is unpaired or in no group; the container queues a playable copy by name, the way Discord’s /play does, and says why when there is none. A same-titled song by someone else is never taken for it.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | music-player/scripts/make-shell.py docker-container/src/api/routes/groups.ts<br>docker-container/src/group/command-service.ts                                                                                                                                                                                                                                                                                                                                                                                                     |
| CHECKED BY | music-player/tests/e2e/np/radio-air.spec.ts — “right-click keeps the song: Up Next, a new playlist, and it survives a reload”<br>music-player/tests/e2e/np/radio-air.spec.ts — “paired: the container reads the title, and the song goes to the group’s queue by name”<br>music-player/tests/e2e/np/radio-air.spec.ts — “touch: a long press on a station opens the same menu, and the lift does not retune”<br>docker-container/tests/integration/radio-and-requests.test.ts — “never queues a same-titled song by someone else” |

A player that can reach a companion takes its Live TV channels from the playlists kept in the companion’s Live TV tab; nothing has to be loaded by hand in the player, and a playlist the page could not read itself is the companion’s to read. A channel with a guide entry shows what is on and what follows; one without says Live and no programme is invented. A guide refresh changes the words and never resets the list or the listener’s place in it. A companion with no playlists, or one that has gone, leaves the guide as it was.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | music-player/scripts/make-shell.py local-helper/src/server.ts windows-companion/src/main/live-tv/index.ts<br>docker-container/src/live-tv/service.ts                                                                                                                                                                                                                                                                      |
| CHECKED BY | music-player/tests/e2e/np/live-tv.spec.ts — “the companion’s channels fill the guide, with what is on and what follows”<br>music-player/tests/e2e/np/live-tv.spec.ts — “a companion with no playlists leaves the guide as it was, and one that is gone breaks nothing”<br>music-player/tests/e2e/np/live-tv.spec.ts — “with no companion on this machine, a paired hub’s copy of the companion’s Live TV fills the guide” |

PLAYER SHELL — PREFERENCES

Below 720px Settings is an iPhone OS drill-down: an index of 44px rows, a back button that names where it returns to. From 720px it is a preferences window with a toolbar of five panes. The browser Back button walks the same history.

|            |                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OWNED BY   | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                    |
| CHECKED BY | music-player/tests/e2e/np/prefs.spec.ts — “media playlists live behind the library button”<br>music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation” |

NP-PREF-002 **A preference applies when it is changed**

ADOPTED

No Save and no Apply. Theme, motion, artwork, defaults, group mode and every algorithm setting apply immediately and persist (player:prefs). Revert, in the footer, restores everything to how it was when Settings opened (NPD-016).

- OWNED BYmusic-player/index.htmlmusic-player/scripts/make-shell.py
- CHECKED BYmusic-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”  
music-player/tests/e2e/np/prefs.spec.ts — “player preferences, the equalizer and group mode; the equalizer survives a reload”  
music-player/tests/e2e/np/discover.spec.ts — “Revert puts back the weight, the algorithm, and removes the copy”

NP-PREF-003 **The algorithm is editable and cannot be made invalid**

ADOPTED

Every recommendation setting has a control. Rules the engine enforces (weights summing to 100, ranges) are checked as the user edits and offered a fix, never silently clamped.

- OWNED BYmusic-player/index.htmlmusic-player/scripts/make-shell.py
- CHECKED BYmusic-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-PREF-004 **Built-in algorithms never change**

ADOPTED

Airwave default, Late night and Crate digger are read-only. The first change to one makes a named copy, selects it and says so; built-ins cannot be renamed or deleted. Delete asks twice.

- OWNED BYmusic-player/index.htmlmusic-player/scripts/make-shell.py
- CHECKED BYmusic-player/tests/e2e/np/discover.spec.ts — “built-ins, copies, rename, duplicate, delete, export, import, and a reload”

NP-PREF-005 **Algorithms move as files, and imports say what they read**

ADOPTED

Export writes one algorithm (or all of yours) as an airwave-algorithm JSON file; Import takes those or a bare engine config, reads only keys the engine knows with values of the right type, and reports how many settings it read and how many it ignored.

- OWNED BYmusic-player/index.htmlmusic-player/scripts/make-shell.py
- CHECKED BYmusic-player/tests/e2e/np/discover.spec.ts — “the shelf records what it shows, and plays from it count”  
music-player/tests/e2e/np/discover.spec.ts — “built-ins, copies, rename, duplicate, delete, export, import, and a reload”

NP-PREF-006 **Settings controls are Mac OS X controls, drawn over native elements**

ADOPTED

Push buttons, pop-ups and checkboxes in Settings wear Mac OS X drawings: Snow Leopard (10.6) push buttons — a matte, brushed-metal pill with a graphite hairline, blue when pressed — and 10.4 Aqua pop-ups (white body, blue arrow segment) and gel checkboxes that AquaArt draws from sampled colour profiles and publishes as custom properties. The elements stay native button, select and checkbox, so keyboard, forms and screen readers are unchanged. 22px on desktop, 30px on touch; the drawings and their black labels are the same in both themes (NPD-021).

- OWNED BYmusic-player/index.htmlmusic-player/scripts/make-shell.py
- CHECKED BYmusic-player/tests/e2e/np/aqua.spec.ts — “buttons, pop-ups and checkboxes wear the Aqua drawings and still work”  
music-player/tests/e2e/np/aqua.spec.ts — “the 30px drawings on a phone”

Sources ▶ Connections tests the Windows companion app (GET /helper/v1/health; an empty address scans 127.0.0.1:17342–17345) and the Docker container (GET /healthz, /readyz, /api/v1/hub; port 4546), using the AudioWave repo’s own endpoints; and Stream from a PC pairs with the companion’s AWSP sidecar by its ticket and six-digit code (docs/AWSP.md §2), then shows the PC’s name, the connection type (relay-carried in a browser) and how many of its tracks joined the library, with the same lights. Each test shows what came back — version, downloaders, allowed sites, readiness checks, fingerprint, encryption — with a green, amber or red light; every request goes in the connection log with its status and time; and the What’s passed table lists each setting, where it goes, and 'Ready' only once that destination has answered.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/connect.spec.ts — “the companion, the container and backup”

music-player/tests/e2e/awsp.spec.ts — “pairs, plays relay-carried, is bit-identical, seeks by range, and resumes across a relay restart”

Sources ▶ Backup chooses what to copy (Music, TV, Movies; playlists, EQ presets and settings) and where (the Docker container, the Windows companion app, or both), and how often. Media moves between the companion app and the container, so it needs both to have answered a test and its folder to be set; settings need only their destination. Back Up Now stays disabled and says which of those is missing; once it runs, the request is written to the connection log and What’s passed lists a backup row per destination.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/connect.spec.ts — “the companion, the container and backup”

Once the player is paired with the Docker container (the hub’s own /pairing/claim, /status and /complete, confirmed by fingerprint), Settings gains a Profile tab: a username, a picture cropped to 256 × 256, and playlists shared as CSV (title, artist, album, seconds), all saved to the container and never kept in the browser. A username someone else has — compared without case — is caught as you type and refused on save. Search lists people on the container above songs; a profile opens in a sheet with their picture, name and shared playlists, and a song you also have can be played from it. Your picture takes the jewel case’s place on the Music page. Until the container has the profile routes, the tab says so.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/profile.spec.ts — “pair, edit the profile on the container, find someone, open their profile”

Settings ▶ Equalizer ▶ Default volume sets the level Airwave opens at — where you left it, or a fixed default — and Set Volume Now moves the bar’s volume to it. Music, radio, Live TV, TV and films all start at and follow that one volume.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py

**CHECKED BY** music-player/tests/e2e/np/volume.spec.ts — “the default volume: Set Volume Now, then each start mode across a reload”

NP-PREF-011 **A backup says how big it is and whether it fits**

ADOPTED

Settings ▶ Sources ▶ Backup shows the expected size, part by part, and each destination's free space as a bar (in use · this backup · free after it). Figures the player can't measure itself come from the companion app and the container; until they answer, the row names which figure is waiting and who measures it. Back Up Now is unavailable when the backup doesn't fit. Algorithms can be backed up like the rest. Sizes are decimal (1 GB = 10<sup>9</sup> bytes).

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py  
**CHECKED BY** music-player/tests/e2e/np/connect.spec.ts — “the companion, the container and backup”

NP-PREF-012 **Groups and invites live on the Profile tab**

ADOPTED

Settings ▶ Profile lists your groups on the container with their member count and your role, makes a new one (you own it), and — for an owner or admin whose player was paired with the “Manage groups” permission — makes an invite link that works once and expires after the time chosen. Without that permission no invite link is offered, because the container would refuse it, and the message names the missing permission rather than blaming the role. Invites you have received wait there until you join or decline; ones you have sent show who they were for and when they expire. A code the container can't withdraw is never shown as withdrawn.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py  
**CHECKED BY** music-player/tests/e2e/np/groups.spec.ts — “the Profile tab: groups, a new group, an invite link, Withdraw, and Leave”  
music-player/tests/e2e/np/groups.spec.ts — “paired without “Manage groups”, no group offers an invite link — the hub would refuse it”  
tests/journey/cross-app.spec.ts — “the whole pass: hub set up, player paired, group joined, invite declined, name taken, group owned, someone found”

NP-PREF-013 **An invite link opens an invite page**

ADOPTED

Opening an invite link shows a page naming the group, who sent it, the container, the role and when it expires, with Join Group and Decline (Not Now keeps it waiting). What the link says is labelled as the link's until the container confirms it. An unpaired player, or one paired with another container, is asked to pair first and offered Connections. The code is taken out of the address bar as soon as it has been read.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py  
**CHECKED BY** music-player/tests/e2e/np/groups.spec.ts — “the invite page: from a link, Not Now, Join, a used code, a bad code, Decline”  
music-player/tests/e2e/np/groups.spec.ts — “not paired: the invite page asks you to pair first”

PLAYER SHELL — STATISTICS AND CHARTS

NP-DATA-001 **One hue for amounts, named hues for genres, and every chart as numbers**

ADOPTED

Amounts use one hue (--viz-1) with a single-hue ramp for the skyline (--viz-sky-0 → --viz-1 → --viz-peak). Genres use seven token hues (--viz-g1...g7, --viz-g0 for the rest), lifted in dark, and a genre's colour never appears without its name beside it. Every chart has its value written beside it or a table under it, and the 3D views are backed by the same numbers in 2D.

**OWNED BY** music-player/index.html music-player/scripts/make-shell.py  
**CHECKED BY** music-player/tests/e2e/np/stats.spec.ts — “before any play: nothing recorded, and no generated source on offer”

NP-DATA-002 **Every figure is set against the period before**

ADOPTED

For 7, 30, 90 days and the year, each tile compares with the period of the same length just before it, with ▲/▼, the figure, and colour for good or bad news (more skips is up and bad). All time has no period before and shows no comparison. Heard-to-the-end and skips count only plays the transport timed (NPD-018).

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/discover.spec.ts — “statistics against the period before, and the recommendation funnel”

NP-DATA-003 **Recommendations are counted from the shelf's own record**

ADOPTED

What Discover shows is logged once per song per day; a play started there is a recommendation taken. The funnel reads recommended → started → heard to the end → replayed → added or favourited from those logs, never from estimates.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/discover.spec.ts — “the shelf records what it shows, and plays from it count”  
music-player/tests/e2e/np/discover.spec.ts — “statistics against the period before, and the recommendation funnel”

NP-DATA-004 **The statistics page works without its 3D views**

ADOPTED

Statistics loads three.js from the page's import map the first time it opens. If that fails or takes more than eight seconds, each 3D well says it couldn't load and the page carries on; every 3D view has the same numbers in 2D or in a table. A theme change repaints the wells, scenes and labels from the tokens.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/stats.spec.ts — “before any play: nothing recorded, and no generated source on offer”  
music-player/tests/e2e/np/stats.spec.ts — “the wells say so, the tables stay”

PLAYER SHELL — LAYOUT ACROSS DEVICES

NP-RESP-001 **The video controller fits the player, not the window**

ADOPTED

Controller-bar content responds to the player's width (container queries): elapsed only under 560px, download and touch volume hidden, More hidden under 358px; the scrubber keeps at least 60px. Checked at 18 device sizes.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/devices.spec.ts — “: the LIVE readout sits right, idle and playing”  
music-player/tests/e2e/np/livevol.spec.ts — “px LIVE is centred in its box, clear of its neighbours, and the scrubber keeps room”

NP-RESP-002 **Picture and controls fit under the status bar**

ADOPTED

On short screens (landscape phones, short laptops) the video player is capped so it and its controls fit below the status bar without scrolling.

OWNED BY music-player/index.html music-player/scripts/make-shell.py

CHECKED BY music-player/tests/e2e/np/devices.spec.ts — “: the LIVE readout sits right, idle and playing”

PLAYER SHELL — ACCESSIBILITY

|                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                      |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| NP-A11Y-001 <b>Every control is reachable and named</b>                                                                                                                                          |                                                                                                                                                                                                                                                                      | ADOPTED |
| Icon-only controls carry accessible names; groups of choices are radiogroups moved with arrow keys; scrolling regions are focusable; headings give the page an outline (one h1).                 |                                                                                                                                                                                                                                                                      |         |
| OWNED BY                                                                                                                                                                                         | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                           |         |
| CHECKED BY                                                                                                                                                                                       | music-player/tests/e2e/np/func.spec.ts — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”<br>music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation” |         |
| NP-A11Y-002 <b>Motion stops when asked</b>                                                                                                                                                       |                                                                                                                                                                                                                                                                      | ADOPTED |
| With the OS reduced-motion setting, or Preferences > Reduce animation, pulses, barber poles, spinners and transitions stop.                                                                      |                                                                                                                                                                                                                                                                      |         |
| OWNED BY                                                                                                                                                                                         | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                           |         |
| CHECKED BY                                                                                                                                                                                       | music-player/tests/e2e/np/prefs.spec.ts — “opening: Settings is a page with its own address, and the toolbar is the navigation”                                                                                                                                      |         |
| NP-A11Y-003 <b>Text meets 4.5:1; exceptions are recorded</b>                                                                                                                                     |                                                                                                                                                                                                                                                                      | ADOPTED |
| Text/background pairs listed in design/foundations.json meet WCAG 2.2 AA (4.5:1 text, 3:1 marks) in both themes. Period colours that fall short are recorded as decisions, never added silently. |                                                                                                                                                                                                                                                                      |         |
| OWNED BY                                                                                                                                                                                         | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                           |         |
| CHECKED BY                                                                                                                                                                                       | reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.                                                                                                                                    |         |

PLAYER SHELL — VOICE AND COPY

|                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                               |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| NP-VOICE-001 <b>Commands are Title Case; an ellipsis means a dialog follows</b>                                                                                                                       |                                                                                                                                                                                                                                                                                                                                               | ADOPTED |
| Menu commands use Title Case verbs ('Go to Album', 'Add to Playlist'). An ellipsis marks exactly the commands that open a dialog for more input ('New Playlist...', 'Change Artist...').              |                                                                                                                                                                                                                                                                                                                                               |         |
| OWNED BY                                                                                                                                                                                              | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                    |         |
| CHECKED BY                                                                                                                                                                                            | music-player/tests/e2e/np/menu2.spec.ts — “six commands in three groups, and an ellipsis marks exactly the dialog commands”                                                                                                                                                                                                                   |         |
| NP-VOICE-002 <b>The verb matches the thing</b>                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                               | ADOPTED |
| A song or film is Played, a channel is Watched, a series is Opened. Commands that make no sense for a kind of item are absent from its menu, not disabled.                                            |                                                                                                                                                                                                                                                                                                                                               |         |
| OWNED BY                                                                                                                                                                                              | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                    |         |
| CHECKED BY                                                                                                                                                                                            | music-player/tests/e2e/np/lists.spec.ts — “Live TV: a channel is watched, not played, and its playlist still reads as a guide”<br>music-player/tests/e2e/np/lists.spec.ts — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”<br>music-player/tests/e2e/np/lists.spec.ts — “the song menu is untouched” |         |
| NP-VOICE-003 <b>Explanations are one plain sentence with a way forward</b>                                                                                                                            |                                                                                                                                                                                                                                                                                                                                               | ADOPTED |
| When something cannot happen, say what to do instead in one sentence, without apology or error codes: 'Pick another station from the list', 'Couldn't reach search — open it on a platform instead:'. |                                                                                                                                                                                                                                                                                                                                               |         |
| OWNED BY                                                                                                                                                                                              | music-player/index.html music-player/scripts/make-shell.py                                                                                                                                                                                                                                                                                    |         |
| CHECKED BY                                                                                                                                                                                            | reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.                                                                                                                                                                                                             |         |

## Accessibility

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Not bolted on afterwards. These are parts of the design.

- **One tab stop per group, arrows to move.** The section strip, the mode switch, the music list, the context menu and the tools of an Airwave window all use a roving `tabIndex`, and the focus ring travels with the selection.
- **Focus is always visible.** An end-to-end test tabs the whole page and fails on any element that shows no indicator.
- **Roles must be true.** A rail that cannot be dragged is `role="img"` with no value pair, not a slider that refuses every change. A button that opens a section does not claim `aria-haspopup="menu"`.
- **Names say what will happen**, including the refusal.
- **Colour is never the only signal.** The playing row has a glyph as well as a tint; status dots carry text.
- **Text meets 4.5:1.** Where the Airwave design's status colours fall short as text on white, the hub reads sentences in darker inks of the same hues (DEC-022); an axe pass runs on every hub tab, the sign-in window and the confirmation sheet.

## Every screen

Every surface found in the products' own navigation, with the states it has to answer and the rules that govern it. `pnpm styleguide:check` rebuilds this list from the `ViewId` declarations, the sheet files and the Discord command table, and fails when something appears that is not recorded in `design/coverage.json`.

**Seen running:** opened in the running product for this revision (see Coverage). **Mockup only:** rendered at every size in the mockups with fixture data — the hub and the companion from their own stylesheets and kits, the player from the component library's page skin. **Not run here:** the product could not be started where this guide was built. **Source only:** checked against the code, not looked at.

### AIRWAVE — THE PLAYER

The shell generated from `design/frontends/airwave-now-playing.html`: the PWA, the single-file build, the Android app and the local helper all show it. Its surfaces are the player-shell rows (adopted). The rows marked proposed are the React interface in `music-player/src`, which is kept and tested but no longer served (DEC-019).

| SURFACE                                                                                                                                                                                                             | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                     | RULES                                                   | SEEN                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------|
| <b>Status bar, mode switch, hero and section strip</b><br>(React specimen, not served — DEC-019) every section<br><code>music-player/src/App.tsx</code><br><code>packages/aqua-ui/src/components/PageBar.tsx</code> | <ul style="list-style-type: none"><li>nothing playing</li><li>playing</li><li>paused</li><li>live stream (read-only scrubber)</li><li>unknown duration</li><li>shared unavailable (reason shown)</li><li>narrow (strip scrolls)</li><li>transport: volume beneath when it cannot flank the keys (DEC-016)</li><li>transport: keys wrapped, previous/play/next alone on the first line (DEC-016)</li></ul> | UX-PRIN-002<br>UX-KEY-001<br>UX-KEY-002<br>UX-TOUCH-001 | <a href="#">○ mockup only</a> |
| <b>Music Library</b><br>(React specimen, not served — DEC-019) section strip → Music Library<br><code>music-player/src/views/Library.tsx</code><br><code>packages/aqua-ui/src/components/MusicList.tsx</code>       | <ul style="list-style-type: none"><li>empty (how to add music)</li><li>scanning</li><li>loaded</li><li>sorted</li><li>selected row</li><li>playing row</li><li>unplayable format</li></ul>                                                                                                                                                                                                                | UX-KEY-001<br>UX-STATE-001<br>UX-A11Y-001<br>VOICE-001  | <a href="#">○ mockup only</a> |
| <b>Now playing</b><br>(React specimen, not served — DEC-019) section strip → Now playing<br><code>music-player/src/views/NowPlaying.tsx</code><br><code>packages/domain/src/visualisers/spectrum.ts</code>          | <ul style="list-style-type: none"><li>nothing playing</li><li>playing (spectrum moving)</li><li>paused (spectrum stopped)</li><li>reduced motion (spectrum at 4 per second)</li><li>equalizer not applied (reason shown)</li><li>shared (group queue)</li></ul>                                                                                                                                           | UX-STATE-001<br>UX-MOTION-001                           | <a href="#">○ mockup only</a> |
| <b>Queue</b><br>(React specimen, not served — DEC-019) section strip → Queue<br><code>music-player/src/views/Queue.tsx</code>                                                                                       | <ul style="list-style-type: none"><li>nothing queued</li><li>queued</li><li>shuffled</li><li>shared (hub revision)</li></ul>                                                                                                                                                                                                                                                                              | UX-STATE-001<br>UX-SHARED-001<br>VOICE-001              | <a href="#">○ source only</a> |
| <b>Playlists</b><br>(React specimen, not served — DEC-019) section strip → Playlists<br><code>music-player/src/views/Playlists.tsx</code>                                                                           | <ul style="list-style-type: none"><li>no playlists</li><li>list</li><li>playlist open</li><li>empty playlist</li></ul>                                                                                                                                                                                                                                                                                    | UX-STATE-001<br>VOICE-001                               | <a href="#">○ source only</a> |
| <b>Search</b><br>(React specimen, not served — DEC-019) status-bar search → Enter<br><code>music-player/src/views/Search.tsx</code><br><code>music-player/src/components/SearchPopover.tsx</code>                   | <ul style="list-style-type: none"><li>prompt (device only / with hub)</li><li>searching hub</li><li>no local matches</li><li>results</li><li>hub unreachable</li></ul>                                                                                                                                                                                                                                    | UX-STATE-001<br>UX-PRIN-002                             | <a href="#">○ source only</a> |
| <b>Listening history</b><br>(React specimen, not served — DEC-019) section strip → Listening history<br><code>music-player/src/views/Metrics.tsx</code>                                                             | <ul style="list-style-type: none"><li>nothing yet</li><li>history</li></ul>                                                                                                                                                                                                                                                                                                                               | UX-STATE-001<br>UX-SAFE-004                             | <a href="#">○ source only</a> |

| SURFACE                                                                                                                                                                                                                               | STATES IT MUST ANSWER                                                                                                                                                                                                                           | RULES                                                                                   | SEEN                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------|
| <b>Constellation</b><br>(React specimen, not served — DEC-019)<br>section strip → Constellation<br>music-player/src/views/Constellation.tsx<br>packages/domain/src/visualisers/constellatio<br>n-layout.ts                            | <ul style="list-style-type: none"> <li>nothing to map</li> <li>field</li> <li>table (equivalent view)</li> <li>reduced motion (table)</li> <li>WebGL unavailable</li> </ul>                                                                     | UX-MOTION-001<br>UX-STATE-001                                                           | <a href="#">O mockup only</a> |
| <b>Settings and equalizer</b><br>(React specimen, not served — DEC-019)<br>profile button in the status bar<br>music-player/src/views/Settings.tsx<br>music-player/src/views/Equalizer.tsx                                            | <ul style="list-style-type: none"> <li>no folders</li> <li>folders</li> <li>hub paired</li> <li>hub not paired</li> <li>equalizer on/off</li> <li>solfeggio preset (dimmed bands)</li> <li>crossfade</li> <li>car/lock-screen report</li> </ul> | UX-PRIN-001<br>UX-PRIN-003<br>UX-PRIN-004<br>UX-LINK-001<br>UX-TOUCH-002<br>UX-SAFE-004 | <a href="#">O mockup only</a> |
| <b>Add to playlist</b><br>(React specimen, not served — DEC-019) +<br>beside the transport, or row menu<br>music-<br>player/src/components/AddToPlaylistSheet.tsx                                                                     | <ul style="list-style-type: none"> <li>open</li> <li>no playlists</li> </ul>                                                                                                                                                                    | UX-KEY-001                                                                              | <a href="#">O source only</a> |
| <b>New playlist</b><br>(React specimen, not served — DEC-019)<br>Playlists → New, or row menu<br>music-<br>player/src/components/NewPlaylistSheet.tsx<br>packages/aqua-ui/src/components/music-list-<br>behaviours.tsx                | <ul style="list-style-type: none"> <li>empty name (validation)</li> <li>valid</li> </ul>                                                                                                                                                        | VOICE-002                                                                               | <a href="#">O source only</a> |
| <b>Download from the hub</b><br>(React specimen, not served — DEC-019) row<br>menu → Download<br>music-<br>player/src/components/DownloadSheet.tsx                                                                                    | <ul style="list-style-type: none"> <li>choose format</li> <li>basis required</li> <li>in progress</li> <li>failed with reason</li> </ul>                                                                                                        | UX-FEED-001<br>UX-PRIN-002                                                              | <a href="#">O source only</a> |
| <b>Fetch with a local tool</b><br>(React specimen, not served — DEC-019)<br>Library → Fetch (when a helper or the Android<br>tools are present)<br>music-player/src/components/FetchSheet.tsx<br>music-player/src/lib/fetch-helper.ts | <ul style="list-style-type: none"> <li>no tool (control hidden)</li> <li>url entry</li> <li>running</li> <li>finished (files)</li> <li>failed (tool's own reason)</li> </ul>                                                                    | UX-PRIN-001<br>UX-FEED-001<br>VOICE-002                                                 | <a href="#">O source only</a> |
| <b>Share</b><br>(React specimen, not served — DEC-019) share<br>button beside the transport, or a playlist<br>music-player/src/components/ShareSheet.tsx                                                                              | <ul style="list-style-type: none"> <li>no hub (explains why)</li> <li>link created</li> <li>copied</li> </ul>                                                                                                                                   | UX-PRIN-002                                                                             | <a href="#">O source only</a> |
| <b>Search popover</b><br>(React specimen, not served — DEC-019)<br>typing in the status-bar search<br>music-<br>player/src/components/SearchPopover.tsx                                                                               | <ul style="list-style-type: none"> <li>counting</li> <li>results</li> <li>none</li> <li>dismissed (arrow brings back)</li> </ul>                                                                                                                | UX-FEED-001<br>UX-KEY-001                                                               | <a href="#">O source only</a> |
| <b>Notice bar and toasts</b><br>(React specimen, not served — DEC-019) any<br>warning the store raises<br>music-player/src/components/NoticeBar.tsx<br>packages/aqua-ui/src/components/Toast.tsx                                      | <ul style="list-style-type: none"> <li>info</li> <li>warning</li> <li>dismissed</li> </ul>                                                                                                                                                      | UX-FEED-001<br>VOICE-002                                                                | <a href="#">O source only</a> |
| <b>Row context menu</b><br>(React specimen, not served — DEC-019) right-<br>click or the row's menu key<br>packages/aqua-ui/src/components/music-list-<br>behaviours.tsx                                                              | <ul style="list-style-type: none"> <li>open</li> <li>disabled items with reason</li> </ul>                                                                                                                                                      | UX-KEY-001<br>UX-PRIN-004                                                               | <a href="#">O source only</a> |
|                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                 |                                                                                         |                               |

| SURFACE                                                                                                                                    | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                  | RULES                                       | SEEN           |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------|
| <b>Library</b><br>the list under the player; Music view<br>music-player/index.html<br>music-player/scripts/make-shell.py                   | <ul style="list-style-type: none"> <li>empty (nothing indexed)</li> <li>rows from this device</li> <li>link rows from search</li> <li>sorted</li> <li>narrowed by scope chip</li> <li>filtered by the bar</li> </ul>                                                                                                                                                                                                   | NP-LIST-001<br>UX-KEY-001<br>UX-STATE-001   | ● seen running |
| <b>Now Playing</b><br>the jewel case and transport at the top of the page<br>music-player/index.html<br>music-player/scripts/make-shell.py | <ul style="list-style-type: none"> <li>nothing playing</li> <li>a track from this device playing (the bar follows the element)</li> <li>paused</li> <li>a link row chosen (does not pretend to play)</li> <li>a station (LIVE)</li> <li>a channel (video bar, LIVE)</li> </ul>                                                                                                                                         | NP-TRANS-001<br>UX-KEY-002                  | ● seen running |
| <b>Search popover</b><br>the header search field<br>music-player/index.html<br>music-player/scripts/make-shell.py                          | <ul style="list-style-type: none"> <li>empty</li> <li>results from companion / iTunes</li> <li>pasted link resolved</li> <li>people on the hub</li> <li>pasted link filled in by the companion (duration, date, album, art)</li> <li>pasted playlist, set or album: its songs, "Title: N songs", and "(the first 200 of M)" when capped</li> <li>set entry named by its address until its page is on screen</li> </ul> | NP-PRIN-002<br>NP-FIND-002                  | ○ source only  |
| <b>Row menu</b><br>right-click or long-press a row<br>music-player/index.html<br>music-player/scripts/make-shell.py                        | <ul style="list-style-type: none"> <li>one row</li> <li>several rows (marquee)</li> <li>a station with a known song: the On air section (Up Next, Group Queue, Playlist)</li> <li>touch: a long press on a station opens the same menu</li> </ul>                                                                                                                                                                      | NP-MENU-001<br>NP-RADIO-002                 | ○ source only  |
| <b>New playlist sheet</b><br>row menu ▶ Add to Playlist ▶ New Playlist...<br>music-player/index.html<br>music-player/scripts/make-shell.py | <ul style="list-style-type: none"> <li>empty name refused</li> <li>created</li> </ul>                                                                                                                                                                                                                                                                                                                                  | NP-MENU-002                                 | ○ source only  |
| <b>Fetch sheet</b><br>transport ▶ Download, on a link row<br>music-player/index.html<br>music-player/scripts/make-shell.py                 | <ul style="list-style-type: none"> <li>no basis chosen (refused, said why)</li> <li>helper found / not found</li> <li>fetching</li> <li>refused by the helper (its reason)</li> <li>fetchd and playing</li> <li>the date beside the title, when the link row has one</li> </ul>                                                                                                                                        | UX-SAFE-001<br>UX-FEED-001<br>NP-FIND-002   | ○ source only  |
| <b>Mini player</b><br>the header while Settings is open<br>music-player/index.html<br>music-player/scripts/make-shell.py                   | <ul style="list-style-type: none"> <li>hidden</li> <li>shown for music and radio</li> </ul>                                                                                                                                                                                                                                                                                                                            | NP-CHROME-001                               | ○ source only  |
| <b>Radio</b><br>toolbar ▶ Radio<br>music-player/index.html<br>music-player/scripts/make-shell.py                                           | <ul style="list-style-type: none"> <li>directory unreachable (offline shelf)</li> <li>a station playing</li> <li>station list</li> <li>the song on the air, read from the stream (paired or with the companion)</li> <li>no feed and nothing to read the stream: the programme format, no invented title</li> </ul>                                                                                                    | NP-LIVE-001<br>NP-RADIO-001<br>NP-RADIO-002 | ● seen running |
|                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                             |                |

| SURFACE                                                                                                                                                                                               | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RULES                                      | SEEN           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------|
| <b>Live TV</b><br>toolbar ▶ Live TV<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                                  | <ul style="list-style-type: none"> <li>no playlist loaded (empty guide)</li> <li>channels from a loaded M3U</li> <li>a channel playing (LIVE)</li> <li>channels and guide from the companion (now and next where a guide has them; Live where it has none)</li> <li>no companion, or one with no playlists: the guide is as it was</li> <li>no companion on this machine: a paired hub's copy of the companion's Live TV fills the guide</li> <li>Until: when the programme on now ends; blank where a channel has no guide</li> </ul> | NP-LIVE-002<br>NP-TV-001                   | ● seen running |
| <b>TV</b><br>toolbar ▶ TV<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                                            | <ul style="list-style-type: none"> <li>no catalogue (nothing invented)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NP-PRIN-001                                | ○ source only  |
| <b>Movies</b><br>toolbar ▶ Movies<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                                    | <ul style="list-style-type: none"> <li>no catalogue (nothing invented)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NP-PRIN-001                                | ○ source only  |
| <b>Settings ▶ Statistics</b><br>profile button ▶ Statistics, or #settings/stats<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                      | <ul style="list-style-type: none"> <li>nothing recorded yet</li> <li>this player's history</li> <li>imported file</li> <li>3D views unavailable (tables still carry the numbers)</li> </ul>                                                                                                                                                                                                                                                                                                                                            | NP-DATA-001                                | ● seen running |
| <b>Settings ▶ Recommendations</b><br>#settings/rec<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                   | <ul style="list-style-type: none"> <li>defaults</li> <li>edited</li> <li>saved algorithm</li> <li>reverted</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  | NP-PREF-001                                | ○ source only  |
| <b>Settings ▶ Sources</b><br>#settings/src<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                           | <ul style="list-style-type: none"> <li>music on this device (folders/files)</li> <li>companion found / missing downloaders</li> <li>companion downloader setting up (with %), not counted as missing</li> <li>hub tested / paired</li> <li>backup measured (nothing sent)</li> <li>backup will not fit (reason)</li> </ul>                                                                                                                                                                                                             | NP-PREF-002<br>UX-FEED-001<br>UX-SETUP-001 | ○ source only  |
| <b>Stream from a PC</b><br>#settings/src ▶ Connections ▶ Stream from a PC; the indicator beside the transport while a PC track plays<br>music-player/index.html<br>music-player/scripts/make-shell.py | <ul style="list-style-type: none"> <li>not paired</li> <li>not a ticket / not six digits (refused before sending, said why)</li> <li>pairing</li> <li>wrong code / code expired or used (the PC's answer, in words)</li> <li>paired and connected: PC name, relay-carried, track count</li> <li>reconnecting (the PC or its relay is away)</li> <li>refused (revoked on the PC: pair again)</li> <li>a PC track playing: indicator reads relay-carried</li> <li>single-file copy: unavailable, and says so</li> </ul>                  | NP-PREF-007<br>NP-PRIN-002<br>UX-FEED-001  | ○ source only  |
| <b>Settings ▶ Player</b><br>#settings/player<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                         | <ul style="list-style-type: none"> <li>appearance</li> <li>listening</li> <li>defaults</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      | NP-PREF-003                                | ○ source only  |
| <b>Settings ▶ Equalizer</b><br>#settings/eq<br>music-player/index.html<br>music-player/scripts/make-shell.py                                                                                          | <ul style="list-style-type: none"> <li>preset</li> <li>custom</li> <li>default volume</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                       | NP-PREF-004                                | ○ source only  |

| SURFACE                                                                                                                             | STATES IT MUST ANSWER                                                                                                                                                                              | RULES                      | SEEN          |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|
| <b>Settings ▶ Profile</b><br>#settings/profile (shown when paired)<br>music-player/index.html<br>music-player/scripts/make-shell.py | <ul style="list-style-type: none"> <li>• name available / taken</li> <li>• picture</li> <li>• shared playlists</li> <li>• groups</li> <li>• invites received / sent</li> <li>• declined</li> </ul> | NP-PREF-005<br>UX-SAFE-001 | ○ source only |
| <b>Profile viewer</b><br>search ▶ a person, or a group member<br>music-player/index.html<br>music-player/scripts/make-shell.py      | <ul style="list-style-type: none"> <li>• their name, picture and shared playlists</li> <li>• a playlist opened</li> </ul>                                                                          | NP-PREF-005                | ○ source only |
| <b>Invite page</b><br>a #invite/<code> link<br>music-player/index.html<br>music-player/scripts/make-shell.py                        | <ul style="list-style-type: none"> <li>• previewed (group, sender, role, expiry)</li> <li>• unknown or used code</li> <li>• joined</li> <li>• declined</li> </ul>                                  | NP-PREF-005                | ○ source only |

## AIRWAVE HUB — ADMIN WINDOW AND PUBLIC PAGES

The Airwave window of design/frontends/airwave-hub.html, served by the container on port 4546. Sign-in and the first-run gate are drawn in the same window.

| SURFACE                                                                                                                                                                                                                               | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RULES                                                    | SEEN           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------|
| <b>Sign in</b><br>any admin URL without a session — drawn in the same window, every tab locked<br>docker-container/src/web/App.tsx<br>DEC-017                                                                                         | <ul style="list-style-type: none"> <li>• connecting to the hub</li> <li>• hub not answering (Try Again)</li> <li>• first run (admin / admin hint)</li> <li>• set up (no hint)</li> <li>• username or password missing</li> <li>• wrong password (the hub's sentence)</li> <li>• rate limited</li> </ul>                                                                                                                                                                                                                                          | UX-SAFE-002<br>VOICE-002<br>UX-CHROME-001                | ● seen running |
| <b>First-run gate: choose a real password</b><br>first sign-in — the amber gate at the top of Overview, the other five tabs locked<br>docker-container/src/web/App.tsx<br>DEC-017                                                     | <ul style="list-style-type: none"> <li>• new password asked twice (the current one is remembered from sign-in)</li> <li>• current password asked for too, after a reload</li> <li>• under twelve characters refused</li> <li>• the two entries differ</li> <li>• refused by the hub (its sentence)</li> <li>• saved: every tab unlocks</li> </ul>                                                                                                                                                                                                | UX-SAFE-002<br>UX-CHROME-001                             | ○ not run here |
| <b>Section error boundary</b><br>any section that throws while rendering<br>docker-container/src/web/ViewBoundary.tsx                                                                                                                 | <ul style="list-style-type: none"> <li>• one sentence where the section would have been, with Try Again; the window, tools and status strip stay up</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                   | UX-STATE-001                                             | ○ not run here |
| <b>The window: title, six tools, pane and status strip</b><br>every admin URL, in every state<br>docker-container/src/web/App.tsx<br>docker-container/src/web/icons.tsx<br>DEC-017                                                    | <ul style="list-style-type: none"> <li>• signed out (every tool locked)</li> <li>• first run (only Overview open; the rest locked with the reason)</li> <li>• signed in (six tools; arrows, Home and End move between them)</li> <li>• a section opened by its #id in the address</li> <li>• attention badge on Overview</li> <li>• status strip: hub name, bind address and reach; what the window just did replaces it for a few seconds</li> <li>• below 640px: tools scroll sideways, tiles two-up, tables drop secondary columns</li> </ul> | UX-CHROME-001<br>UX-KEY-001<br>UX-KEY-002<br>UX-FEED-001 | ● seen running |
| <b>Confirmation sheet</b><br>any destructive action: revoke, remove, restore, import, withdraw, archive, cancel a download, remove the bot token<br>docker-container/src/web/ui.tsx<br>docker-container/src/web/styles.css<br>DEC-021 | <ul style="list-style-type: none"> <li>• open: the question, what will happen, Cancel and the verb</li> <li>• Cancel has the focus</li> <li>• Tab moves between the two buttons only</li> <li>• Escape cancels</li> <li>• focus returns to the button that asked</li> </ul>                                                                                                                                                                                                                                                                      | UX-SAFE-001<br>UX-FOCUS-001                              | ○ not run here |

| SURFACE                                                                                                                     | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RULES                                       | SEEN           |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------|
| <b>Overview</b><br>Overview tab<br>docker-container/src/web/views/Overview.tsx<br>docker-container/src/web/views/common.tsx | <ul style="list-style-type: none"> <li>loading (tiles read — and Loading...)</li> <li>loaded (tiles: hub, devices, groups, library, providers, storage)</li> <li>first run (Devices, Groups and Library read Off, with the reason)</li> <li>error (tiles read Not available; Try Again in Needs attention)</li> <li>alerts (count on the Overview tool)</li> <li>nothing needs you</li> <li>signed-in line and Sign Out at the foot</li> <li>the badge on the Overview tool and the Needs attention list come from one reading, so they always agree</li> </ul>                                                                                                             | UX-STATE-001<br>UX-KEY-001<br>UX-CHROME-001 | ○ not run here |
| <b>Devices and pairing</b><br>Devices tab → Devices<br>docker-container/src/web/views/Devices.tsx                           | <ul style="list-style-type: none"> <li>no devices</li> <li>permissions ticked before a code is made</li> <li>pairing code shown on the LCD plate (with QR)</li> <li>code expired</li> <li>claim waiting for the verification code</li> <li>pending pairing refused</li> <li>revoke (confirm sheet)</li> </ul>                                                                                                                                                                                                                                                                                                                                                               | UX-SAFE-001<br>UX-SAFE-003<br>UX-STATE-001  | ○ not run here |
| <b>Groups</b><br>Groups tab<br>docker-container/src/web/views/Groups.tsx<br>DEC-017                                         | <ul style="list-style-type: none"> <li>no groups</li> <li>group with queue and members (drift per device)</li> <li>archived</li> <li>new group</li> <li>invite made (code shown once)</li> <li>invite listed by who it is for, its role and state (never its code)</li> <li>invite withdrawn (confirm sheet)</li> <li>archive (confirm sheet)</li> <li>a new invite opens under its own row with its code, folded away with its button or Escape</li> </ul>                                                                                                                                                                                                                 | UX-SHARED-001<br>UX-STATE-001               | ● seen running |
| <b>Providers</b><br>Music tab → Providers<br>docker-container/src/web/views/Providers.tsx                                   | <ul style="list-style-type: none"> <li>working</li> <li>limited</li> <li>needs setting up</li> <li>off</li> <li>down</li> <li>details open (capabilities, limits, credit)</li> <li>set-up form (secrets write-only; blank keeps what is stored)</li> <li>saved and tested (result beside the button)</li> <li>external tool on without setup (DEC-024)</li> <li>external tool row names the tools on this hub: (yt-dlp, spotDL) (DEC-036)</li> <li>requests and limits</li> <li>details open under their own row; Escape closes them and puts the caret back on the button</li> <li>Requests and limits lists the providers in use, the rest behind a disclosure</li> </ul> | UX-PRIN-002<br>UX-A11Y-001                  | ○ not run here |
| <b>Library</b><br>Music tab → Library<br>docker-container/src/web/views/Library.tsx                                         | <ul style="list-style-type: none"> <li>no folders</li> <li>folder scanning</li> <li>folder scan failed</li> <li>tracks</li> <li>no matches</li> <li>remove folder (confirm sheet)</li> <li>Scan on a folder's row scans that folder alone</li> <li>Scan Now scans every folder</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                   | UX-STATE-001<br>UX-SAFE-001                 | ● seen running |
|                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                             |                |

| SURFACE                                                                                                                                                                                                                   | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | RULES                                                                    | SEEN           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------|
| <b>Downloads</b><br>Music tab → Downloads<br>docker-container/src/web/views/Downloads.tsx                                                                                                                                 | <ul style="list-style-type: none"> <li>• none (who asks for one)</li> <li>• waiting</li> <li>• running (stage as words or a bar)</li> <li>• paused</li> <li>• failed with reason</li> <li>• cancelled</li> <li>• done</li> <li>• cancel (confirm sheet)</li> <li>• playlist entry: one row each, From names the list and the position (DEC-036)</li> <li>• formats this hub cannot write struck through, with why</li> <li>• storage line</li> </ul>                                                                                 | UX-FEED-001<br>UX-PRIN-002<br>UX-SAFE-001                                | ○ not run here |
| <b>Live TV from the companion</b><br>Music tab → Live TV from the companion (drawn after Downloads, inside the downloads section)<br>docker-container/src/web/views/LiveTv.tsx<br>docker-container/src/live-tv/service.ts | <ul style="list-style-type: none"> <li>• loading</li> <li>• nothing yet: a sharing companion sends its channels when it syncs</li> <li>• kept: which companion sent it, how many channels, the guide, when it was updated</li> <li>• remove (confirm sheet); the companion sends them again when it next syncs</li> <li>• failed to load (Try Again)</li> <li>• read-only: channels are added and removed in the companion</li> </ul>                                                                                                | UX-TV-001<br>UX-STATE-001<br>UX-SAFE-001                                 | ○ not run here |
| <b>Profiles</b><br>Devices tab → Profiles<br>docker-container/src/web/views/Profiles.tsx                                                                                                                                  | <ul style="list-style-type: none"> <li>• none</li> <li>• listed</li> <li>• renaming</li> <li>• name taken</li> <li>• remove picture (confirm sheet)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       | UX-SAFE-001<br>UX-FEED-001                                               | ● seen running |
| <b>Shared links</b><br>Sharing tab → Shared links<br>docker-container/src/web/views/Shares.tsx<br>DEC-028                                                                                                                 | <ul style="list-style-type: none"> <li>• none</li> <li>• active (expiry in words)</li> <li>• revoked</li> <li>• revoke (confirm sheet)</li> <li>• Create Link: a hub playlist or album from the picker and how long the link works; the hub builds the item list (DEC-028)</li> <li>• link made: the full link shown once, under the form, with Copy</li> <li>• nothing to share yet: the picker says so</li> </ul>                                                                                                                  | UX-SAFE-001                                                              | ● seen running |
| <b>Recommendations</b><br>Music tab → Recommendations<br>docker-container/src/web/views/Recommendations.tsx                                                                                                               | <ul style="list-style-type: none"> <li>• defaults</li> <li>• edited</li> <li>• nothing sent until Save; Revert puts back what the hub has</li> <li>• a number outside its range refused before sending</li> <li>• refused by the hub: its sentence under the form</li> </ul>                                                                                                                                                                                                                                                         | UX-PRIN-003<br>UX-FORM-001                                               | ● seen running |
| <b>Discord</b><br>Sharing tab → Discord<br>docker-container/src/web/views/Discord.tsx<br>docker-container/src/discord/service-interface.ts                                                                                | <ul style="list-style-type: none"> <li>• no token</li> <li>• has a token, switched off</li> <li>• worker not running</li> <li>• connected</li> <li>• voice playing</li> <li>• last error</li> <li>• template preview</li> <li>• remove token (confirm sheet)</li> <li>• warnings and errors as plain sentences</li> <li>• server IDs checked before anything is sent</li> <li>• Save and Start sends the lot; Revert puts back what the hub has</li> <li>• Invite the Bot disabled with its reason until a token is saved</li> </ul> | UX-PRIN-002<br>UX-DISCORD-003<br>VOICE-002<br>UX-SAFE-001<br>UX-FORM-001 | ● seen running |

| SURFACE                                                                                          | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RULES                                      | SEEN           |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------|
| <b>Network</b><br>System tab → Network<br>docker-container/src/web/views/Network.tsx             | <ul style="list-style-type: none"> <li>• localhost</li> <li>• lan</li> <li>• remote (restart required)</li> <li>• edited: nothing sent until Save; the public address checked first; Revert puts back what the hub has</li> <li>• saved: Saved. in the status strip</li> <li>• what works where: this machine, your network, the internet</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | UX-SAFE-002<br>UX-FORM-001                 | ● seen running |
| <b>Backup</b><br>System tab → Backup<br>docker-container/src/web/views/Backup.tsx<br>DEC-029     | <ul style="list-style-type: none"> <li>• no backups</li> <li>• list</li> <li>• Back Up Now (said in the status strip)</li> <li>• restore (confirm sheet)</li> <li>• restore failed with reason</li> <li>• import: checked first, then confirmed in a sheet</li> <li>• space measured</li> <li>• space unknown</li> <li>• next backup will not fit</li> <li>• settings: the folder inside the data volume, the parts included, how often (every day or week, at a time) and how many to keep — sent together with Save; Revert puts them back</li> <li>• a folder outside the data volume refused before sending</li> <li>• the next scheduled backup said in words, on the hub's clock; a slot missed while the hub was off is taken when it starts (DEC-029)</li> <li>• Download on each archive (an attachment, admin only)</li> </ul> | UX-SAFE-001<br>UX-STATE-001<br>UX-FORM-001 | ● seen running |
| <b>Diagnostics</b><br>System tab → Diagnostics<br>docker-container/src/web/views/Diagnostics.tsx | <ul style="list-style-type: none"> <li>• loaded</li> <li>• log level chosen (the list follows it)</li> <li>• support bundle saved as a file, with what it leaves out said beside the button</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | UX-STATE-001                               | ● seen running |
| <b>Public share page</b><br>/s/<token><br>docker-container/src/api/routes/shares.ts<br>DEC-008   | <ul style="list-style-type: none"> <li>• track or playlist</li> <li>• stream allowed</li> <li>• link unavailable</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | UX-SAFE-004                                | ○ not run here |

## AIRWAVE COMPANION — WINDOWS

The Airwave window of design/frontends/airwave-companion.html in an Electron window whose chrome is its title bar.

| SURFACE                                                                                                                                                         | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                             | RULES                                    | SEEN           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------|
| <b>Folders (Saved Music, Saved TV, Saved Movies)</b><br>Library tab → Saved Music · Saved TV · Saved Movies<br>windows-companion/src/renderer/views/Folders.tsx | <ul style="list-style-type: none"> <li>• no folders (one line per kind)</li> <li>• loading (the list holds its place)</li> <li>• watched / kept / N songs</li> <li>• scanning (n of m)</li> <li>• drive not connected (kept in the list, said under it)</li> <li>• couldn't scan (reason)</li> <li>• added (said beside the button)</li> <li>• remove (confirm sheet; files untouched)</li> </ul> | UX-SAFE-001<br>VOICE-001<br>UX-STATE-001 | ○ not run here |
|                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                   |                                          |                |

| SURFACE                                                                                                                            | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RULES                                                                                 | SEEN           |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------|
| <b>Music on This PC</b><br>Library tab → Music on This PC<br>windows-<br>companion/src/renderer/views/Library.tsx                  | <ul style="list-style-type: none"> <li>nothing indexed: no music folder yet</li> <li>nothing indexed: the scan is still reading</li> <li>no matches</li> <li>tracks</li> <li>selection: click, Ctrl/Shift-click, arrows, Space, Ctrl+A; Enter or double-click shows the file</li> <li>Scan Now disabled with reason (no music folder)</li> <li>Send to Hub disabled with reason (no hub paired / nothing chosen)</li> <li>Send to Hub: queued, progress under Remote ▶ Transfers</li> <li>a tempo read from a tag is plain; a measured one wears ≈</li> <li>tempo waiting: Setting up FFmpeg</li> <li>tempo unavailable: FFmpeg could not be set up (Try again in Settings)</li> <li>tempo unavailable: no FFmpeg (install it and scan again)</li> <li>a windowed list: only the rows in view are drawn and every song is counted; End, Shift+Home and Ctrl+A reach every song, fetching only what they need</li> </ul> | UX-STATE-001<br>VOICE-001<br>NP-PRIN-002<br>UX-SETUP-001<br>UX-KEY-001<br>UX-PRIN-002 | ○ not run here |
| <b>Hub connection and what is shared</b><br>Remote tab → Hub connection<br>windows-companion/src/renderer/views/Hub.tsx<br>DEC-023 | <ul style="list-style-type: none"> <li>not paired (address and pairing code)</li> <li>pairing (verification code on the LCD plate; waiting for the hub to confirm)</li> <li>pairing refused (the hub's sentence)</li> <li>connected</li> <li>paired but not reachable</li> <li>sharing on from pairing when the hub grants it (DEC-023)</li> <li>sharing as it was left, not "off" each time the window opens</li> <li>sharing off: Sync Now disabled with reason</li> <li>plain-http warning (notice bar)</li> <li>forget (confirm sheet; turns sharing off for the next hub)</li> </ul>                                                                                                                                                                                                                                                                                                                               | UX-SAFE-001<br>UX-SAFE-003<br>UX-PRIN-002                                             | ○ not run here |
| <b>Transfers</b><br>Remote tab → Transfers<br>windows-<br>companion/src/renderer/views/Transfers.tsx                               | <ul style="list-style-type: none"> <li>nothing moving (paired / not paired)</li> <li>sending (bar and bytes)</li> <li>waiting</li> <li>sent</li> <li>cancelled</li> <li>didn't send (reason under the list)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | UX-FEED-001<br>VOICE-001                                                              | ○ not run here |
| <b>Backup</b><br>Settings tab → Backup<br>windows-<br>companion/src/renderer/views/Backup.tsx                                      | <ul style="list-style-type: none"> <li>no folder chosen (Back Up Now disabled, reason shown)</li> <li>measuring</li> <li>fits / does not fit (space bar: used · this backup · free)</li> <li>running with progress</li> <li>archives listed</li> <li>restore (confirm sheet)</li> <li>delete (confirm sheet)</li> <li>what goes in: music, TV, movies, playlists, EQ presets, recommendation algorithms (from the paired hub, named; the reason when there is none) and these settings with the Live TV links</li> <li>how often: by hand, every day or every week while the companion is open; a run that is due is taken when the companion next opens</li> <li>how many to keep</li> <li>a scheduled run that could not happen is reported, not dropped</li> </ul>                                                                                                                                                   | UX-SAFE-001<br>VOICE-002                                                              | ● seen running |

| SURFACE                                                                                                                                               | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RULES                                                                                  | SEEN           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------|
| <b>About</b><br>Settings tab → About<br>windows-<br>companion/src/renderer/views/About.tsx                                                            | <ul style="list-style-type: none"> <li>• where its data is kept, with Open</li> <li>• cache by part, Clear Cache (confirm sheet), then what was cleared</li> <li>• logs: Open, Export (secrets, tokens and folder paths taken out), detailed logs switch</li> <li>• build, signing, and how updates are checked</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | VOICE-003<br>UX-PRIV-001                                                               | ● seen running |
| <b>Live TV</b><br>Live TV tab → Channel playlists (M3U) ·<br>Programme guides (EPG)<br>windows-<br>companion/src/renderer/views/LiveTv.tsx<br>DEC-018 | <ul style="list-style-type: none"> <li>• no links (two empty lists, in the design's words)</li> <li>• checking...</li> <li>• kept: ✓ N channels / ✓ N-day guide</li> <li>• refused (the reason under the field; what was typed is kept; nothing added)</li> <li>• stopped answering (keeps what it last held; Check Again)</li> <li>• remove (the round minus)</li> <li>• Add disabled with reason until a link is pasted</li> <li>• while sharing is on, its channels and guide are kept on the hub too (links with a sign-in left out); turning sharing off removes them</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | UX-TV-001<br>UX-STATE-001<br>VOICE-001<br>UX-PRIN-001                                  | ○ not run here |
| <b>Settings</b><br>Settings tab → Settings<br>windows-<br>companion/src/renderer/views/Settings.tsx<br>DEC-018                                        | <ul style="list-style-type: none"> <li>• downloader setting up (progress bar and %)</li> <li>• downloader waiting its turn (Setting up...)</li> <li>• downloader ready (set up automatically, or found on this PC)</li> <li>• downloader couldn't be set up (amber, reason, Try Again; badge on the Settings tool)</li> <li>• downloader cannot be set up on this PC (amber, with what to install)</li> <li>• downloaders not looked at yet: found / missing (helper not running)</li> <li>• Check All (last checked ...)</li> <li>• a preference written back as it changes, said at the foot of the pane</li> <li>• helper running / not running with reason</li> <li>• port invalid</li> <li>• token shown / hidden / copied</li> <li>• restore defaults (confirm)</li> <li>• a tool: Check asks about that tool; Update when a newer version is out (badge on the tool); Install when it is missing; Update waits, with the reason, while the helper is busy</li> <li>• Update downloaders automatically: on by default</li> <li>• Downloads: the folder from the system's picker, format, how many at once and a speed limit</li> <li>• new versions of the companion: checked by default, asked at once when switched on, said when one is out; Download opens the release page</li> <li>• notifications when a download finishes: one preference with "When one finishes"</li> <li>• helper on your network: off by default; what it exposes said before it is switched on (DEC-032)</li> </ul> | UX-PRIN-001<br>UX-FEED-001<br>UX-SETUP-001<br>UX-SAFE-001<br>UX-NET-001<br>UX-PRIV-001 | ● seen running |
| <b>Notice bar</b><br>any main-process notice<br>windows-<br>companion/src/renderer/views/NoticeBar.tsx                                                | <ul style="list-style-type: none"> <li>• info</li> <li>• warning</li> <li>• error (announced at once)</li> <li>• dismiss</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | VOICE-002                                                                              | ○ not run here |
|                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |                |

| SURFACE                                                                                                                                                                                                                                        | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RULES                                                      | SEEN           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------|
| <b>The window: title bar chrome, four tools, pane and status line</b><br>the app window<br>windows-companion/src/renderer/App.tsx<br>windows-companion/src/renderer/icons.tsx<br>DEC-020                                                       | <ul style="list-style-type: none"> <li>four tools (arrows, Home and End move between them, choosing as they go)</li> <li>a pane is built the first time its tool is chosen, then kept</li> <li>attention badge on Settings (downloaders that need the person)</li> <li>status line: connected and last synced / can't reach the hub / no hub paired; folders, playlists, guides and devices counted</li> <li>opened in a browser, without the bridge: says so and offers nothing</li> <li>Windows draws minimise, maximise and close over the chrome's top-right corner</li> </ul>                                                                                                                                                                               | UX-CHROME-001<br>UX-KEY-001<br>VOICE-003                   | ● seen running |
| <b>Confirmation sheet</b><br>removing a folder, forgetting the hub, revoking a device, restoring or deleting a backup, restoring the defaults<br>windows-companion/src/renderer/ui.tsx<br>windows-companion/src/renderer/styles.css<br>DEC-021 | <ul style="list-style-type: none"> <li>destructive: the action, then Cancel as the default button</li> <li>not destructive: Cancel, then the action as the default button</li> <li>Escape cancels</li> <li>a second question answers the first with no</li> <li>focus returns to where it was</li> <li>the sheet dims only the pane under the toolbar, and stays modal</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                | UX-SAFE-001<br>UX-FOCUS-001                                | ○ not run here |
| <b>Stream to your devices (Pair a device)</b><br>Remote tab → Stream to your devices<br>windows-companion/src/renderer/views/Streaming.tsx<br>windows-companion/src/main/awsp.ts<br>DEC-018                                                    | <ul style="list-style-type: none"> <li>off (default; the pairing controls say so)</li> <li>starting</li> <li>running: pairing code on the LCD plate with its time left, ticket and QR</li> <li>code expired</li> <li>sidecar missing or crashed (reason, restarting)</li> <li>devices paired, each direct / through a relay / not connected, with its quality cap</li> <li>revoke (confirm sheet)</li> <li>port invalid</li> <li>Wi-Fi and Ethernet / metered connections: two switches, both on by default; paused with the reason on a connection that is switched off (DEC-031)</li> <li>the connection switches are off while streaming itself is off</li> <li>each paired device says when it was last seen, and how it is connected while it is</li> </ul> | UX-SAFE-001<br>UX-FEED-001<br>UX-PRIN-002<br>UX-STREAM-001 | ○ not run here |

## DISCORD BOT

Messages rendered from templates the operator can edit.

| SURFACE                                                                                                                                                     | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                     | RULES                                              | SEEN           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------|
| <b>Bot replies</b><br>/play, /queue, /skip ... or the prefix form<br>docker-container/src/discord/gateway.ts<br>packages/domain/src/templates.ts<br>DEC-005 | <ul style="list-style-type: none"> <li>queued</li> <li>now playing</li> <li>skipped / vote counted</li> <li>refused (ephemeral)</li> <li>no results</li> <li>cannot be played here</li> <li>joined / left voice</li> <li>not in a voice channel</li> <li>missing Connect/Speak</li> </ul> | UX-DISCORD-001<br>UX-DISCORD-002<br>UX-DISCORD-003 | ○ not run here |

## ANDROID SHELL

A WebView around the player, plus system UI.

| SURFACE                                                                                                                                                                                                                                                 | STATES IT MUST ANSWER                                                                                                                                                                   | RULES       | SEEN                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------|
| <b>WebView shell</b><br>launcher icon<br>android/app/src/main/java/com/nowplaying/player/MainActivity.kt<br>android/app/src/main/res/values/strings.xml<br>DEC-012                                                                                      | <ul style="list-style-type: none"> <li>player loaded from assets</li> <li>renderer crashed (reloads)</li> <li>back = history, else background</li> <li>hub over http refused</li> </ul> | UX-SAFE-004 | <input type="radio"/> not run here |
| <b>Notification, picker and save dialog</b><br>a fetch starts; a file input opens; an export is saved<br>android/app/src/main/java/com/nowplaying/player/FetchService.kt<br>android/app/src/main/java/com/nowplaying/player/BlobDownloads.kt<br>DEC-010 | <ul style="list-style-type: none"> <li>fetching (foreground notification)</li> <li>permission prompt (API 33+)</li> <li>saved</li> <li>could not save</li> </ul>                        | VOICE-002   | <input type="radio"/> not run here |

### LOCAL HELPER

A console program that serves the player and runs download tools.

| SURFACE                                                                                                     | STATES IT MUST ANSWER                                                                                                                                                                                                                 | RULES                                  | SEEN                               |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------|
| <b>Local helper console</b><br>now-playing-helper<br>local-helper/src/cli.ts<br>local-helper/src/options.ts | <ul style="list-style-type: none"> <li>tools found / missing with install hint</li> <li>missing tools setting up (verified download)</li> <li>--no-auto-tools: found / missing only</li> <li>token shown</li> <li>stopping</li> </ul> | VOICE-002<br>UX-SETUP-001<br>UX-DL-001 | <input type="radio"/> not run here |

## Journeys

The 13 things people come to do, from where they start to where they can fail. Each step list is what the code does today; each failure is one the product answers in words.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>First run of the hub</b></p> <p>A fresh data volume; the container is up on 127.0.0.1:4546.</p> <ol style="list-style-type: none"> <li>Open the admin page: the window, every tool locked, the sign-in form in the pane</li> <li>Sign in with admin / admin</li> <li>Overview opens with the amber gate; the other five tools stay locked</li> <li>Choose a real password</li> <li>Every tool unlocks</li> </ol> <p><b>WHEN IT GOES WRONG</b></p> <ul style="list-style-type: none"> <li>Fewer than twelve characters, or two entries that differ: refused in a sentence</li> <li>Login rate-limited after repeated failures</li> <li>A deep link to a locked tab is ignored</li> </ul> <p>UX-SAFE-002 UX-CHROME-001</p> <p>docker-container/tests/integration/setup-and-pairing.test.ts</p> <p>docker-container/tests/e2e/first-run.setup.ts</p> <p>docker-container/tests/dom/admin-shell.dom.test.tsx</p> | <p><b>Pair a player or the companion</b></p> <p>Setup complete; the device can reach the hub.</p> <ol style="list-style-type: none"> <li>Admin ticks the permissions and starts pairing (Devices): a code on the LCD plate</li> <li>Device enters the hub's address and the code</li> <li>Both screens show the same verification code</li> <li>Admin types it and confirms</li> <li>Device receives a scoped credential; a companion granted library:share starts sharing (DEC-023)</li> </ol> <p><b>WHEN IT GOES WRONG</b></p> <ul style="list-style-type: none"> <li>Code expired</li> <li>Fingerprint differs: do not confirm</li> <li>Wrong codes lock out that address for a while</li> </ul> <p>UX-SAFE-003</p> <p>docker-container/tests/integration/setup-and-pairing.test.ts</p> <p>windows-companion/tests/integration/companion-and-hub.test.ts</p> <p>windows-companion/tests/dom/companion-shell.dom.test.tsx</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Pair with a PC and play its music, relay-carried

The companion app runs its AWSP sidecar on a PC; the player is the served build (a service worker is the bridge).

1. On the PC, Settings ▶ Remote shows a ticket and a six-digit code
2. In the player, Sources ▶ Connections ▶ Stream from a PC: paste the ticket, type the code, Pair
3. The card shows the PC's name, relay-carried, and its track count; the tracks join the library
4. Play one; the bar follows the element and the indicator beside the transport reads relay-carried
5. The relay drops mid-track; the player reconnects and carries on from the last byte it held

### WHEN IT GOES WRONG

- A wrong or expired code is refused in words, and nothing is stored
- A revoked player is told to pair again, and stops reconnecting
- The single-file copy has no service worker, and says streaming needs the served player

NP-PREF-007 NP-PRIN-002

music-player/tests/e2e/awsp.spec.ts

windows-companion/awsp-server/tests/awsp.rs

## Add music on this device and play it

The player is open (installed or not).

1. Library shows how to add music
2. Choose a folder or files
3. Tracks appear; unplayable formats say so
4. Play; the hero and the list mark the playing row

### WHEN IT GOES WRONG

- Browser without folder access falls back to file choice
- A file that disappeared says it cannot be found

UX-STATE-001 UX-A11Y-001

music-player/tests/e2e/local-file.spec.ts

music-player/tests/e2e/player.spec.ts

## Listen together

Player paired with a hub; a group exists.

1. Switch the mode to Shared
2. Join or create a group
3. Anyone queues; everyone hears it at the same point
4. Skips and pauses apply to the group

### WHEN IT GOES WRONG

- No hub: the switch explains why
- Stale command: refused with the current revision
- Member without the DJ role: vote-skip instead of skip

UX-PRIN-002 UX-SHARED-001

docker-container/tests/integration/groups-and-queue.test.ts

## Play music into a Discord voice channel

Bot token installed, bot enabled, default group chosen, worker container running.

1. Join a voice channel
2. /play <search or link>
3. The bot joins your channel and replies Queued
4. Audio follows the group timeline; pause, skip and seek from any device apply

### WHEN IT GOES WRONG

- Not in a voice channel: the reply says so
- Missing Connect/Speak: the reply names them
- Source that cannot stream in Discord: skipped with the reason
- Worker not running: the admin panel says so

UX-DISCORD-001 UX-DISCORD-002 UX-DISCORD-003

docker-container/tests/integration/discord-worker.test.ts

docker-container/tests/unit/discord-voice.test.ts

docker-container/tests/integration/discord-parity.test.ts

## Download a track the hub may give you

Paired player; the provider allows it.

1. Row menu → Download
2. Choose format and the basis (e.g. content I own)
3. Progress in words
4. The file saves as a link download

### WHEN IT GOES WRONG

- Format unavailable without FFmpeg
- Provider refuses the basis
- Cancelled: the partial file is removed

UX-FEED-001 UX-LINK-001

docker-container/tests/security/ssrf-and-downloads.test.ts  
docker-container/tests/integration/downloads-queue.test.ts

## Share a track or playlist

Paired player.

1. Share
2. Choose whether streaming is allowed
3. Copy the link
4. The recipient opens the public page

### WHEN IT GOES WRONG

- No hub: the sheet explains why
- Link revoked or used up: the page says it is unavailable

UX-PRIN-002

docker-container/tests/integration/shares-and-sync.test.ts  
docker-container/tests/security/share-limits.test.ts

## Send files from Windows to the hub

Companion paired; a folder indexed.

1. Add a music folder (Library)
2. Choose songs in Music on This PC
3. Send to Hub
4. Remote ▶ Transfers shows progress; Cancel stops it

### WHEN IT GOES WRONG

- Drive not connected
- No hub paired, or nothing chosen: Send to Hub is disabled and says why
- Hub unreachable: the transfer fails with the reason under the list

UX-FEED-001

windows-companion/tests/integration/companion-and-hub.test.ts

## Downloaders set themselves up

The companion (or the standalone helper) starts on a machine without yt-dlp, spotDL or FFmpeg.

1. The helper starts; the window opens without waiting
2. Each missing tool is fetched from its project's GitHub release, one at a time, and checked against the published SHA-256
3. Settings ▶ Downloaders shows Setting up... with progress, then Ready — Set up automatically
4. When FFmpeg lands, tempos are measured for the library without a rescan

### WHEN IT GOES WRONG

- No published SHA-256, or a mismatch: nothing is installed; Couldn't set up with the reason and Try Again
- Offline: retried on the next start or after six hours; Try Again retries at once
- No build for this platform (FFmpeg on macOS/Linux): the install advice is shown instead

UX-SETUP-001 UX-FEED-001

local-helper/tests/unit/install.test.ts  
local-helper/tests/unit/provision.test.ts  
local-helper/tests/integration/helper-server.test.ts  
windows-companion/tests/integration/main-process.test.ts  
windows-companion/tests/dom/companion-shell.dom.test.tsx

## Add a Live TV playlist and watch the guide fill in

The companion is running on the PC the player is opened on.

1. Companion ► Live TV: paste an M3U link and choose Add
2. The row reads checking... while the main process reads the link
3. Kept: the row reads ✓ and how many channels it holds
4. Paste an XMLTV link the same way for now and next
5. Open the player's Live TV: the guide lists the companion's channels, with what is on and what follows

### WHEN IT GOES WRONG

- Not a playlist, not a guide, too large or on this PC's own network: refused in a sentence under the field; what was typed stays; nothing is kept
- A kept link stops answering: it keeps what it last held, says so, and offers Check Again
- No companion, or one with no playlists: the player's guide is as it was

```
UX-TV-001 NP-TV-001
music-player/tests/e2e/np/live-tv.spec.ts
windows-companion/tests/integration/live-tv.test.ts
local-helper/tests/integration/helper-tv.test.ts
windows-companion/tests/dom/companion-shell.dom.test.tsx
```

## Back up and restore

Setup complete.

1. Hub: System ► Backup → Back Up Now
2. Restore → confirm in the sheet
3. The hub reports success or the reason it failed
4. Companion: Settings ► Backup → choose a folder, Back Up Now; restore and delete ask first

### WHEN IT GOES WRONG

- Invalid backup refused
- Swap failed: previous database put back

```
UX-SAFE-001
docker-container/tests/integration/backup-and-search-
cursor.test.ts
windows-companion/tests/integration/backup.test.ts
```

## Fetch with yt-dlp or spotDL

The local helper is running, or the Android app has unpacked its tools.

1. Library → Fetch
2. Paste a link and pick a format
3. Progress
4. Files are added to the library

### WHEN IT GOES WRONG

- Tool missing: the control is not shown
- Tool error: its own message, without paths
- FFmpeg missing: the file is saved as the site sent it, and the job says it has no cover art or tags

```
UX-PRIN-001 VOICE-002 UX-DL-001 NP-FIND-002
music-player/tests/e2e/helper.spec.ts
local-helper/tests/integration/helper-server.test.ts
local-helper/tests/unit/download-tags.test.ts
local-helper/tests/integration/helper-resolve.test.ts
music-player/tests/e2e/np/links.spec.ts
```

## The Discord bot

The bot is the one surface Aqua does not draw: Discord does. What the suite controls is the words, the embed colour, whether a reply is private, and when the bot moves in and out of voice. Replies come from templates an operator can edit in the hub (Admin → Discord); these are the defaults, rendered with sample values by the same function the bot uses.

Commands: `/play` `/nowplaying` `/queue` `/skip` `/pause` `/resume` `/stop` `/shuffle` `/clear` `/join` `/leave` `/settings`. Every one also works as a prefix command (`!play`) once the Message Content intent is granted, and the two forms are identical by construction ([UX-DISCORD-001](#)). Embed colours are the system's own: selection blue, success, warning and danger.

*Illustration — Discord renders its own interface. This shows content and colour, not Discord's layout.*

|                                                                                                                       |                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>success</b><br>A command with nothing more specific to say                                                         | Done.                                                                                                         |
| <b>queued</b><br><code>/play</code> found something playable                                                          | <b>Queued #3</b><br><b>Signal Fade</b> — Cassette Bloom<br>Requested by alex · 0:04 · Public domain           |
| <b>nowPlaying</b><br><code>/nowplaying</code>                                                                         | <b>Now Playing</b><br><b>Signal Fade</b> — Cassette Bloom<br>Live from Pier 9<br>Requested by alex in Kitchen |
| <b>skipped</b><br><code>/skip</code> by a DJ or the requester                                                         | Skipped <b>Signal Fade</b> (skipped).                                                                         |
| <b>permissionDenied</b><br>The caller lacks the DJ or admin role<br>only the caller sees this                         | You don't have permission to do that (the DJ role is required).                                               |
| <b>noResults</b><br><code>/play</code> found nothing<br>only the caller sees this                                     | No results for "No results".                                                                                  |
| <b>unavailableSource</b><br>The match cannot stream in Discord<br>only the caller sees this                           | <b>Signal Fade</b> cannot be played here: YouTube only plays in its own player.                               |
| <b>emptyQueue</b><br><code>/queue</code> or <code>/nowplaying</code> with nothing queued<br>only the caller sees this | The queue is empty. Use <code>/play</code> to add something.                                                  |
| <b>joined</b><br><code>/join</code> , or <code>/play</code> when the bot was not in voice                             | Joined #music.                                                                                                |
| <b>left</b><br><code>/leave</code> , or the idle timeout                                                              | Left the voice channel.                                                                                       |
| <b>error</b><br>Anything unexpected; the reason is the error message<br>only the caller sees this                     | Something went wrong: Queue is at revision 12                                                                 |
| <b>wrongChannel1</b><br>A command outside the designated channel<br>only the caller sees this                         | Music commands work in #music.                                                                                |
|                                                                                                                       |                                                                                                               |

|                             |                                   |
|-----------------------------|-----------------------------------|
| <b>paused</b><br>/pause     | Paused.                           |
| <b>resumed</b><br>/resume   | Resumed.                          |
| <b>stopped</b><br>/stop     | Stopped and cleared the queue.    |
| <b>shuffled</b><br>/shuffle | Shuffled 12 tracks.               |
| <b>cleared</b><br>/clear    | Cleared 12 tracks from the queue. |

#### VOICE-CHANNEL REPLIES

Added to the reply when the bot cannot move into voice. They are sentences in the gateway, not templates.

- “Join a voice channel first (or use join <channel>), so I know where to play.”
- “I need View Channel, Connect and Speak in <channel> to play there.”
- “There is no voice channel called “<name>” that I can see.”
- “Could not connect to the voice channel. Check that the bot has Connect and Speak there, and that outbound UDP is not blocked.”

**Known gap (DEC-005).** The default `noResults` template quotes `{{reason}}`, but the command service passes the search text as `title`, so a failed search currently reads “No results for “No results”.”

# Platforms

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## WEB AND PWA — PLAYER AND HUB ADMIN

- **Breakpoints follow content, not a device list.** The Airwave window changes at 640px — the tools scroll sideways, tiles go two-up, the key-value list stacks and tables drop their secondary columns — and at 560px, where side-by-side groups stack. The library's page skin changes at 480, 560, 620, 720 and 900px. Check at the mockup sizes: 320, 390, 768, 1280 and 1680px.
- **The pointer decides density, not the width.** (`pointer: coarse`) switches the touch layer on at any width; a narrow desktop window keeps desktop density ([UX-TOUCH-001](#)). The Airwave window's touch layer is small — list rows and the round minus grow, buttons do not — which is an open question for the hub on a phone (DEC-027).
- **Installable and offline.** The player ships a manifest, icons and a service worker, opens with the network off once it has loaded online, and loads nothing from another origin. The single-file build, `now-playing.html`, runs straight from disk.
- **Themes.** The player follows the system colour scheme. The Airwave window is light only, in both OS appearances (DEC-003).
- **Keyboard.** In the player, Space toggles playback and Escape closes a popover without losing what was typed. In the Airwave window the tools are one tab stop: arrows, Home and End move the selection, and locked tools are skipped ([UX-KEY-001](#)).

## ANDROID

- The app is the player inside a WebView, loaded from its own assets, so every web rule applies unchanged. The player is built with `NP_BASE_PATH=/assets/app/`.
- Back goes back in the page's history; with nowhere to go it moves the app to the background, so music keeps playing.
- System UI carries its own copy (`strings.xml`): the “Fetching music” notification while a download runs, the document picker for file inputs, and the save dialog for exports (“Saved ...” / “Could not save ...”). Notification permission is asked for the first time a fetch starts (DEC-010).
- Hubs are reached over https only (DEC-012).

## WINDOWS COMPANION — ELECTRON

- One window with four tools, 640 × 760 as it opens and never smaller than 520 × 440. Its chrome is its title bar: the operating system's own is hidden, the page draws the title, Windows draws minimise, maximise and close over the top-right corner, and the chrome is the drag region while the tools stay clickable (DEC-020, [UX-CHROME-001](#)).
- A tray icon (`resources/tray.svg`) whose menu has Open, Scan Library Now and Quit. With “Keep running in the notification area when the window closes” on, closing the window keeps the companion running.
- It is the only product that reads the filesystem; its copy says so and never implies files are copied or uploaded in the background.
- Pairing shows the hub's fingerprint, and a notice warns when the hub address is plain http off the local network.
- Live TV links are read by the main process, never by the page: public addresses only, with a size cap and a deadline, and kept only when they hold channels or programmes ([UX-TV-001](#)).

## TERMINAL — LOCAL HELPER

- Takes the first free port in a small range, prints the token to paste into Settings → Platforms, and names each missing tool with the command that installs it.
- Ctrl+C removes only the helper's own working folder, never the folder it was pointed at.

## ENGINEERING APPENDIX — DOCKER DELIVERY

Docker is how the hub is delivered, not a design surface. What it means for a design change:

- The image (`docker-container/Dockerfile`) builds the admin window from `docker-container/src/web` with the generated Airwave stylesheets and AquaArt from `@now-playing/aqua-ui`, so a design change reaches the hub only when the image is rebuilt: `docker compose up -d --build` or `./nowplaying update`.
- The hub listens on port 4546, published to `127.0.0.1` by default, with `/healthz` for the health check. Host names other than IP addresses, `localhost` and `.local/.lan` names must be listed in `NP_ALLOWED_HOSTS` or set as `NP_PUBLIC_ENDPOINT`.
- The Discord worker is the same image under `--profile discord`. Its default replies change when `packages/domain/src/templates.ts` changes and the image is rebuilt, or immediately when an operator edits a template in the hub.
- The player, the Android app and the companion are built separately; each needs its own rebuild to pick up a design change.

## Every product, every size — and editable

---

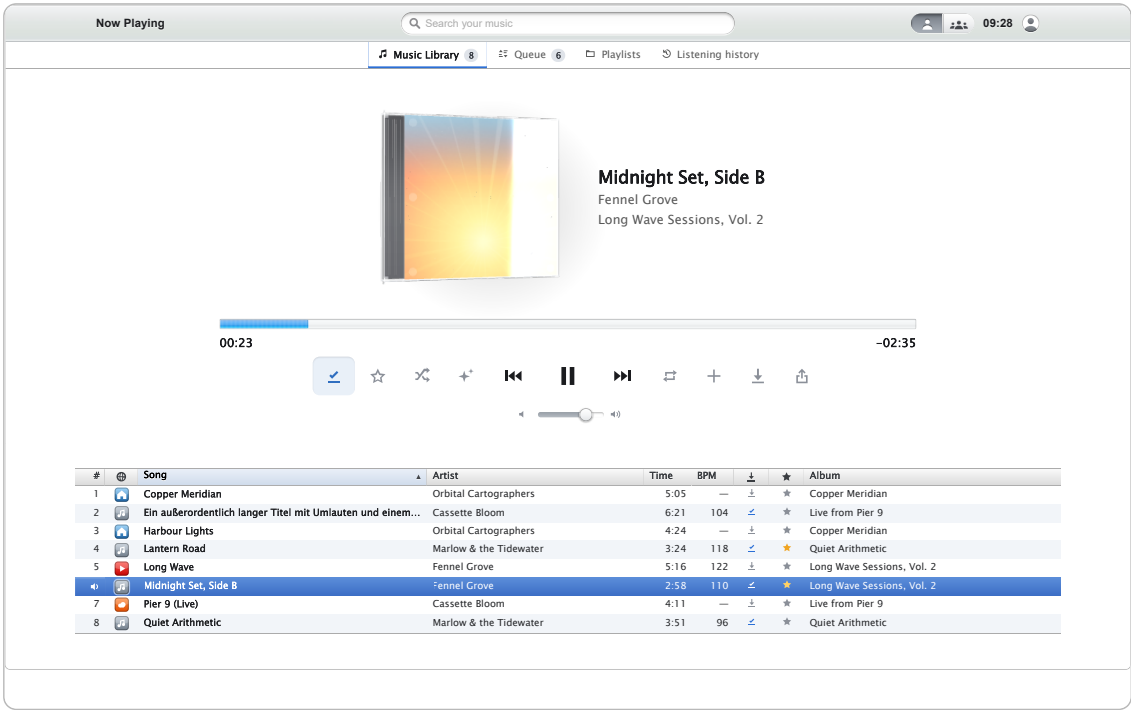
Each frame below is an iframe with its own viewport, so a 320px column is a real 320px viewport and the media queries behave exactly as they do on a phone. The hub's and the companion's frames hold only what those products load — the generated Airwave stylesheets, their own `styles.css`, AquaArt and their own kits — and the player's hold the component library's page skin. Under each one is a measurement taken *inside* that frame: what runs off the side, the smallest text, the smallest thing you could tap.

A frame marked *touch layer emulated* has its (`pointer: coarse`) rules re-applied inside it, read back out of the real stylesheets — an iframe inherits the desktop's pointer, so without that a phone frame would quietly show desktop sizes. The Airwave window has almost no touch layer (list rows and the round minus grow; buttons do not), so the hub's phone frames report their target sizes without being failed on them (DEC-027). At the bottom you can edit any custom property the stylesheets define and watch every frame repaint at once; what you copy out says which file each change belongs in.

In print, every screen is shown at two of its product's sizes with the fit measured inside each frame. The interactive version — every size, the product switch and the token editor — is <docs/design/styleguide.html>.

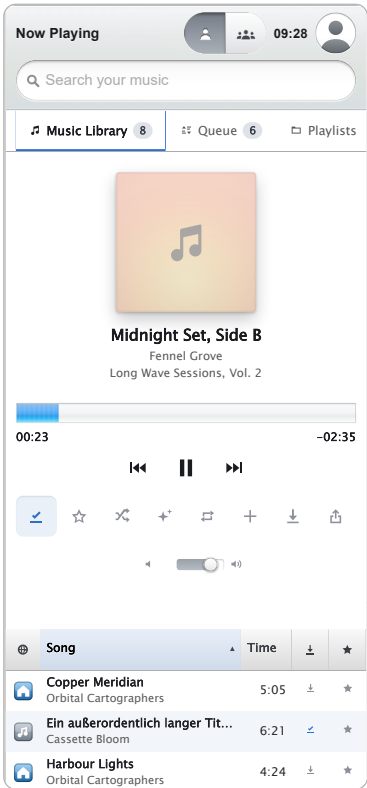
Now playing and library

The whole page skin at once. Narrow it and watch the list drop its number and artist columns and put the artist under the title instead.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px

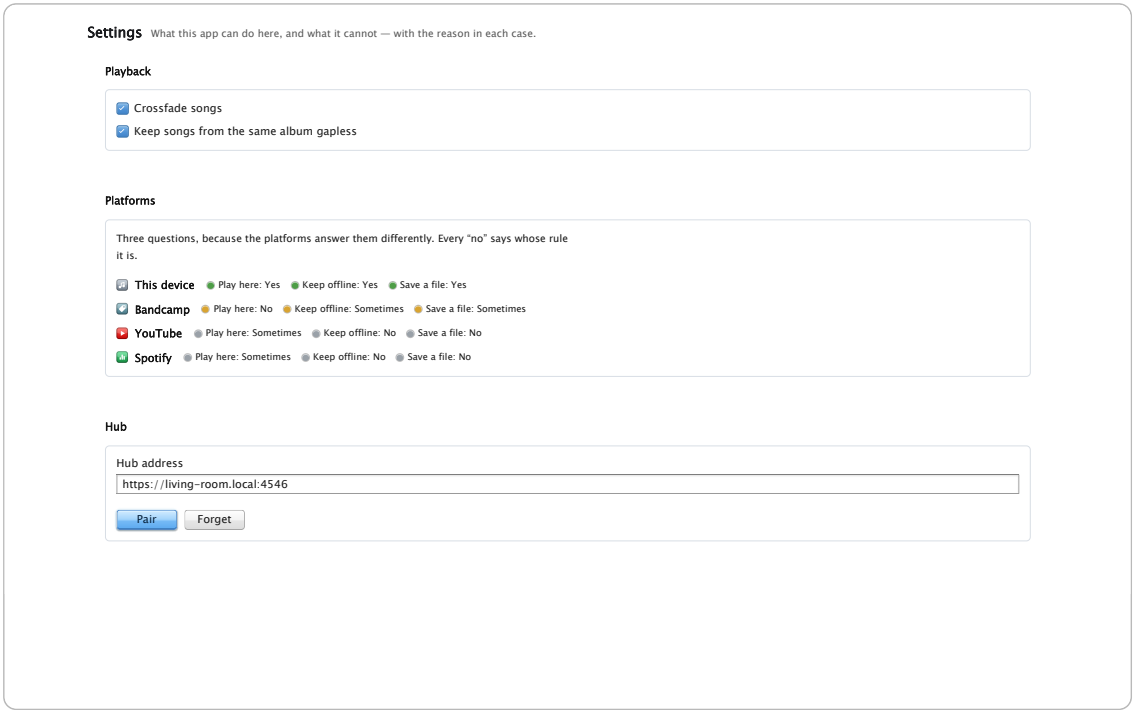


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 12px · smallest target 44px

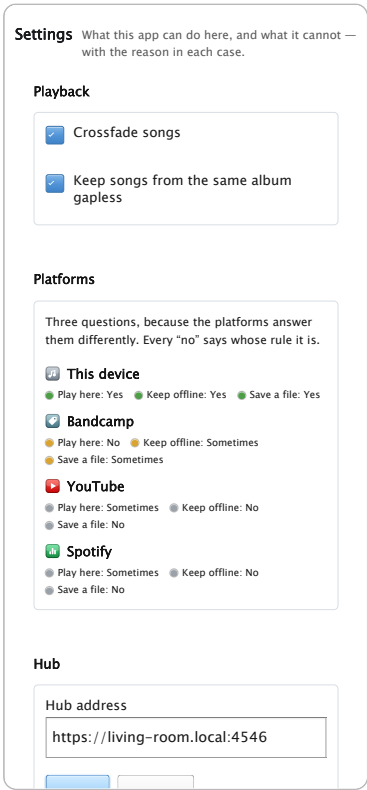
Settings

Where most of the app's explaining happens, so it is the screen where text size and wrapping matter most.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 12px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 13px · smallest target 44px

Now playing (audio chain)

What the audio chain is doing to the song, and the spectrum visualiser under it — drawn by the player's own spectrum code from a generated analyser frame.

Now playing

What the audio chain is doing to this song, measured rather than assumed.

How this is being played

Song

Midnight Set, Side B — Fennel Grove

Position

0:23 of 2:58

Source

A file on this device

Equalizer

Bass Lift — the preset chosen for this album

Limiter

On — prevents clipping when the equalizer adds gain

Headroom

3.0 dB trim applied to keep the boosted signal below full scale

Retuning

Off — playing at the original pitch

Added latency

11.6 ms



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

Now playing

What the audio chain is doing to this song, measured rather than assumed.

How this is being played

Song

Midnight Set, Side B — Fennel Grove

Position

0:23 of 2:58

Source

A file on this device

Equalizer

Bass Lift — the preset chosen for this album

Limiter

On — prevents clipping when the equalizer adds gain

Headroom

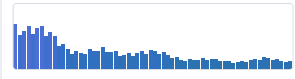
3.0 dB trim applied to keep the boosted signal below full scale

Retuning

Off — playing at the original pitch

Added latency

11.6 ms

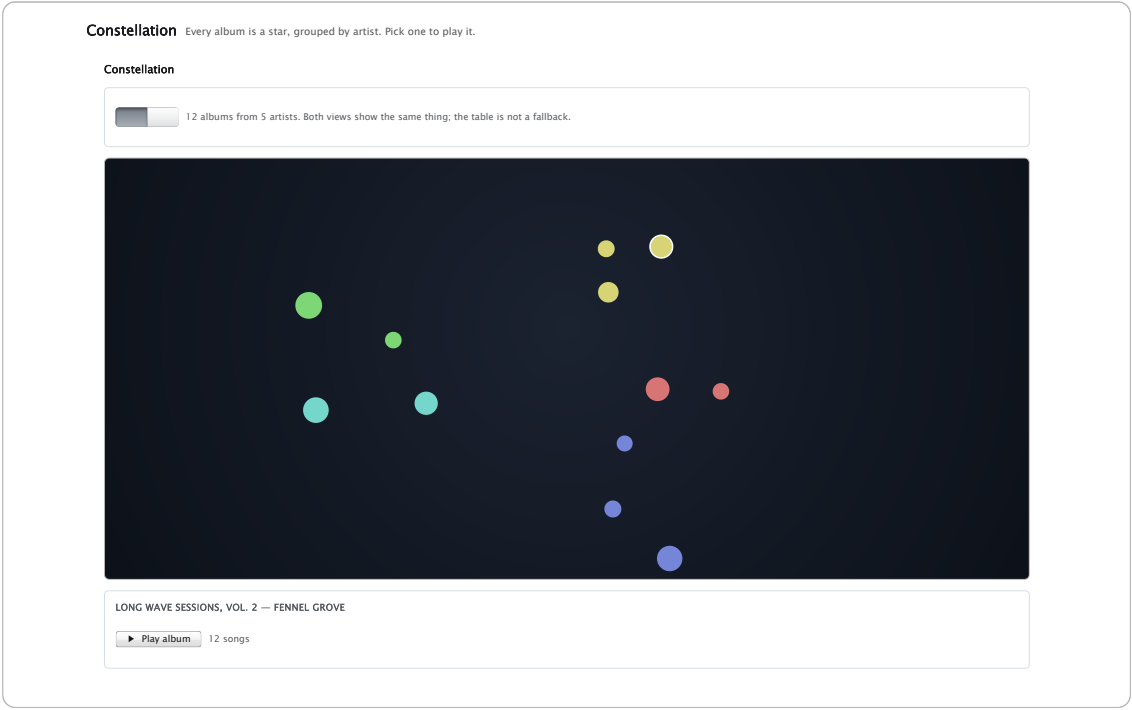


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 13px · smallest target —

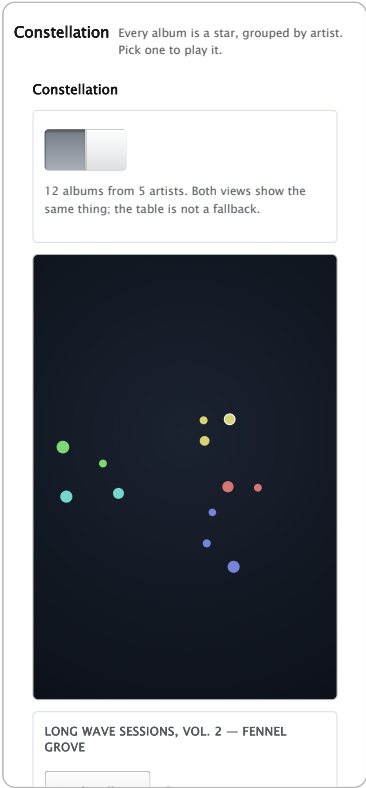
Constellation

The library as a star field, placed by the player’s own layout code. The table beside it is the same information, not a fallback.



Laptop 1280 × 800, shown at 53%

● Fits · nothing off the side · smallest text 11px

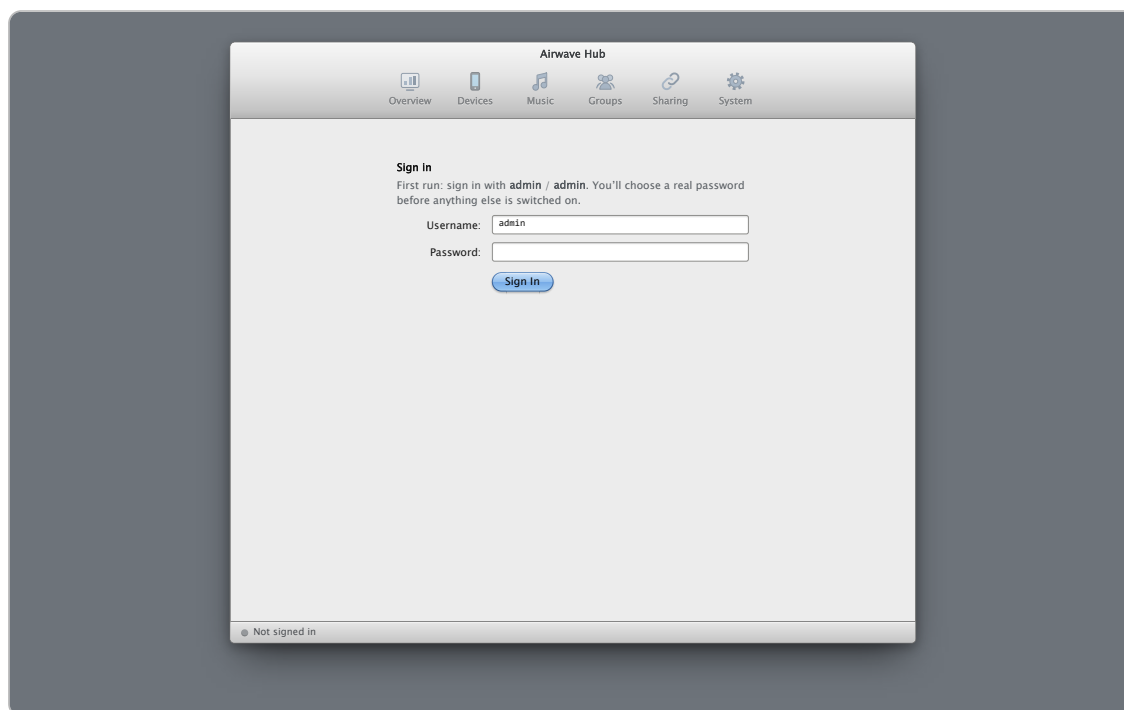


Phone 390 × 844, shown at 56% touch layer emulated

● Fits · nothing off the side · smallest text 13px · smallest target 44px

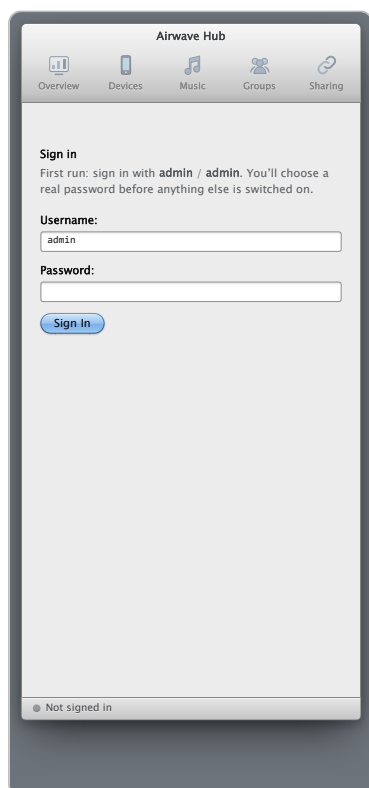
## Sign in

Not signed in: the same window with every tab locked and the form in the pane. There is no separate sign-in page.



**Laptop** 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

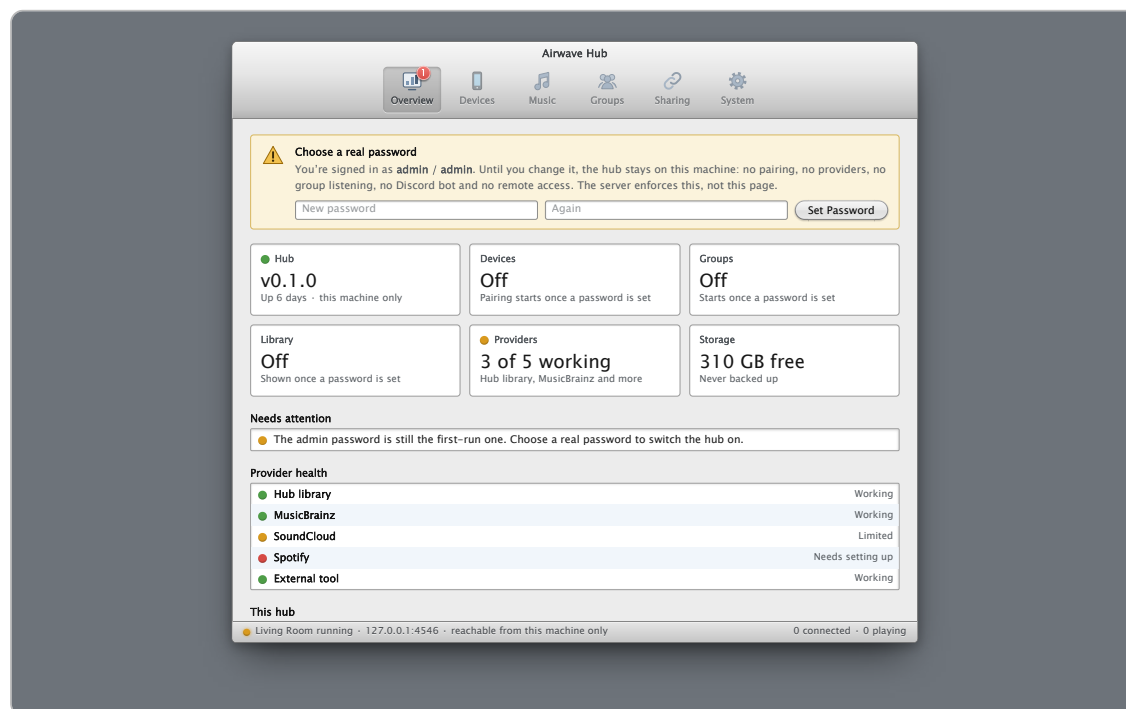


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

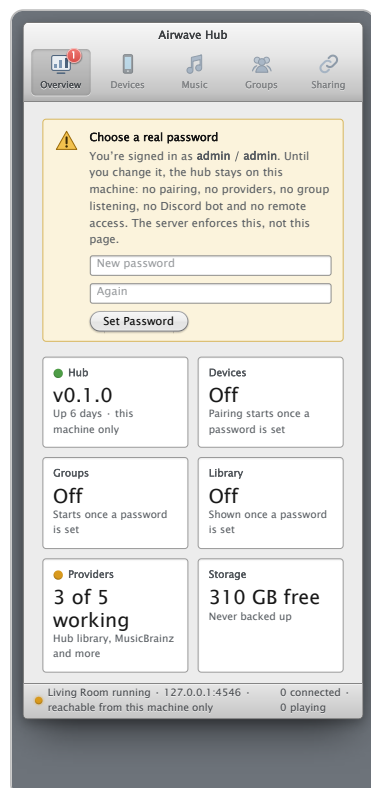
## First run

Signed in with the first-run password: the amber gate on Overview, the other five tabs locked. The tiles that depend on gated routes say they are off instead of showing a number.



**Laptop** 1280 × 800, shown at 53%

● Fits · nothing off the side · smallest text 11px

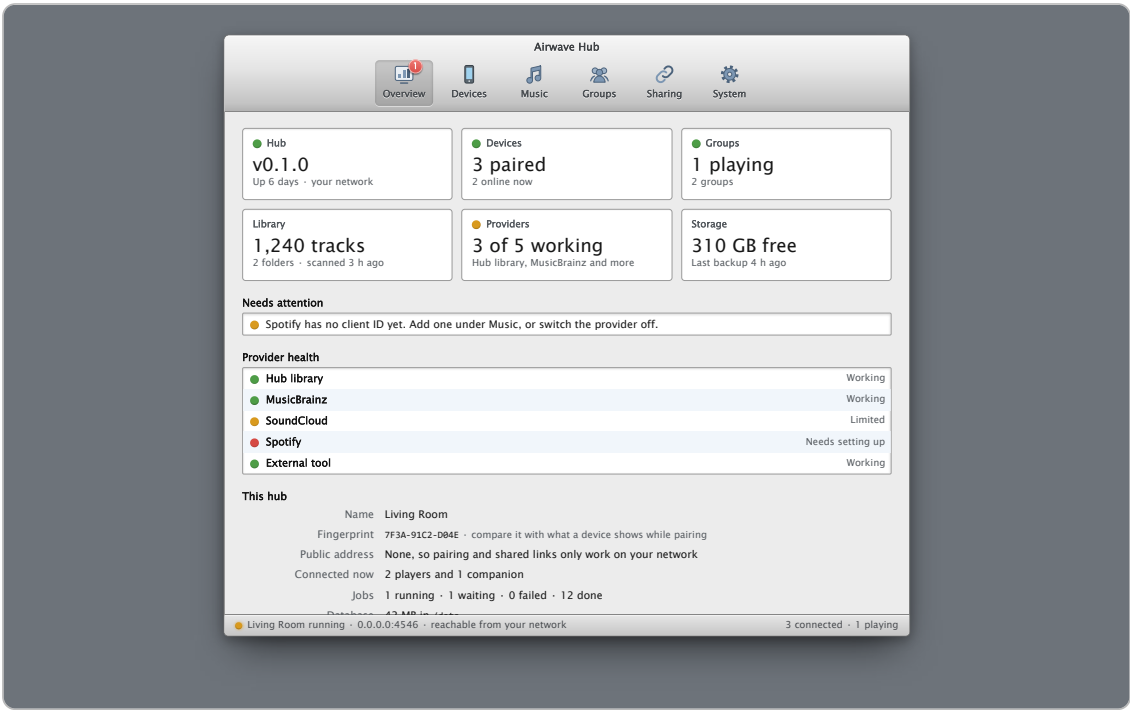


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits · nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

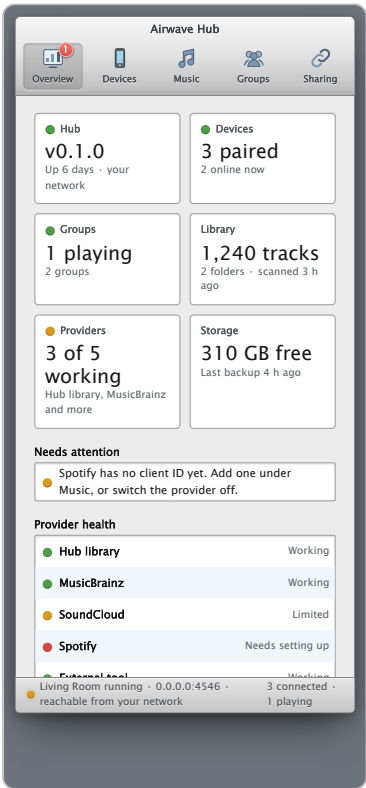
Overview

Six tiles, what needs attention, each provider's health as a word, and the hub's own facts. Below 640px the tiles go two-up and the key–value list stacks.



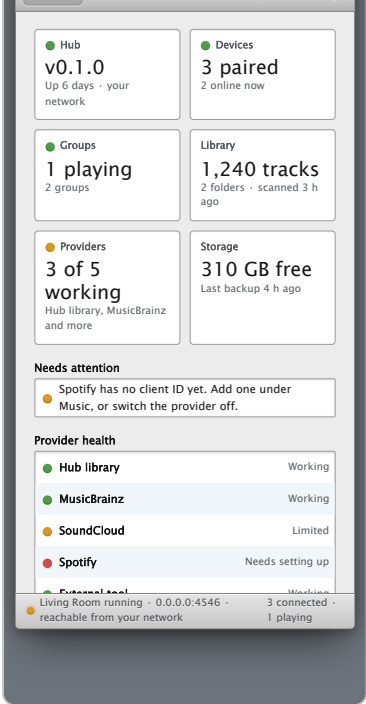
Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)



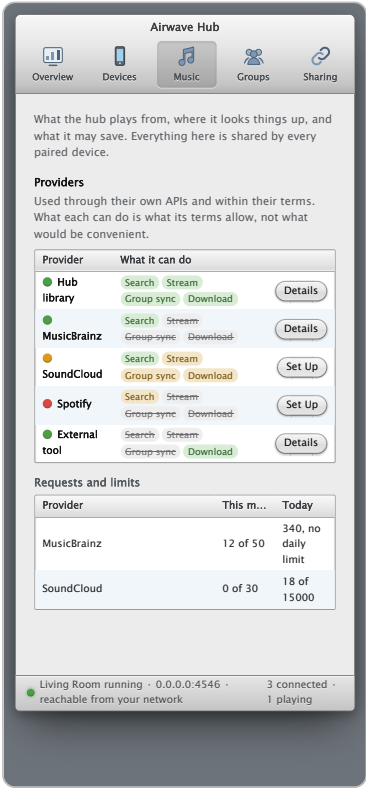
Music ▸ Providers

A table inside a well. Below 640px the Role column and the two request counters are dropped rather than scrolled sideways.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px

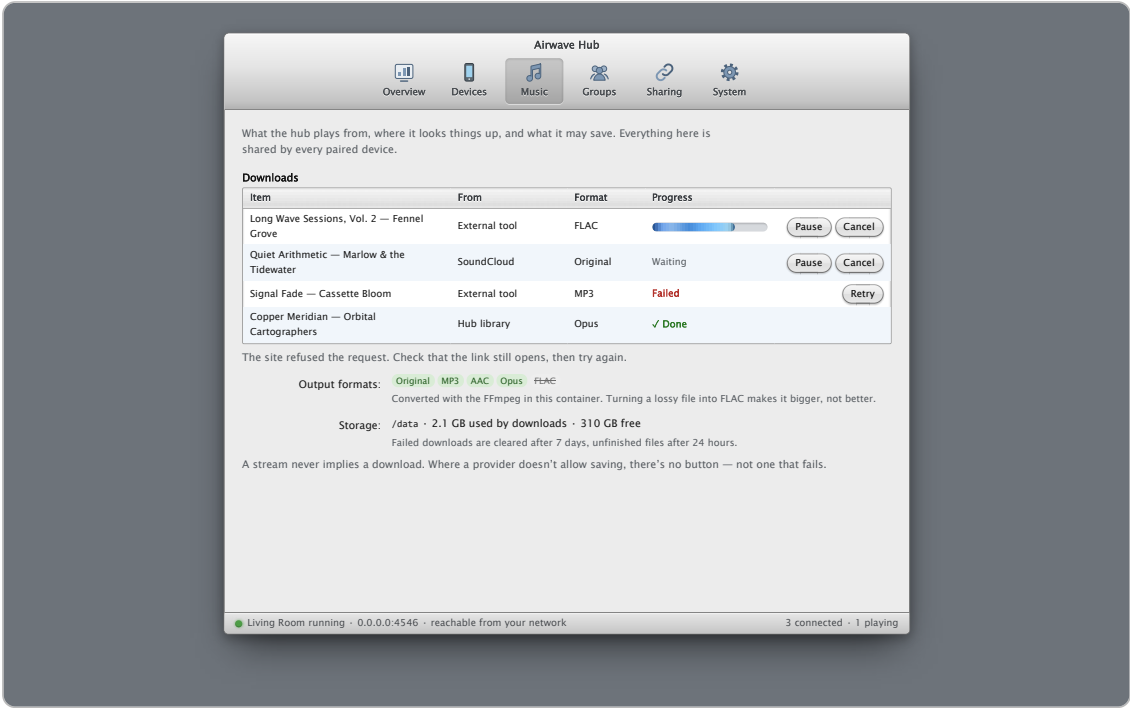


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

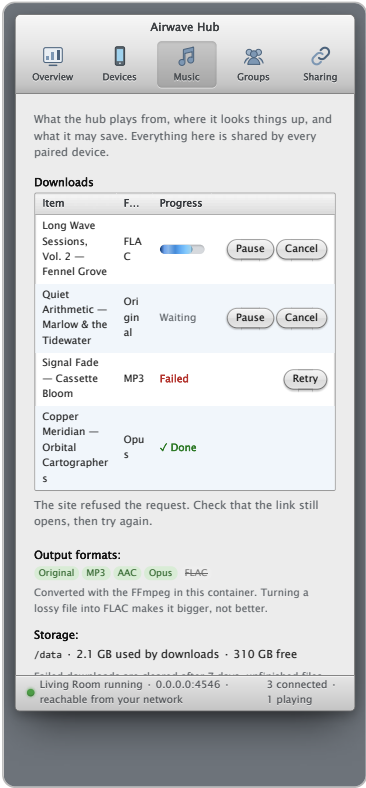
Music ▶ Downloads

Every state a job can be in, as words or a bar; the formats this hub can write, struck through where it cannot; and the storage line.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px

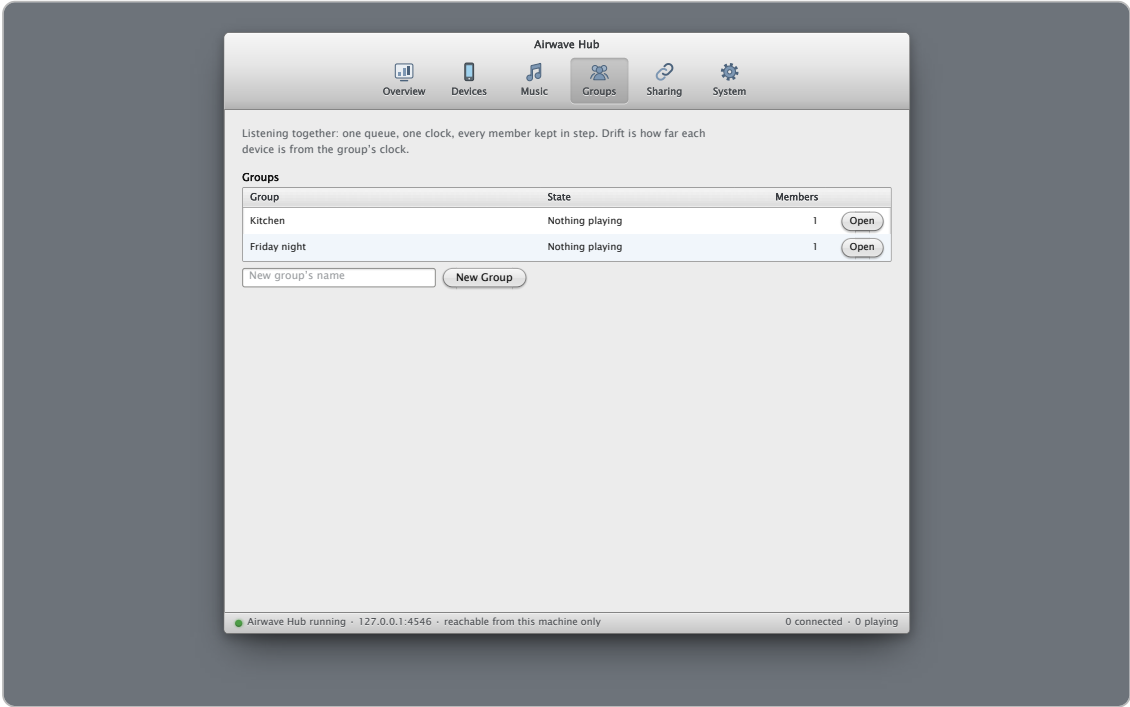


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

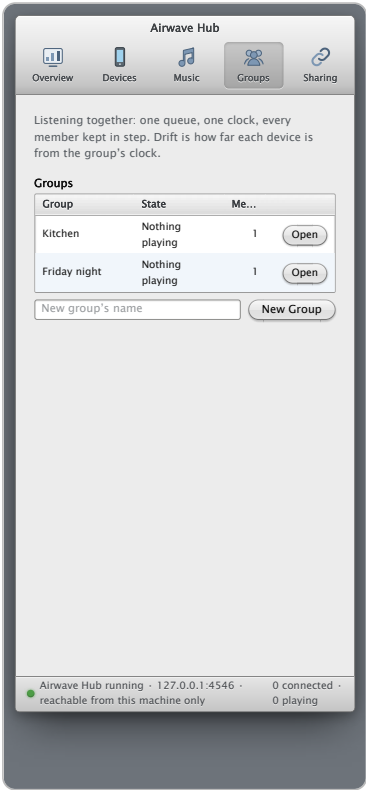
Groups

The hub's own Groups view, answered by a recorded hub: two groups, nothing playing, and the New Group row. Open one to see its transport, queue and invites.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

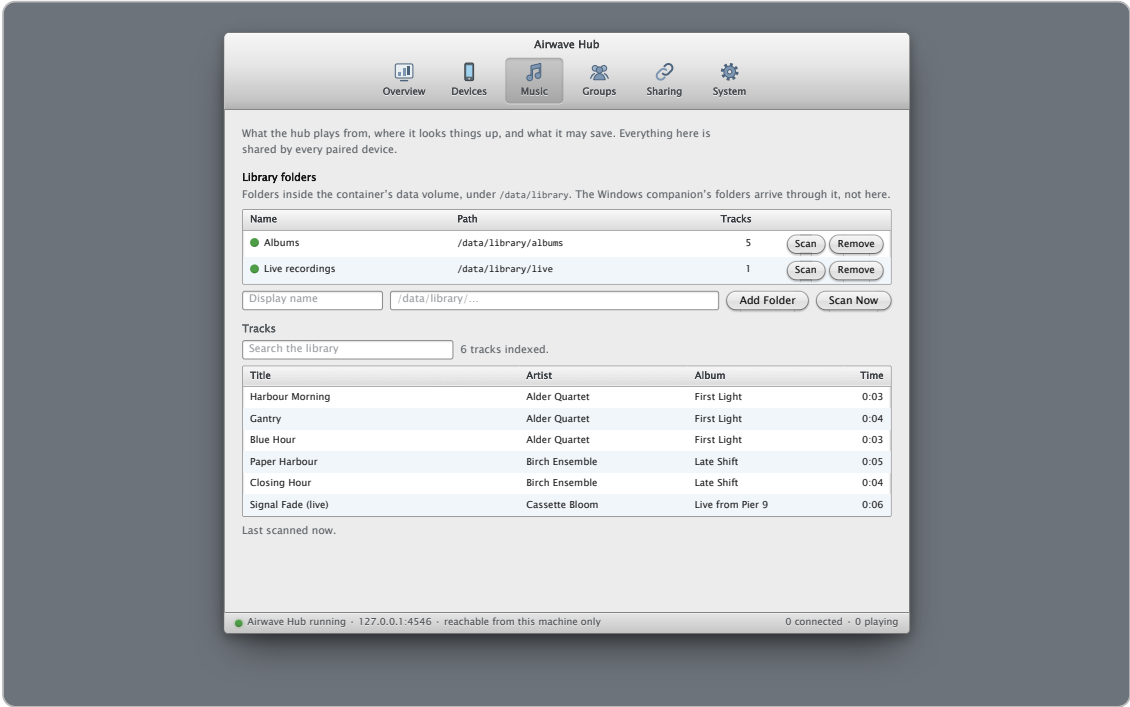


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

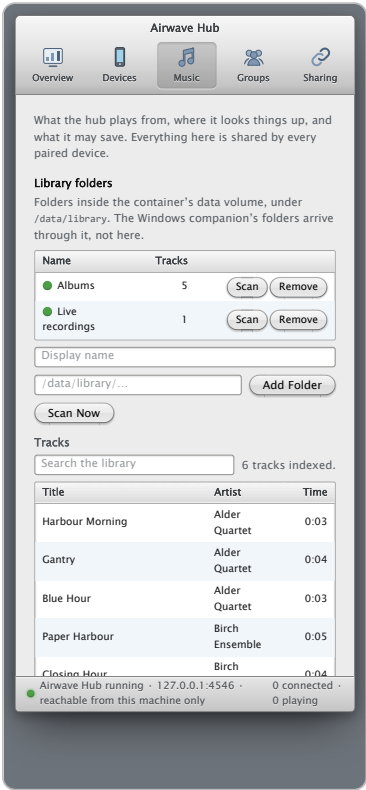
Music ▶ Library

The hub's own Library view: two folders in the data volume, each with its own Scan (that folder alone) and Remove, the fields to add a folder beside Scan Now, and the songs found.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

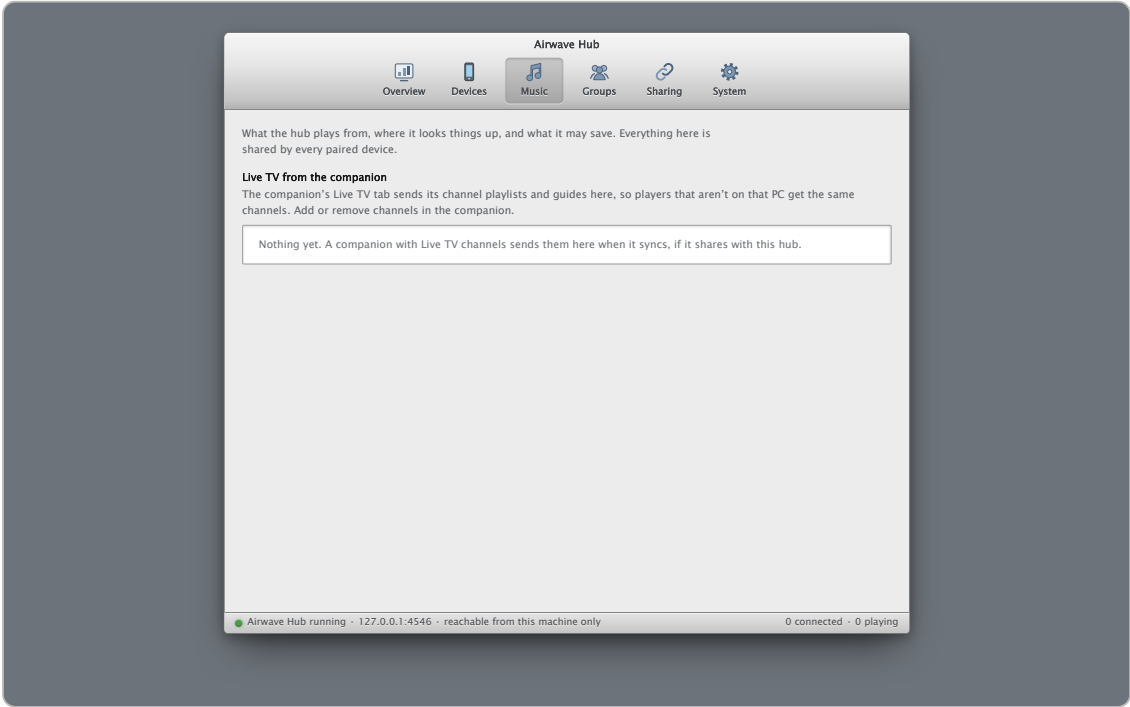


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

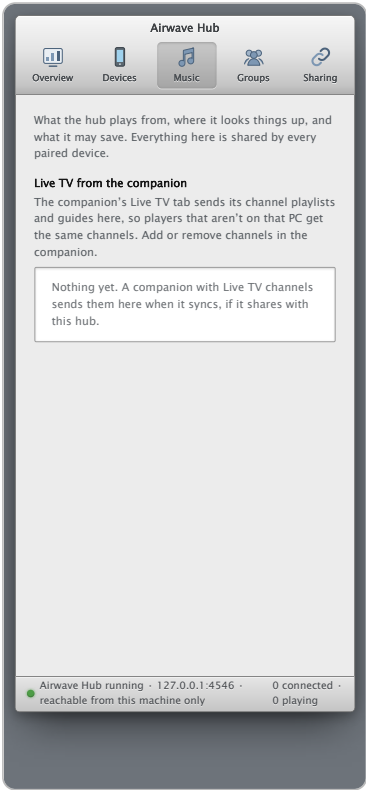
Music ▶ Live TV from the companion

The hub's own Live TV row, read-only: what a sharing companion sent, or — as recorded here — the sentence saying nothing has arrived yet. Channels are added and removed in the companion.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

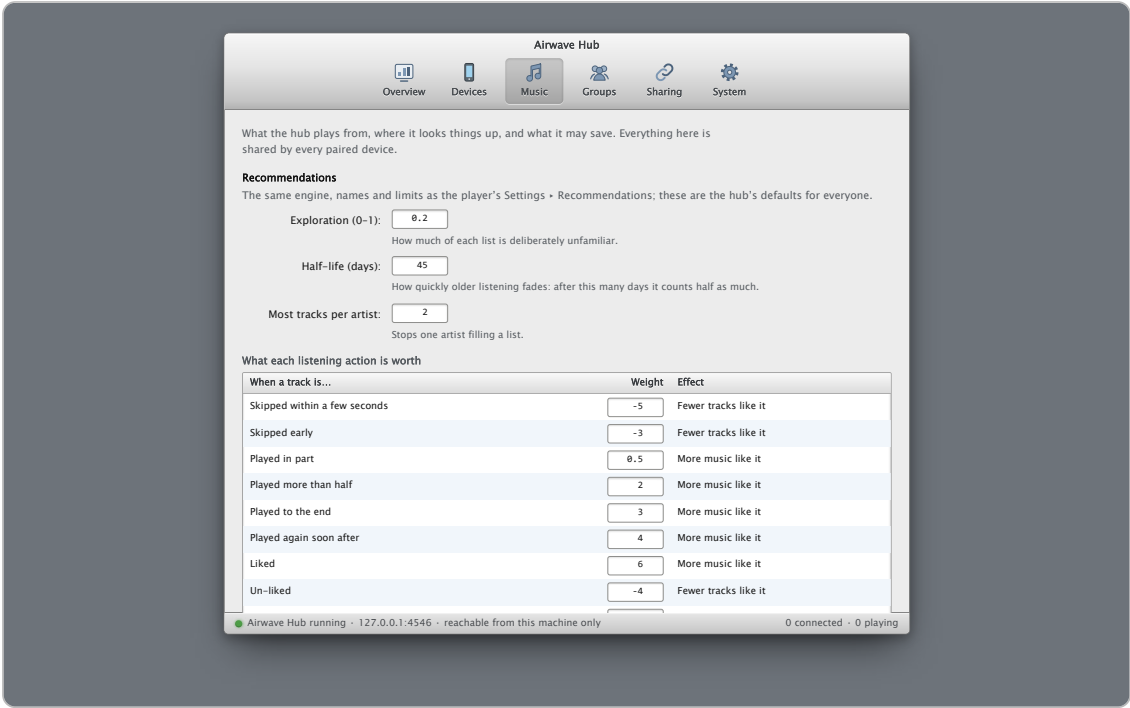


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 50px · desktop density, not held to 44px (DEC-027)

Music ▶ Recommendations

The hub's own Recommendations view: the defaults every player's engine starts from, with Save and Revert for the whole form.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

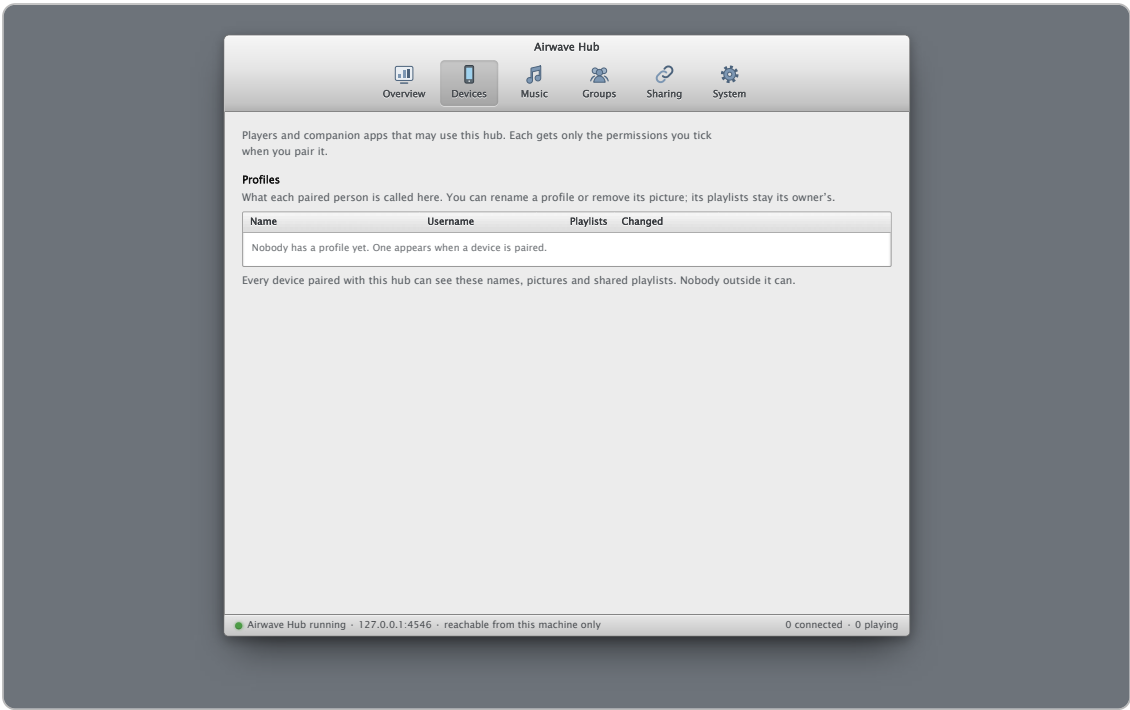


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

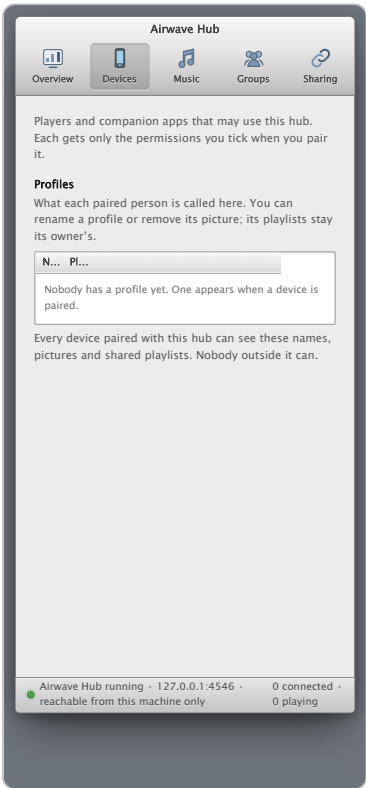
Devices ▶ Profiles

The hub's own Profiles view on a hub with no paired players yet: the empty list says what fills it.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

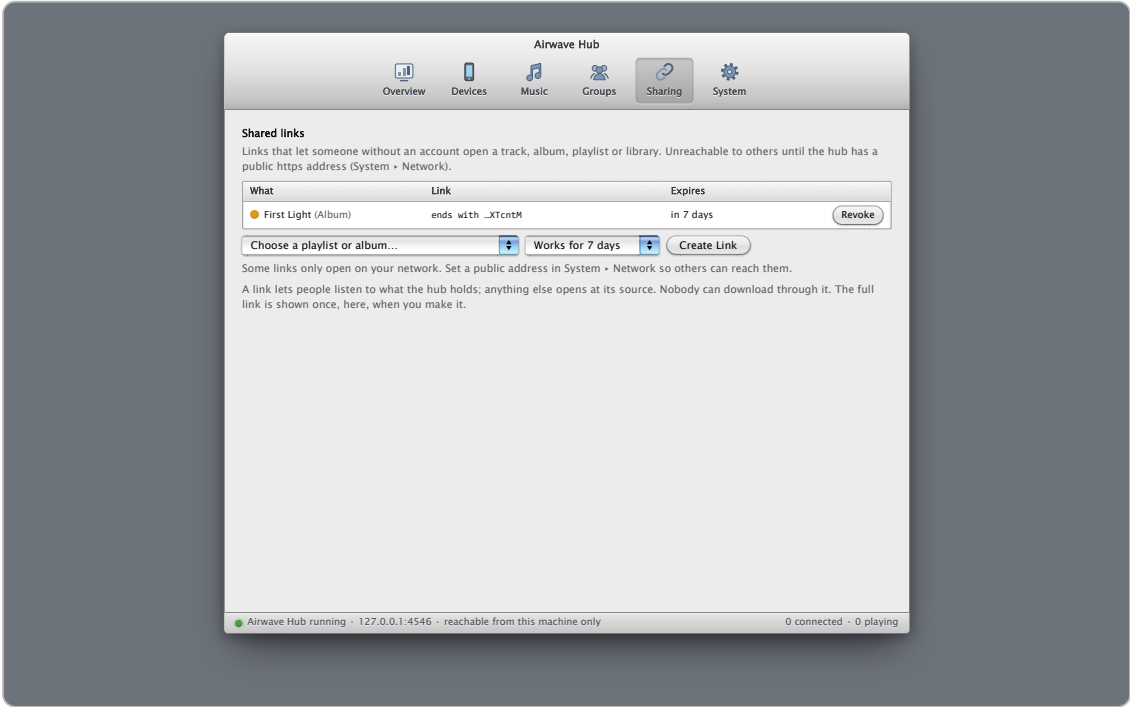


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 50px · desktop density, not held to 44px (DEC-027)

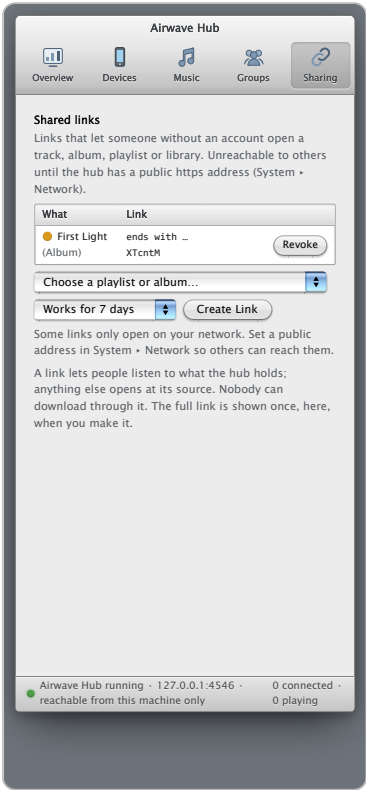
Sharing ▶ Shared links

The hub's own Shared links view: one album link with its expiry and counts, and Create Link with the picker of hub playlists and albums and how long the link lasts.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

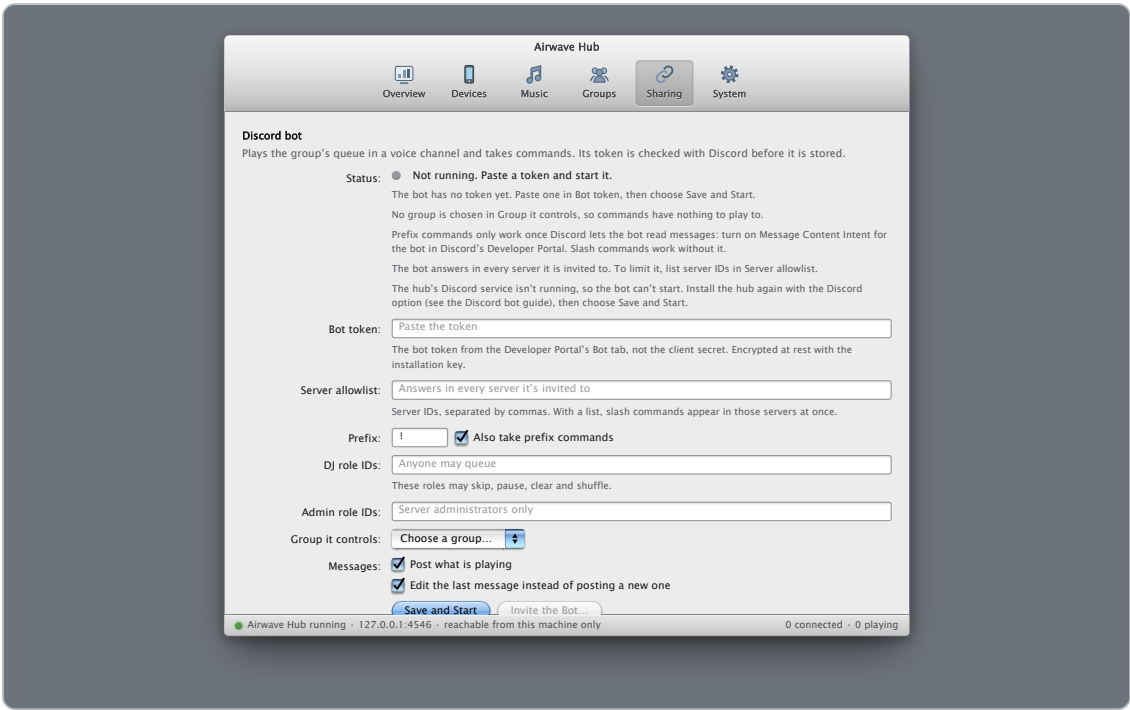


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

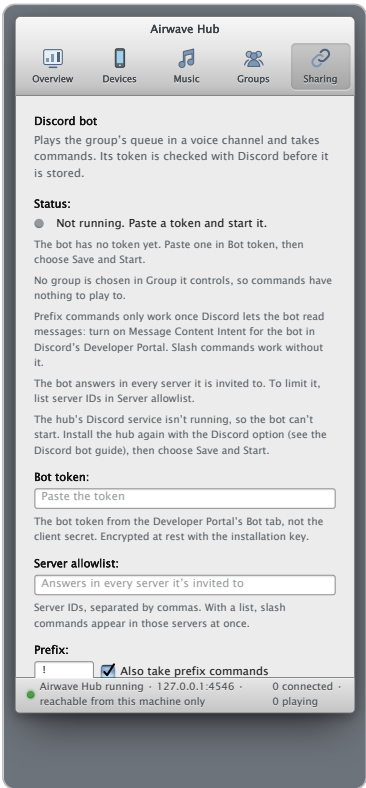
Sharing ▶ Discord

The hub's own Discord view with no bot token yet: the form, Save and Start with Revert, and Invite the Bot disabled with its reason until a token is saved.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

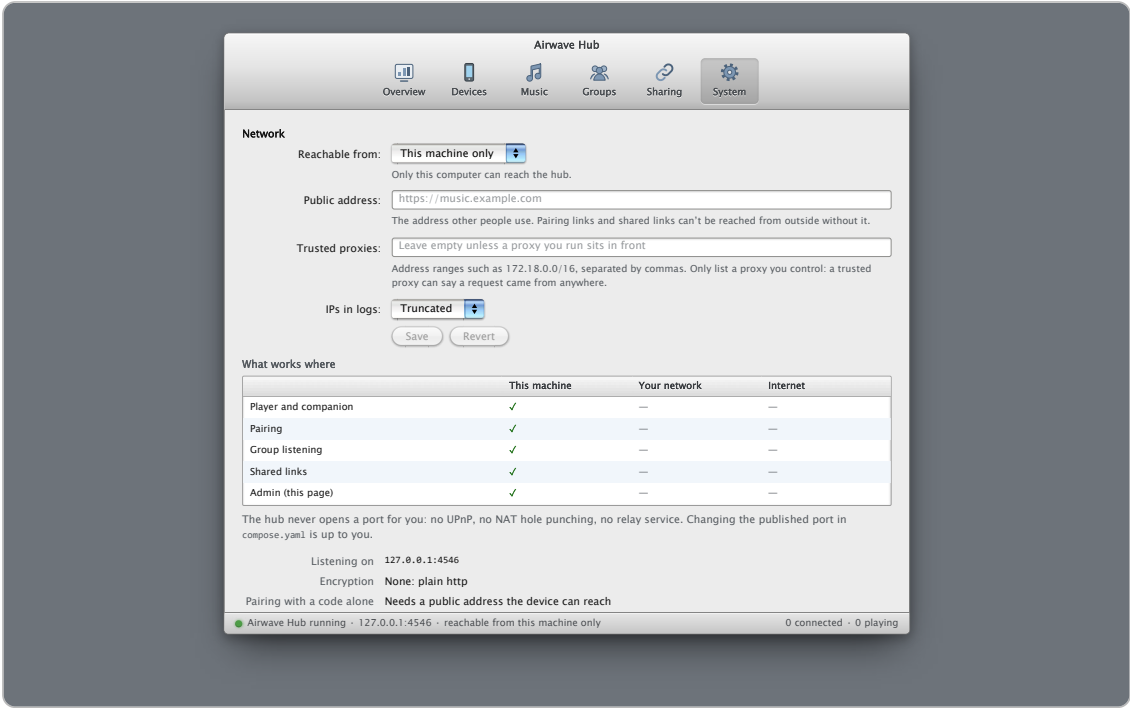


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 18px · desktop density, not held to 44px (DEC-027)

System ▶ Network

The hub's own Network view: how it listens and where it is reached from, with an explicit Save and Revert — nothing applies as you type.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

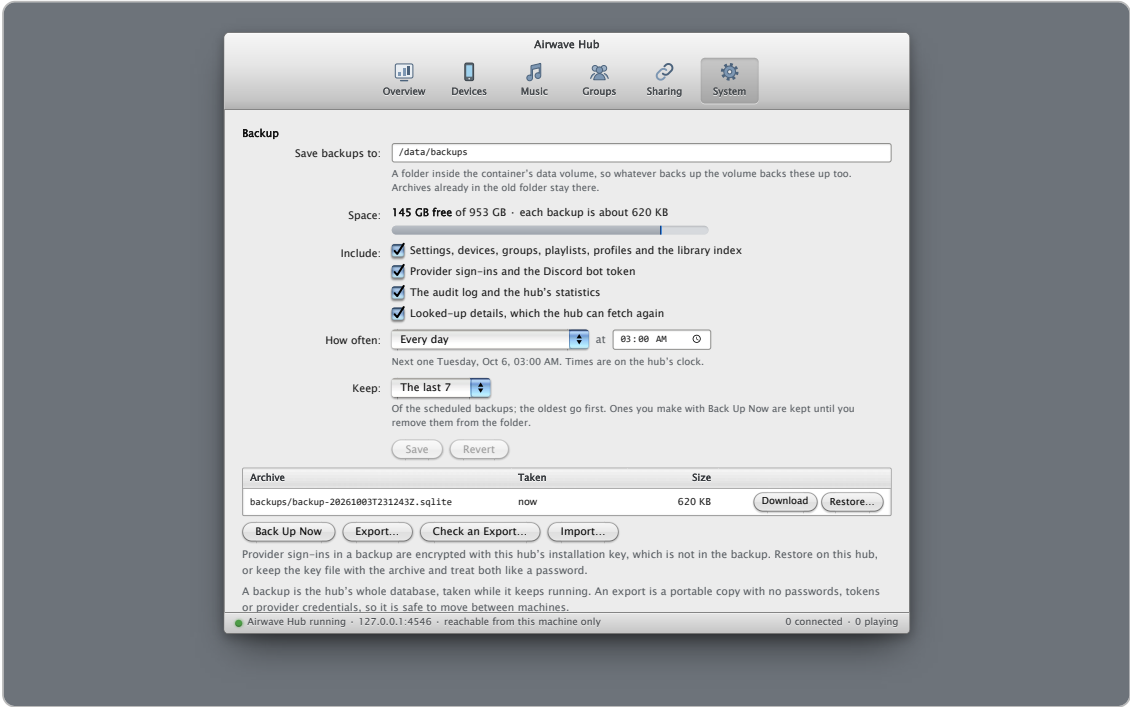


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

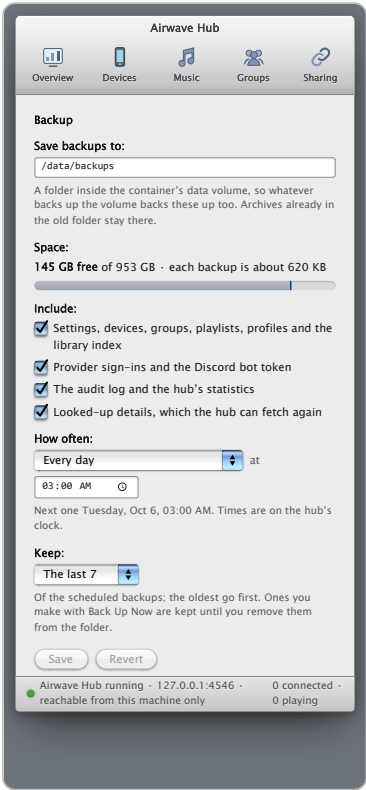
System ▸ Backup

The hub's own Backup view: one archive with its own Download, the space it has, and the settings — where, what goes in, the daily schedule and how many to keep.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

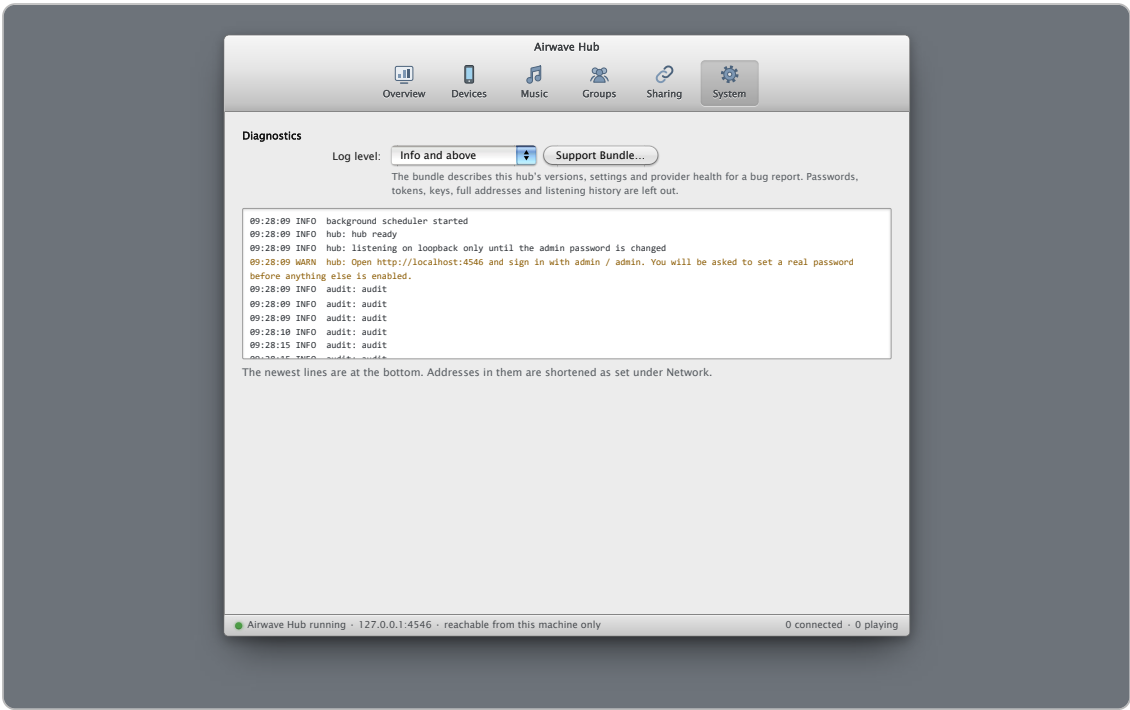


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 18px · desktop density, not held to 44px (DEC-027)

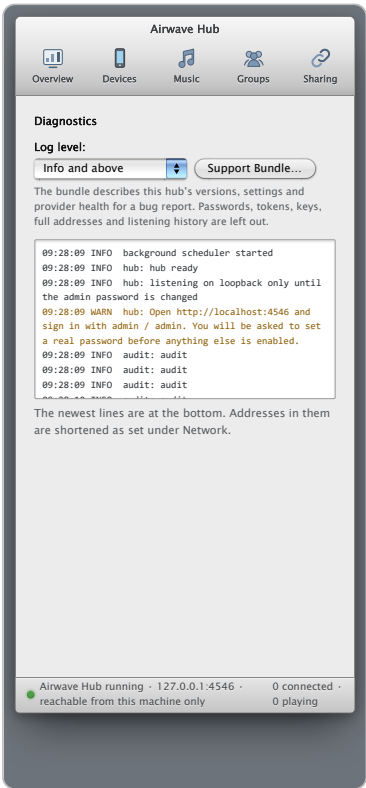
System ▶ Diagnostics

The hub's own Diagnostics view: the recent log at a chosen level and the support bundle, which says what it leaves out.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

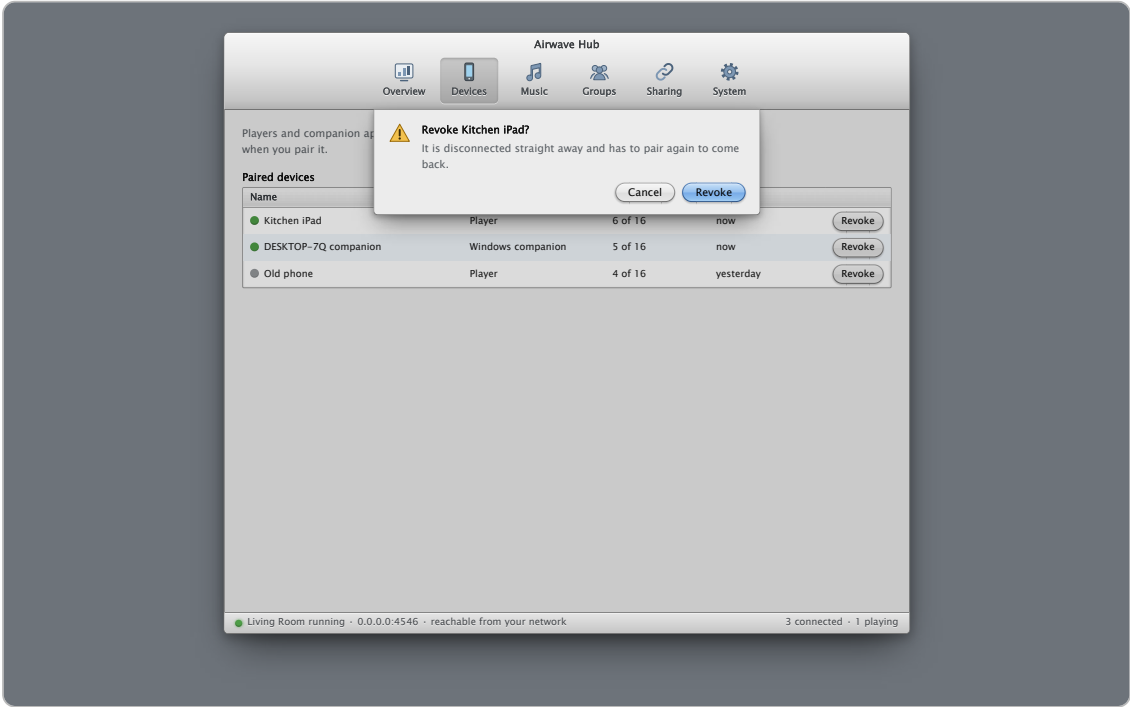


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

Confirmation sheet

A destructive action asks in a sheet that drops over the pane. Cancel has the focus, Escape cancels, and the window behind it is inert. This is the hub kit's own sheet, asked for real.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

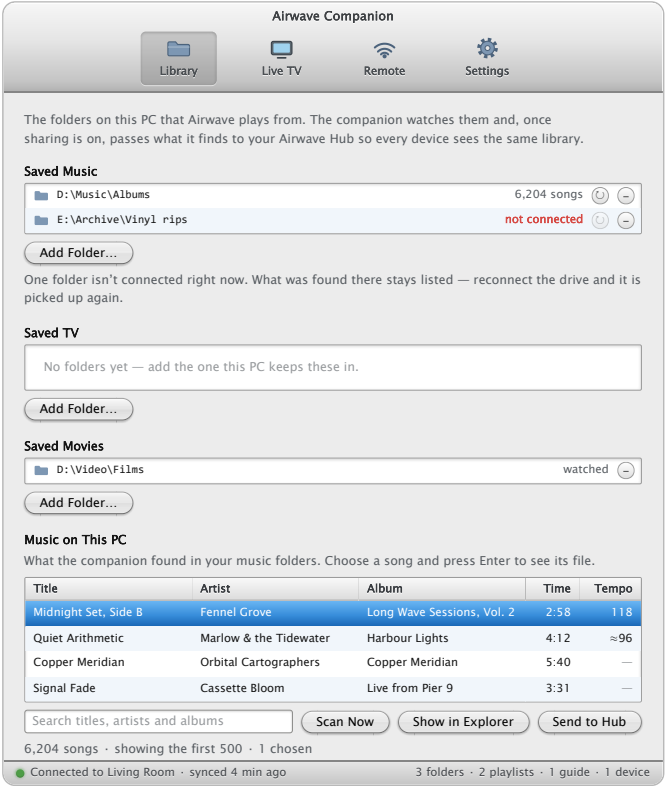


Phone 390 × 844, shown at 56% touch layer emulated

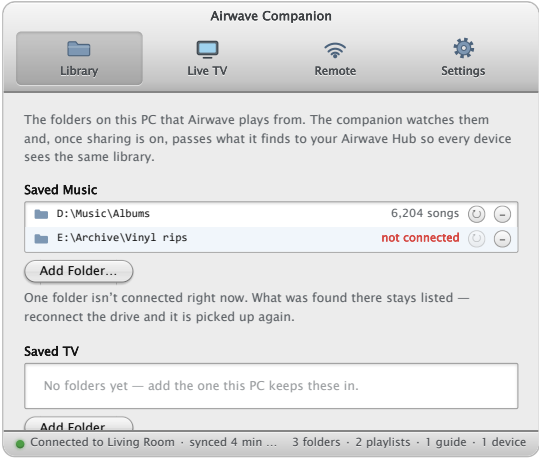
● Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

Library

Saved Music, Saved TV and Saved Movies, then the songs found. A folder whose drive is away is said so in its row and under the list.



As it opens 640 × 760, shown at 62%  
● Fits nothing off the side · smallest text 11px



Smallest window 520 × 440, shown at 62%  
● Fits nothing off the side · smallest text 11px

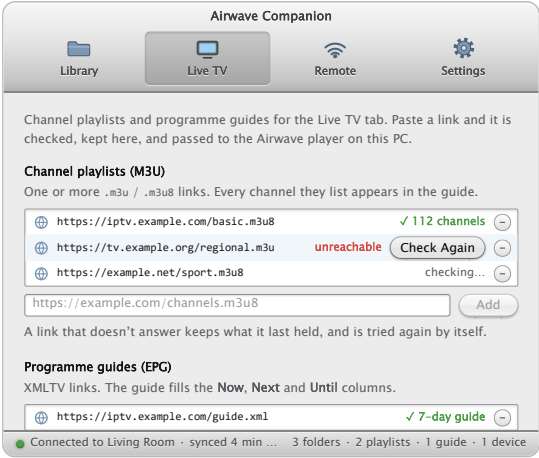
Live TV

A kept link, one that stopped answering (Check Again), one being checked, and a refused one: the reason under the field, the typed text left in it.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

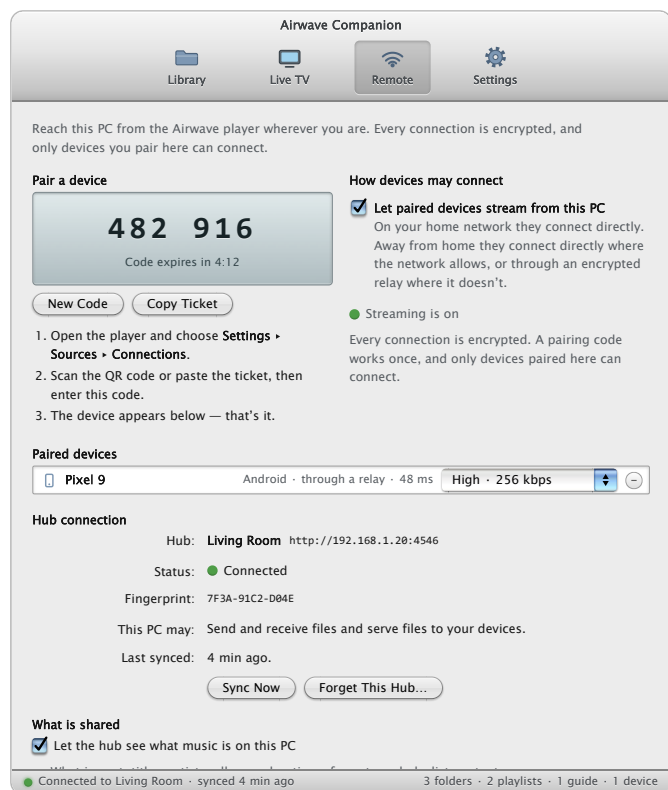


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

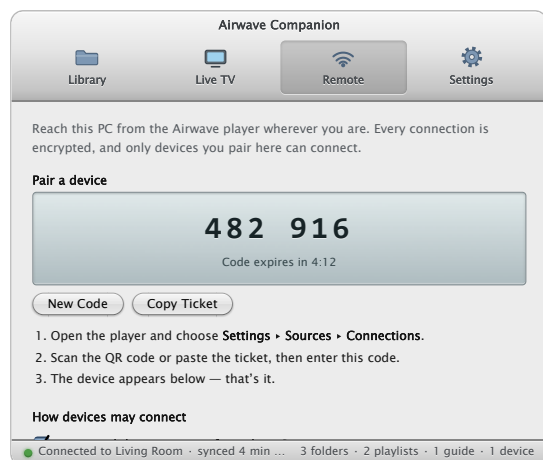
## Remote

Pairing a device, the hub connection and what is shared, and transfers. Untick sharing and Sync Now is disabled with its reason.



**As it opens** 640 × 760, shown at 62%

● Fits · nothing off the side · smallest text 11px



**Smallest window** 520 × 440, shown at 62%

● Fits · nothing off the side · smallest text 11px

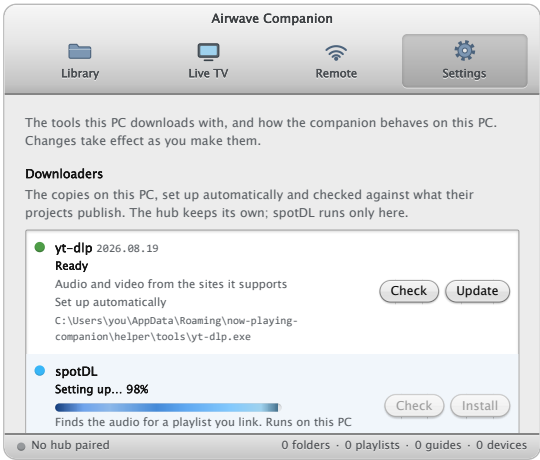
Settings

The companion's own Settings view, answered by a recorded companion: one downloader ready, one still being set up (a bar, and the badge on the tool), FFmpeg found on the PATH; then General with update checks and notifications, Downloads, and Network with the helper on this PC only.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

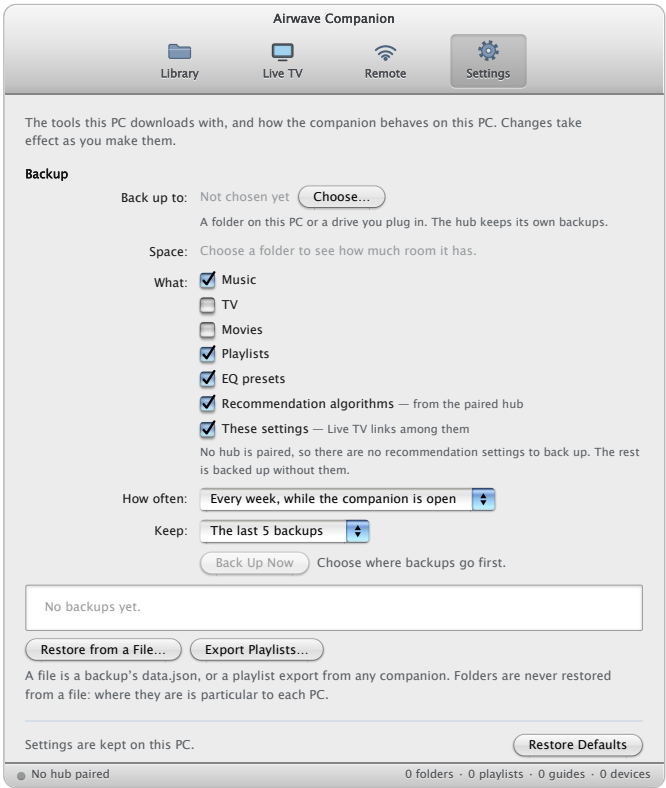


Smallest window 520 × 440, shown at 62%

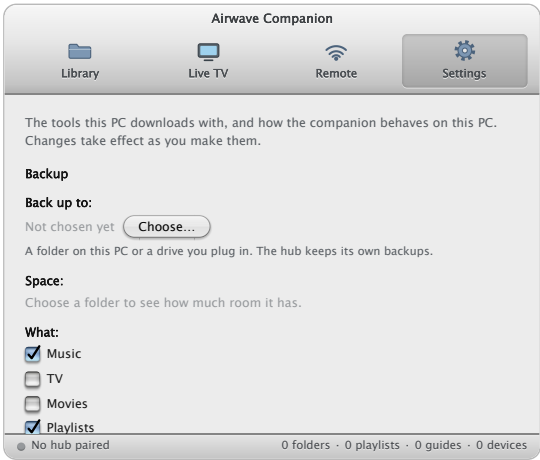
Fits nothing off the side · smallest text 11px

Settings ▸ Backup

The companion's own Backup view, recorded on a new install: a weekly schedule, five kept, what goes in (recommendation settings only once a hub is paired), and Back Up Now waiting for a folder to be chosen.



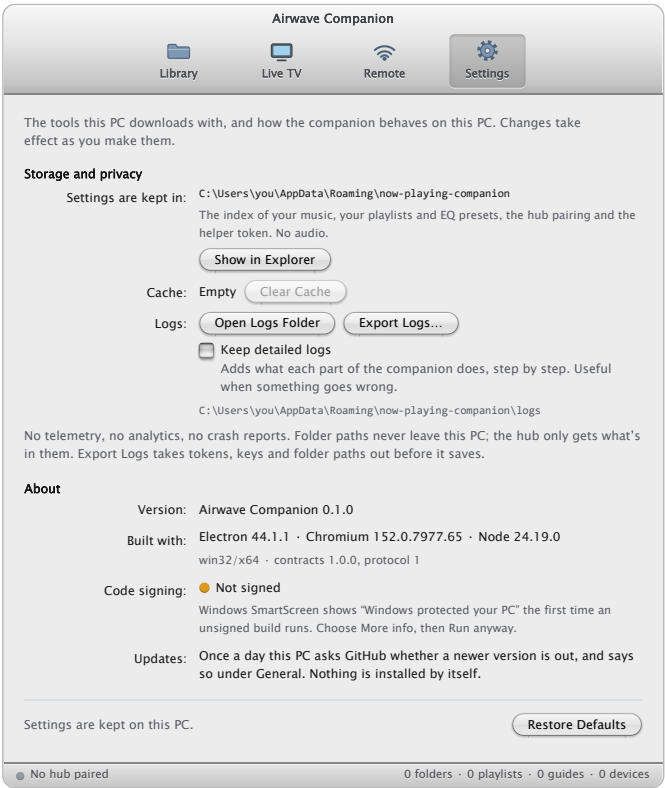
As it opens 640 × 760, shown at 62%  
● Fits nothing off the side · smallest text 11px



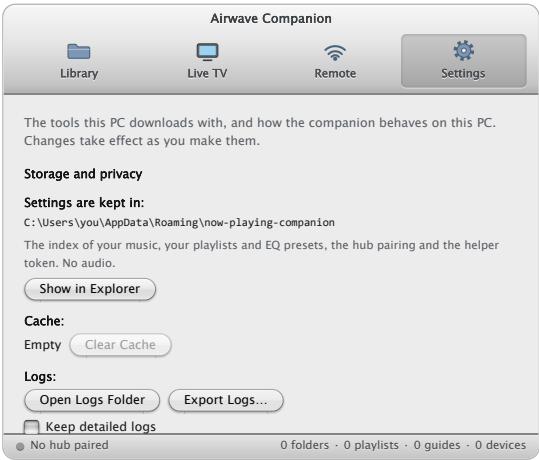
Smallest window 520 × 440, shown at 62%  
● Fits nothing off the side · smallest text 11px

Settings ▸ Storage and About

The companion's own About view: where its data lives, the cache and logs with their controls, and the build, with how updates are checked. The paths are the installed ones.



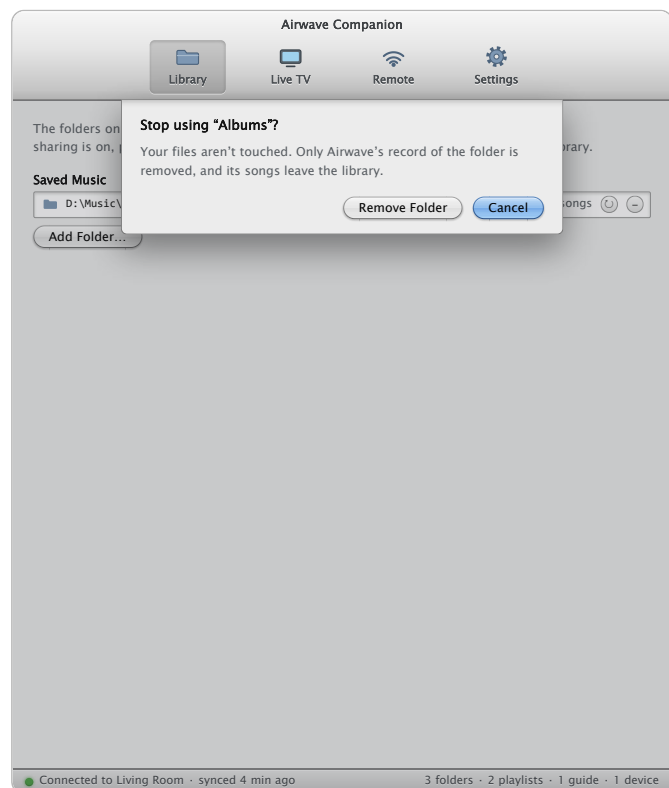
As it opens 640 × 760, shown at 62%  
● Fits nothing off the side · smallest text 11px



Smallest window 520 × 440, shown at 62%  
● Fits nothing off the side · smallest text 11px

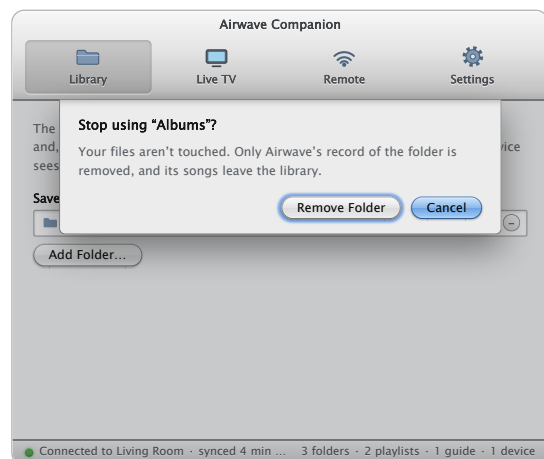
## Confirmation sheet

The companion's sheet is a modal dialog. For a destructive action Cancel is the default button, so Return never removes anything. This is the companion kit's own sheet, asked for real.



**As it opens** 640 × 760, shown at 62%

● Fits nothing off the side · smallest text 11px



**Smallest window** 520 × 440, shown at 62%

● Fits nothing off the side · smallest text 11px

## Sources and commands

This page is generated. The sources below are the design authority: change them, then rebuild. The exports — this page and its PDF — are views, so an edit or annotation on either is a request to change a source, not a change in itself.

| CONCERN           | SOURCE                                                                                                                                                                                                                                         | OWNS                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| designs           | design/frontends/airwave-now-playing.html<br>design/frontends/airwave-hub.html<br>design/frontends/airwave-companion.html                                                                                                                      | How each app looks. The player is the first file served as a shell (music-player/scripts/make-shell.py writes music-player/index.html from it); the hub and the companion are drawn with stylesheets generated from the other two.                                                                                                                                                                                                            |
| windowStylesheets | packages/aqua-ui/src/styles/airwave-window.css<br>packages/aqua-ui/src/styles/airwave-hub.css                                                                                                                                                  | What the hub and the companion wear: the window designs' own CSS, copied out by packages/aqua-ui/scripts/make-window-css.py (pnpm build:window-css). Generated — change the design file, never these.                                                                                                                                                                                                                                         |
| windowArt         | packages/aqua-ui/src/airwave/aqua-art.ts                                                                                                                                                                                                       | AquaArt: the drawn push button, pop-up and checkbox both windows wear, published as --aq-* custom properties. One module, imported as @now-playing/aqua-ui/airwave-art.                                                                                                                                                                                                                                                                       |
| windowKits        | docker-container/src/web/ui.tsx<br>docker-container/src/web/icons.tsx<br>docker-container/src/web/styles.css<br>windows-companion/src/renderer/ui.tsx<br>windows-companion/src/renderer/icons.tsx<br>windows-companion/src/renderer/styles.css | Each window's own kit, icons and remaining styles: the design's markup as components, the states the designs never drew, and (hub) the contrast inks. Per product on purpose (DEC-026).                                                                                                                                                                                                                                                       |
| tokens            | packages/aqua-ui/src/styles/tokens.json                                                                                                                                                                                                        | Documented values: colour, type, space, size, shadow, motion and the page skin for the component library; the 'airwave' group records the Airwave window's palette, which the design files own.                                                                                                                                                                                                                                               |
| stylesheets       | packages/aqua-ui/src/styles/aqua.css<br>packages/aqua-ui/src/styles/now-playing.css                                                                                                                                                            | The component library's custom properties (light scheme on :root, dark and touch layers after it). Mirror tokens.json by hand; see tokenMap. Loaded by the player's React source, the gallery and this guide — not by the hub or the companion.                                                                                                                                                                                               |
| tokenMap          | design/token-map.json                                                                                                                                                                                                                          | Which custom property carries each token, and recorded exceptions.                                                                                                                                                                                                                                                                                                                                                                            |
| overrides         | packages/aqua-ui/src/styles/overrides.css                                                                                                                                                                                                      | Temporary trials written by the styleguide's token editor, for the component library only. Empty in the repository.                                                                                                                                                                                                                                                                                                                           |
| components        | packages/aqua-ui/src                                                                                                                                                                                                                           | The component library: every shared React control, shell, overlay and state. Imported by the player's React source in music-player/src (kept, not served — DEC-019), the gallery and this guide; it exports only what the player imports, and the styleguide's usage scan lists any component no product imports so it can be removed. The hub and the companion import only the Airwave stylesheets and AquaArt from this package (DEC-026). |
| rules             | design/ux-rules.json                                                                                                                                                                                                                           | Interaction and content rules with stable IDs, their owning code and their tests.                                                                                                                                                                                                                                                                                                                                                             |
| coverage          | design/coverage.json                                                                                                                                                                                                                           | Every surface, state and journey, with discovery sources.                                                                                                                                                                                                                                                                                                                                                                                     |
| decisions         | design/decisions.md                                                                                                                                                                                                                            | Adopted, proposed and unresolved decisions and exceptions (DEC-###).                                                                                                                                                                                                                                                                                                                                                                          |
| prose             | docs/design/STYLEGUIDE.md                                                                                                                                                                                                                      | The reasoning behind the visuals, in long form.                                                                                                                                                                                                                                                                                                                                                                                               |
| conformance       | docs/AQUA_CONFORMANCE.md                                                                                                                                                                                                                       | Spec §17/§18 MUSTs mapped to tests, reviewer checks or deviations.                                                                                                                                                                                                                                                                                                                                                                            |
| specification     | docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md                                                                                                                                                                                             | The reconstruction target (Snow Leopard / iTunes 9).                                                                                                                                                                                                                                                                                                                                                                                          |

### CHANGING SOMETHING

1. Find the owner: a token, a component, a rule by ID, or a surface in the ledger.
2. Change it there — and, for a rule, the code that implements it and the test that proves it.
3. Run `pnpm styleguide:check` and the tests the rule names; then `pnpm styleguide:build` and `pnpm styleguide:pdf`.

4. Look at the result in the mockups at every size, and at every page of the PDF.
5. A deliberate exception goes in `design/decisions.md` with its reason. Never widen an exception to make a check pass.

## COMMANDS

|                               |                                                                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>BUILD:WINDOW-CSS</code> | regenerates <code>airwave-window.css</code> and <code>airwave-hub.css</code> from <code>design/frontends/airwave-companion.html</code> and <code>airwave-hub.html</code> .                                                                       |
| <code>STYLEGUIDE:BUILD</code> | rebuilds <code>docs/design/styleguide.html</code> from the sources above (same as <code>pnpm build:styleguide</code> ).                                                                                                                          |
| <code>STYLEGUIDE:CHECK</code> | sources and evidence exist, rule IDs are unique, every discovered screen is in the ledger, <code>tokens.json</code> matches the stylesheets (the library's and the Airwave window's), and the committed HTML was built from the current sources. |
| <code>STYLEGUIDE:SHELL</code> | after <code>pnpm build:player</code> , retakes the served player's screenshots for The served shell ( <code>packages/aqua-ui/styleguide/shell</code> ) and records which shell they show.                                                        |
| <code>STYLEGUIDE:PDF</code>   | prints the committed HTML to <code>docs/design/styleguide.pdf</code> with Playwright's Chromium (set <code>NP_STYLEGUIDE_BROWSER</code> to use an installed Chrome or Edge).                                                                     |
| <code>E2E</code>              | measures every mockup at every size.                                                                                                                                                                                                             |
| <code>CONFORMANCE</code>      | includes <code>packages/aqua-ui/tests/unit/aqua-conformance.test.ts</code> and <code>design-sources.test.ts</code> .                                                                                                                             |

## Changing any of this

---

- **Tokens first.** In the component library a new colour or size goes in `tokens.json`, then into the stylesheet as a custom property. In the Airwave window it goes in the design file's `:root`, then in the `airwave` group of `tokens.json`, which the check compares with the generated stylesheet. A literal hex in a component is a bug.
  - **Both schemes.** Light on bare `:root`; redefine only what changes inside the dark guard.
  - **The Airwave window changes in its design file.** Edit `design/frontends/airwave-companion.html` or `airwave-hub.html`, run `pnpm build:window-css`, and carry any change to the design's AquaArt script into `src/airwave/aqua-art.ts`. The player's shell changes in `airwave-now-playing.html` and is rebuilt by `make-shell.py`.
  - **Both skins, if it is a library control.** Button, Slider and friends are drawn in the window skin and the page skin.
  - **Say why in the CSS.** Every unobvious value carries a comment explaining what it reconstructs.
  - **The gates.** `AQUA_CONFORMANCE.md` maps every §17/§18 MUST to a test, a named reviewer check, or a recorded deviation, and `pnpm verify` runs the lot. A deviation is legitimate; an undocumented one is not.
  - **Rebuild this page.** `pnpm styleguide:build` regenerates `docs/design/styleguide.html` from the same sources the products use, `pnpm styleguide:check` fails if it no longer matches its sources, and `pnpm styleguide:pdf` prints it.
- 

The correction log in `DEVIATIONS.md` is worth reading before a large change. It records the mistakes this system has already made — buttons a decade too glossy, a list redesigned instead of expanded, a picker that indexed music it could never play — and every one was found by building the thing and looking at it.

## Coverage and decisions

What this guide covers and how each part was checked, for source revision 14aac31baf1723ec. The denominator is every surface in design/coverage.json, which the check keeps in step with the products' navigation.

| PRODUCT                                     | SURFACES | IN<br>MOCKUPS | SEEN<br>RUNNING | NOT RUN BECAUSE                                                                                                                                                                                            |
|---------------------------------------------|----------|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Airwave — the player                        | 37       | 5             | 5               | —                                                                                                                                                                                                          |
| Airwave Hub — admin window and public pages | 20       | 18            | 11              | Seen at runtime on 2026-10-03 (see runtime). Overview, Devices, Providers and Downloads were not captured in that run; they are covered by the hub's DOM, e2e, a11y and journey suites.                    |
| Airwave Companion — Windows                 | 12       | 11            | 4               | Electron-only; its renderer needs the preload bridge. Its Settings tab was seen in the packaged app on 2026-10-03 (see runtime); the other tabs are covered by its DOM and contract tests and the mockups. |
| Discord bot                                 | 1        | 0             | 0               | Needs a bot token and a Discord server; covered by the voice and parity tests.                                                                                                                             |
| Android shell                               | 2        | 0             | 0               | No Android SDK or JDK 17 on the machine that built this revision.                                                                                                                                          |
| Local helper                                | 1        | 0             | 0               | Console output was not captured for this revision; covered by its integration tests.                                                                                                                       |

Automatic checks when this page was built: 85 rules with evidence, 57 surfaces discovered from navigation, 12 Discord commands, 237 tokens compared with the stylesheets (1 recorded exceptions). No open problems.

recorded exception: font.family ↔ --aqua-font: Lucida Grande, Lucida Sans Unicode, Helvetica Neue, Arial, sans-serif vs "Lucida Grande", "Aqua Fallback", "Lucida Sans Unicode", "Helvetica Neue", Arial, sans-serif (DEC-002)

### SEEN IN A RUNNING PRODUCT

| WHEN       | SCENARIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | VIEWPORTS<br>· THEME | SURFACES |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|
| 2026-10-03 | A scratch hub (pnpm build:hub, then dist/server.js on port 4564 with a fresh data folder, stopped by its PID) seeded through its API the way an administrator would: sign in with admin/admin, set a real password, two library folders with six tagged songs, two groups and an invite, an album link valid for a week, a daily backup schedule keeping 7, one backup. The admin window was signed into in Chrome and each section opened by its #id and captured (outerHTML and a 1280×900 screenshot each), and the same hub's answers to every route those sections read were recorded (packages/aqua-ui/styleguide/fixtures/hub-api.json). Seen: the sign-in in the window, the six tools, and each section with its new controls — Create Link with the playlist-or-album picker and how long it works, the backup folder, parts, schedule and Keep with Save and Revert and a Download per archive, a Scan per library folder, Network and Recommendations with Save and Revert, Discord's plain-sentence warnings with Invite the Bot withheld, and Diagnostics' log and support bundle.<br>Playwright driving installed Chrome (DOM capture and screenshots); recorded API answers | 1280×900 · light     | 11       |
| 2026-10-03 | The packaged companion (win-unpacked, built 2026-10-03 17:37) started with PORTABLE_EXECUTABLE_DIR in a scratch folder, so no person's data was read or written, attached over CDP on port 9335 and stopped by its PID. The backup schedule was set to weekly keeping 5 through the bridge; the Settings tab was opened and the Backup, Storage and privacy and About groups captured (outerHTML and screenshots), and the answers to every channel the Settings tab reads were recorded (packages/aqua-ui/styleguide/fixtures/companion-ipc.json). Seen: yt-dlp ready with Check and Update, spotDL setting itself up, FFmpeg found on the PATH; Update downloaders automatically on; the update check on; the helper on this PC only; Backup with no folder yet (Back Up Now disabled with the reason), recommendation algorithms unavailable with the reason (no hub paired), Live TV links among these settings; the cache by part, the logs, and the build with how updates are checked.<br>Playwright over CDP against the packaged app                                                                                                                                               | 640×760 · light      | 4        |
|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      |          |

| WHEN       | SCENARIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | VIEWPORTS<br>· THEME            | SURFACES |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------|
| 2026-10-04 | pnpm build:player, then pnpm styleguide:shell: the production build served by vite preview on port 4183 (stopped by its PID), every outside host refused, eight tagged songs added through the shell's own import path, Gantry played, Radio opened with the bundled stations, Live TV filled from a stubbed companion helper (three channels, two with a guide), and Settings opened — at 1280×860 and 390×844. The screenshots are the guide's The served shell (packages/aqua-ui/styleguide/shell). scripts/styleguide-shell-shots.mjs (Playwright, installed Chrome) | 1280×860,<br>390×844 ·<br>light | 5        |

## EARLIER RECORDS, SUPERSEDED

What was seen running before an interface was replaced. Kept as history; these surfaces are not counted as seen.

- **2026-09-16** · Player dev server, empty library, no hub: each section, Settings, search with no match, the Shared segment (toast by pointer and keyboard), and one token change (--np-accent) seen in the player and the rebuilt guide, then reverted. These are the React interface's surfaces, served then and not now: the player serves the Airwave shell (DEC-019).
- **2026-09-16** · Hub admin dev server against a fresh data folder: the first-run sign-in screen. Signing in was not attempted; it needs a person to enter the password. The sign-in screen seen here was replaced by the Airwave Hub window, which draws sign-in inside the window (DEC-017).

## DECISIONS AND EXCEPTIONS

| ID      | STATUS       | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-001 | ● ADOPTED    | Lucida Grande is not shipped (not redistributable). The window skin names it first and falls back through a tuned chain.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DEC-002 | ● ADOPTED    | --aqua-font inserts the bundled "Aqua Fallback" face after Lucida Grande; tokens.json font.family does not list it. The CSS is authoritative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DEC-003 | ● ADOPTED    | The windows (the Airwave window of the hub and the companion, and the library's window-skin tokens) have no dark palette, because Snow Leopard had none: airwave-window.css declares its values for both OS appearances and renders light in each. The player and the styleguide follow the system colour scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DEC-005 | ● UNRESOLVED | Discord noResults renders No results for "{{reason}}", but the command service passes the search text as title and No results as reason, so the bot says "No results for "No results"." Proposed fix: the template uses {{title}}. Changing a default template changes what existing servers see.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DEC-006 | ● ADOPTED    | Cover Flow is not implemented; the 3D jewel-case stage stands in for it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DEC-007 | ● ADOPTED    | The player replaces the inset LCD display with the hero, which states the same information at reading size.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DEC-008 | ● UNRESOLVED | The public share page (docker-container/src/api/routes/shares.ts) is standalone HTML under a strict CSP with its own inline stylesheet. It uses a 12px pill button and weight 600, which the system otherwise forbids. Proposed: restyle it with the page-skin values (4px push button, weights 400/700).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DEC-009 | ● ADOPTED    | The mockups are not the product view files executed verbatim: those are wired to a database, an API or Electron's bridge. The hub's and the companion's frames hold the stylesheets those products load (the generated Airwave sheets and each styles.css), AquaArt, and the products' own kits and icons, arranged as the views arrange them, with fixture data. The player's frames are the component library's page skin, because the served shell is one document with its own scripts and cannot be portalled into a frame. Runtime coverage says which surfaces were seen in the running product and which only in a mockup. Since DEC-030 most hub and companion sections are the products' own views; this still holds for the screens arranged in airwave-screens.tsx and for the player's page-skin frames. |
| DEC-010 | ● ADOPTED    | Android notifications, the document picker and the save dialog are system UI. The guide documents their copy (strings.xml) and when they appear, not their appearance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DEC-011 | ● PROPOSED   | tokens.json is mirrored into the stylesheets by hand; pnpm styleguide:check compares the two through design/token-map.json. Proposed: generate the custom properties from tokens.json so the map becomes unnecessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DEC-012 | ● UNRESOLVED | The Android shell only reaches hubs over https (mixed content is never allowed and the manifest forbids cleartext), but the player's pairing form still accepts an http:// address, so the refusal happens at connection time. Proposed: say so in the pairing form on Android.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DEC-013 | ● ADOPTED    | The Windows companion is Electron-only; its renderer cannot run in a browser without the preload bridge (opened in one, it says so and offers nothing), so its runtime evidence comes from its DOM tests and the mockups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DEC-014 | ● ADOPTED    | The equalizer window (.eqw) has an intrinsic width: max-width: 600px, which holds its rails about 36 px apart the way the iTunes window does. It had been stretching to whatever container held it — 958 px in the styleguide — and read as a strip rather than a window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| ID      | STATUS    | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-015 | ● ADOPTED | The note in every app icon is placed by centring the path's measured box (x 4.4–19.8, y 3–21.7 of its 24-unit grid) on the icon's optical target: the tile centre for the player and maskable icons, the folder mouth (240, 308) for the companion, the 32-square for the tray. The player icon had shipped 27 px left of centre; <code>tests/unit/icon-glyphs.test.ts</code> now pins all four placements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DEC-016 | ● ADOPTED | Play sits on the progress rail's centre line at every width, with previous and next on its line either side. <code>KeyTransport</code> measures its row: the volume line stays beside the keys only while it fits on both sides (otherwise it goes beneath, centred), and the keys wrap only when they do not fit on one line, with previous, play and next alone on the first line. The player's eleven keys never leave room for the volume beside them inside the 820 px hero, so the player shows it beneath at every width. Covered by <code>tests/dom/page-chrome.test.tsx</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DEC-017 | ● ADOPTED | The hub admin GUI is the six-tab window of <code>design/frontends/airwave-hub.html</code> (Overview, Devices, Music, Groups, Sharing, System), drawn with the stylesheets generated from that file ( <code>airwave-window.css</code> , <code>airwave-hub.css</code> ), AquaArt, and the hub's own kit ( <code>docker-container/src/web/ui.tsx</code> ); the thirteen sections keep their data flows and ledger entries and stack inside those tabs, and <code>#section-id</code> in the address still opens the right one. It is one window in every state. Not signed in, the sign-in form is drawn in the same window with every tab locked. On first run the design's amber gate sits at the top of Overview and the other five tabs are locked with the reason: the server refuses every gated route until the password is changed, so the tiles that depend on those routes say they are off rather than showing a number. What the design draws and the hub did not have when the window was first built — a form that creates a shared link, Live TV on the hub, and the Backup tab's path, schedule and include controls — it has now: see DEC-028, DEC-029 and the hub's Live TV row ( <code>hub-live-tv</code> ). The invite table shows who an invite is for, its role and its state, not its code, because the hub keeps only a code's hash; the code appears once, when made. The header and gate colours that joined <code>tokens.json</code> as <code>header*</code> and <code>warningBg/warningEdge</code> for the earlier window skin stay in the library; the hub now reads the design's own <code>--head-*</code> and <code>--warn-*</code> . |
| DEC-018 | ● ADOPTED | The Windows companion is the four-tab window of <code>design/frontends/airwave-companion.html</code> (Library, Live TV, Remote, Settings), drawn with <code>airwave-window.css</code> , AquaArt and its own kit ( <code>windows-companion/src/renderer/ui.tsx</code> ). Live TV is real: a pasted M3U or XMLTV link is read in the main process, kept only when it holds channels or programmes, and served to the player on this PC through the helper routes <code>/helper/v1/tv/channels</code> and <code>/helper/v1/tv/guide</code> (UX-TV-001, NP-TV-001). Remote ▶ Pair a device is the AWSP streaming server's pairing; the hub connection and transfers sit under it. Design controls not adopted, because the app cannot do what they promise: separate Wi-Fi and mobile-data toggles (one switch: paired devices may stream); per-tool Check, Update and Install and an auto-update toggle (tools are set up automatically and re-checked together with Check All); the Downloads fieldset; update checks and update notifications; running the helper without pairing as a choice (it always runs); cache and log controls; and "last seen" on a paired device (the row says how it is connected now). A backup is a dated folder ( <code>data.json</code> + copied folders), measured with <code>local-helper/src/measure.ts</code> so the helper route and this window report one number; sizes are decimal units as the player shows them.                                                                                                                                                                                                         |
| DEC-019 | ● ADOPTED | The player's interface is the Airwave frontend served as the shell (shape (a) of the master prompt): <code>music-player/index.html</code> is generated from <code>design/frontends/airwave-now-playing.html</code> by <code>music-player/scripts/make-shell.py</code> , every edit asserted against the text it replaces, and <code>src/shell/bridge.ts</code> stands behind the file's seams with <code>db.ts</code> , <code>library.ts</code> , <code>PlaybackEngine</code> , <code>HubClient</code> and bundled <code>three.js</code> . Chosen because the edits that make playback real touch one transport block and one boot block, and the file's twenty-one suites come with it. What the shell no longer contains: the sixteen invented rows, the generated Statistics year, the demo history, the invented TV catalogue, the base64 clips, the two invented albums (now "Nothing playing" / "No group session"), and the direct call to <code>api.anthropic.com</code> (a third-party account). The React interface under <code>music-player/src/{App,views,components}</code> is no longer served; its logic layer ( <code>src/lib,src/state</code> ) is what the bridge uses. Back Up Now in the player measures and says nothing was sent: no route lets one machine start a backup on another.                                                                                                                                                                                                                                                                                                                                                     |
| DEC-020 | ● ADOPTED | The companion's chrome is its title bar. The operating system's title bar is hidden (the frame is kept, so the window still snaps, resizes and maximises), the page draws the title, and Windows draws its own minimise, maximise and close over the top-right corner in a clear 26 px overlay. The chrome is the drag region; the tools are not, so they stay clickable. The application menu is removed. The window is named once, from <code>PRODUCT_NAME</code> . Rule UX-CHROME-001; pinned by <code>windows-companion/tests/contract/app-identity.test.ts</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DEC-021 | ● ADOPTED | In the hub and the companion a confirmation is a sheet that drops from under the toolbar over the window it belongs to, never a second window or the browser's own dialog. The safe answer has the keyboard: in the hub Cancel takes the focus; in the companion Cancel is the default button for a destructive action, so Return never removes anything. Escape cancels, focus is held in the sheet and returns to what asked. The two are built differently — the hub's is a <code>role="alertdialog"</code> layer over an inert window, the companion's a modal <code>&lt;dialog&gt;</code> — and each product keeps its own (DEC-026). Rule UX-SAFE-001.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|         |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| ID      | STATUS       | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-022 | ● ADOPTED    | The hub web adds four inks to the design's palette, in <code>docker-container/src/web/styles.css</code> : <code>--ink-quiet #6b6f73</code> , <code>--success-ink #2f7a28</code> , <code>--danger-ink #b3322c</code> and <code>--link #1764b2</code> , and sets a struck-through capability chip's text to <code>#6b6b6b</code> . The design's <code>--ink-3 (#96999c)</code> , <code>--success (#4e9d47)</code> and <code>--danger (#d64a44)</code> measure 2.9:1, 3.4:1 and 3.9:1 as text on white, under WCAG AA's 4.5:1; the hues are kept and only their lightness moves. The design's values still colour lamps, bars and placeholders. All five are recorded in <code>tokens.json</code> ( <code>airwave.inkQuiet</code> , <code>successInk</code> , <code>dangerInk</code> , <code>link</code> , <code>capOffInk</code> ) and checked against the stylesheet.                                                                                                                                                                                                                                                                      |
| DEC-023 | ● ADOPTED    | In the companion, pairing with a hub that grants <code>library:share</code> is the decision to share: sharing is on from pairing until it is switched off under Remote ▶ What is shared, which is how the app behaved before the checkbox was wired. The checkbox shows what was chosen, Sync Now is disabled with the reason while sharing is off, and forgetting a hub turns sharing off for the next one.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DEC-024 | ● ADOPTED    | The hub's downloader is ready without setup (owner decision, 2026-10-03; see docs/DOWNLOADS_AND_LEGAL.md). The external-tool provider is enabled by default and is <code>yt-dlp</code> when nothing is configured; a hub outside the image finds the tool in the image path, its own tools folder or <code>PATH</code> , and fetches a verified copy when there is none. Allowlisted hosts, no cookies and a rights basis on every request are unchanged, and the Providers table still shows on or off as the hub reports it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DEC-025 | ● ADOPTED    | The suite is named Airwave, as its three designs name it: Airwave (the player), Airwave Hub and Airwave Companion, from <code>packages/contracts/src/branding.ts</code> . The player's shell keeps the design's name; "Now Playing" is the player's home page and nothing else. Machine identifiers (the slug, URL scheme, storage namespace, User-Agent, the Windows appid, the package names and the companion's data folder) keep their spelling, so no library, pairing or taskbar pin is orphaned. This guide is the Airwave Style Guide.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DEC-026 | ● ADOPTED    | One style guide covers the three apps, and the guide says what each is built from. AquaArt — the drawn push button, pop-up and checkbox — is one module, <code>packages/aqua-ui/src/airwave/aqua-art.ts</code> , imported by the hub and the companion as <code>@now-playing/aqua-ui/airwave-art</code> ; the two copies it replaces drew identical bytes. The two window kits stay per product: they write the same classes, but with different props and, for the sheet and the checkbox, different markup, so one shared component would change what one of the products renders. The <code>packages/aqua-ui</code> React components and the stylesheets <code>aqua.css</code> and <code>now-playing.css</code> are kept and tested for the player's React source; the styleguide reads from the products' imports which components it uses. The components no product imported, and <code>aqua-window.css</code> and <code>aqua-media.css</code> , which only they wore, were removed on 2026-10-03; the guide names any component that falls out of use so it can go the same way.                                                   |
| DEC-027 | ● UNRESOLVED | The hub's admin window is often opened on a phone, and there it keeps the design's desktop density: 22 px push buttons and 11 px text, under the 44 px and 12 px floors the page skin holds (UX-TOUCH-001). The design's touch layer covers only list rows and the round minus. The styleguide measures and prints those sizes in the hub's phone frames but does not fail the hub on them. Proposed: add a touch layer to <code>design/frontends/airwave-hub.html</code> (taller buttons, fields and tools under <code>pointer: coarse</code> ) and hold the hub to the same floors. The companion is a desktop window and is not affected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DEC-028 | ● ADOPTED    | The admin window makes shared links from what the hub holds. A device shares from its own library and uploads the item list itself; the admin has no library of its own, so Sharing ▶ Create Link offers the playlists devices have synced to the hub and the albums in the hub's library, and for a playlist the hub builds the item list from its synced copy ( <code>docker-container/src/shares/sources.ts</code> ) — the same metadata a device would have sent. What a link grants does not change: items stream only where the hub holds the same file, and nothing can be downloaded through it. The full link is shown once, when made.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DEC-029 | ● ADOPTED    | Hub backups run on a schedule on the hub's own clock: every day, or one day a week, at a time; off is allowed. The next run is the first slot after the schedule was last saved or last ran, so a hub that was off at its slot takes the backup when it next starts — once, not once per missed slot — and a changed schedule counts from the moment it is saved (changing weekly to daily at 02:59 does not mean a backup at 03:00 for every day that was missed). A failed run waits for its next slot rather than retrying every minute. Only scheduled backups count against "keep the last N"; ones made with Back Up Now stay until removed from the folder. The companion's schedule is the same idea while it is open: a day or a week since the last backup makes a run due, and one that could not happen is reported.                                                                                                                                                                                                                                                                                                          |
| DEC-030 | ● ADOPTED    | The style guide runs the hub's and the companion's own view components where it can. The views reach their product through one module each — the hub's API client ( <code>docker-container/src/web/lib/api.ts</code> ) and the companion's bridge ( <code>windows-companion/src/renderer/bridge.ts</code> ) — and the styleguide build, only, swaps those two for <code>packages/aqua-ui/styleguide/fixtures/fake-hub-api.ts</code> and <code>fake-companion-bridge.ts</code> , which answer reads from recordings of a real hub and a real companion ( <code>record-hub-api.mts</code> , <code>record-companion-ipc.mjs</code> ; machine paths written as <code>/data</code> and the installed data folder) and refuse every change with a sentence. Screens that need a state a recording cannot reach stay arranged by hand (DEC-009). The served player cannot be put in a frame, so it is shown as screenshots of the production build ( <code>pnpm styleguide:shell</code> ); <code>shots.json</code> records the hash of the shell they show and <code>pnpm styleguide:check</code> fails when the shell changes without new ones. |

| ID      | STATUS    | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| DEC-031 | ● ADOPTED | The companion streams to paired devices on metered connections by default, as it does on Wi-Fi and Ethernet: the person who switched streaming on asked for it to work away from home, where a phone hotspot is common, and a default that silently refused would read as broken. Each connection kind has its own switch under Remote; on one that is switched off streaming pauses and says why, reading the cost from what Windows reports (UX-STREAM-001).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DEC-032 | ● ADOPTED | The local helper can be opened to the home network, and only to be read. It is off by default. Switched on, other devices may use the four token-free read routes in <code>LAN_READ_ROUTES</code> (health, the radio's now-playing, and the Live TV channels and guide) by this PC's own address only; every other route still refuses them, even with the token, and the radio and TV routes keep their own rule that no page and no token is not enough. A rebound name or a page from the internet is refused. Downloads, files and settings stay on this PC (UX-NET-001).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DEC-033 | ● ADOPTED | The companion's check for a new version is keyless and opens nothing it was told to. It asks the project's public releases on GitHub (no key, no account), compares versions as numbers, and says what it found; Download opens the fixed release page through the main process. No URL from the API's answer is ever opened, so a tampered answer cannot send the person anywhere. The check is on by default, can be switched off, and asks at once when switched on (UX-PRIV-001).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DEC-034 | ● ADOPTED | The companion backs up the paired hub's recommendation settings as seeds, not as the hub's internal profile. The backup keeps the artists and genres the taste profile weighs highest ( <code>seedsFromProfile</code> ), named with the hub they came from; restoring hands them back to a hub as starting points, which that hub weighs with its own data. A raw profile would tie a backup to one hub's engine version and could not be offered to another hub. With no hub paired, or one that cannot give them, the rest is backed up and the reason is said.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DEC-035 | ● ADOPTED | The component library keeps only what a product imports. The 57 components no product imported — the earlier window skin (window frame, toolbar, transport, LCD, scrubber, search field, results popover, source list and their icons), tabs, menus, avatars and the unused state panels — were removed with their tests, gallery entries, specimens and the two stylesheets only they wore ( <code>aqua-window.css</code> , <code>aqua-media.css</code> ), at the owner's request. What remains is imported by the player's React source; the parts MusicList uses (Marquee, its row menu and scroller) stay private to it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DEC-036 | ● ADOPTED | The hub's downloader matches the companion's (owner decision, 2026-10-04): the image ships spotDL beside yt-dlp (verified through the hub's own installer; the image moved from Debian bookworm to trixie because spotDL's only Linux build needs glibc 2.38), downloads carry full tags and the cover, playlist, set and album links are supported, and a link has its real title without any API key. A playlist link is one job per entry — at most 200, duplicates skipped, one rights basis checked against every entry — and Music ▸ Downloads shows it as one row per entry whose From cell names the list and the position ("Soulhack, 2 of 11"), in the existing table markup with no new chrome. The provider's row names the tools that are on the hub ("External media tool (yt-dlp, spotDL)"). Unchanged: allowlisted hosts (open.spotify.com added for spotDL only), no cookies or credentials, a rights basis on every request, and Spotify's own audio is never touched — spotDL fetches a matching recording from YouTube Music. Supersedes the TOOL_PRESETS note that spotDL belonged only to the local helper. |

#### NOT COVERED, ON PURPOSE

- `docker-deployment` — Docker is delivery, not a visual surface. Its ports, health check and how rebuilt assets reach users are in the engineering appendix of the styleguide.
- `styleguide` — The styleguide documents the products; it is not one of them.
- `aqua-ui-gallery` — Developer specimen page (`pnpm --filter @now-playing/aqua-ui dev`); its specimens are rendered in the styleguide.